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Measuring and Closing the Gap Between Declared and Real Urgency in a Supply Chain Digital Twin

*Cognitive adequacy scoring, forecast-assisted declaration,
and an opening towards satellite-observed capacity*

Engineering and Applied Sciences

Thesis

MSc in Computational and Software Techniques in Engineering

Digital Engineering Design

ESTIA Institute of Technology, Bidart

Academic Year: 2025–2026

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August 2026



Research carried out at the Direction Innovation Numérique, ALTEN Group

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Use of software tools

Numerical results were produced with the Python codebase described in Section 3.3, using NumPy, SciPy, Matplotlib, Shapely, PyProj and Ultralytics. Large language models were used in two distinct and clearly separated capacities. First, as an *object of study*: the declarants of both campaigns are language models, an experimental design choice reported and discussed as such in Sections 3.4.4 and 4.9. Second, as a *writing aid* for language correction and $\text{\LaTeX}$ formatting. All scientific claims, all numerical results and all interpretations are my own, and every figure in this thesis was produced either by the project's own code or drawn by hand.

Abstract

In mutualised logistics networks the urgency an operator declares often diverges from the urgency computable from the operational state. Over-declaration wastes shared capacity; under-declaration hides rupture risk. No existing supply chain digital twin measures that divergence or propagates it across a supplier network.

This thesis builds and tests one. Real urgency is aggregated at each node from six KPI blocks through a weighted probabilistic OR and propagated upstream across a directed acyclic supply graph; declared urgency is elicited through a consistency-gated pairwise questionnaire and propagated downstream; their mismatch is scored on $[0, 100]$ with an asymmetric Prospect-Theory penalty. It was evaluated through two pre-registered campaigns with language-model declarants: a nineteen-turn replay of the 2020 to 2022 semiconductor shortage over eight nodes, with a blind control arm and a forecast-assisted arm on one frozen scenario; and an airframer supply base of fifteen nodes carrying no exogenous shock, whose milestone truth is derived from the indicator trajectory rather than scripted.

Three results follow. As a device for deciding which supplier to attend to, the gap works and works early: over the six turns preceding the public onset of the crisis it flags only nodes that later miss a delivery commitment, three of the four that eventually do, and the worst is flagged eight turns before its miss while that node's own manager declares an urgency of 0.001. The second chain bounds that claim: where declarants judge their state correctly and simply withhold escalation, the gap selects nothing, so the instrument detects misperception and not discretion. As a device for attaching a calibrated probability to a named week it fails, at a Brier skill between -3.4 and -6.9 against a base-rate forecaster, with twenty-one confident predictions of which none materialised. One block explains it: the time block reads a lead-time distribution that no shock ever reached. The repair is not the obvious one, since lost time must be added to the completion date rather than multiplied into it, and it halves the Brier deficit without making it positive.

Two methodological findings generalise beyond this system. The definition of an adverse outcome moves the reported calibration by a factor of five while leaving discrimination unchanged. And displaying a forecast to the declarant transfers authority from the shop floor to the model, in whichever direction the model points, on a layer that a two-number calendar ratio outranks at six horizons of eight. Every rate above is bounded by the sample rather than the construction: two scenarios, twenty-three nodes, seven positives. The instrument and its diagnostic method transfer; the rates do not yet.

Keywords: supply chain digital twin, urgency adequacy, cognitive bias, multi-criteria decision analysis, forecast calibration, automation bias, serious game.

Résumé

Dans les réseaux logistiques mutualisés, l'urgence qu'un opérateur déclare diverge souvent de celle que l'on peut calculer à partir de l'état opérationnel. La sur-déclaration gaspille de la capacité partagée ; la sous-déclaration dissimule un risque de rupture. Aucun jumeau numérique de chaîne d'approvisionnement existant ne mesure cette divergence ni ne la propage à l'échelle d'un réseau de fournisseurs.

Ce mémoire en construit un et le met à l'épreuve. L'urgence réelle est agrégée à chaque nœud à partir de six blocs de KPI par un OU probabiliste pondéré, puis propagée vers l'amont sur un graphe orienté acyclique ; l'urgence déclarée est élicitée par un questionnaire de comparaisons par paires soumis à un contrôle de cohérence, puis propagée vers l'aval ; leur écart est noté sur $[0, 100]$ par une pénalité asymétrique issue de la théorie des perspectives. Le système a été évalué par deux campagnes pré-enregistrées dont les déclarants sont des modèles de langage : un rejeu en dix-neuf tours de la pénurie de semi-conducteurs de 2020 à 2022 sur huit nœuds, avec un bras témoin en aveugle et un bras assisté par prévision sur un scénario gelé ; et la base d'approvisionnement d'un avionneur, quinze nœuds sans aucun choc exogène, dont la vérité jalon découle de la trajectoire des indicateurs au lieu d'être écrite à la main.

Trois résultats en découlent. Comme instrument pour décider quel fournisseur regarder, l'écart fonctionne, et tôt : sur les six tours précédant l'irruption publique de la crise, il ne signale que des nœuds qui rateront un engagement de livraison, trois des quatre qui finiront par le faire, et le plus mal noté est signalé huit tours avant son échec alors que son propre responsable déclare une urgence de 0,001. La seconde chaîne borne cette affirmation au lieu de la confirmer : là où les déclarants jugent correctement leur état et diffèrent simplement l'alerte, l'écart ne désigne rien. L'instrument détecte donc une méprise, non une rétention. Comme instrument pour attacher une probabilité calibrée à une semaine nommée, il échoue : *skill* de Brier compris entre $-3,4$ et $-6,9$ face à un prévisionniste annonçant le taux de base, et vingt-et-une prévisions confiantes dont aucune ne s'est réalisée. Un seul bloc l'explique : le bloc temporel lit une distribution de délai qu'aucun choc n'atteint jamais. Le correctif n'est pas celui qu'on croit, puisque le temps perdu doit s'ajouter à la date d'achèvement et non s'y multiplier, et il divise par deux le déficit de Brier sans le rendre positif.

Deux constats de méthode dépassent ce système. La définition de l'issue déplace la calibration rapportée d'un facteur cinq sans toucher à la discrimination. Et afficher une prévision au déclarant déplace l'autorité du terrain vers le modèle, dans le sens où pointe le modèle, sur une couche qu'un rapport calendaire à deux nombres surclasse à six horizons sur huit. Ce qui borne chacun de ces taux est l'échantillon et non la construction : deux scénarios, vingt-trois nœuds, sept positifs, trente-trois jalons observables.

Mots-clés : jumeau numérique de chaîne d'approvisionnement, adéquation d'urgence, biais cognitif, aide multicritère à la décision, calibration de prévisions, biais d'automatisation, jeu sérieux.

Acknowledgements

I would like to thank everyone who supported this research project at the Direction Innovation Numérique of the ALTEN Group.

My particular thanks go to **Émilienne Lardy**, **Selma Ferhat** and **Sébastien Perthuisot**. Émilienne and Selma supervised the scientific side of this work throughout, and their demands, their availability and the quality of their reviews did more than anything else to raise the rigour of what follows. Sébastien, Research Programme Director and Quality & Supply Chain Expert at the DIN, provided the industrial supervision and the confidence to let the scientific direction of the subject develop as it did.

I thank my academic supervisors, **Dr Peter Sherar** of the Centre for Computational Engineering Sciences at Cranfield University, and **Prof. David Gomez** of the ESTIA Institute of Technology, for their methodological guidance and for the academic framing of this thesis.

Finally, I thank the whole Direction Innovation Numérique team, and the other interns in the department, for the welcome they gave me and for the discussions that fed this work from beginning to end.

Acronyms

CI Consistency Index. 94

CR Consistency Ratio. 94

DIN Direction Innovation Numérique. 1

DT Digital Twin. 8

GLEC Global Logistics Emissions Council. 1, 12

HGV Heavy Goods Vehicle. 2

IPG Itinerary Performance Index. 2

LPPL Log-Periodic Power Law. 6

MCDM Multi-Criteria Decision-Making. 5, 8

MTS Manchester Triage System. 11

PI Physical Internet. 5

RI Random Index (Saaty). 94

SC Supply Chain. 8

TMT Temporal Motivational Theory. 6

TTR Time To Recovery. 2

TTS Time To Service. 2

VECTO Vehicle Energy Consumption calculation TOol. 12

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1 Introduction

1.1 Industrial and Technical Context

Supply chain planning is a multi-dimensional assessment problem. An operator deciding which of two late shipments to expedite weighs cost, deadline pressure, supplier reliability and, increasingly, carbon footprint, while most monitoring tools still process one criterion at a time (Pinedo, 2012; Chakraborty et al., 2020). Practitioners close the gap with judgement, which is why the quality of that judgement is an engineering concern.

This thesis was produced during an R&D placement in the Direction Innovation Numérique (DIN) of the ALTEN Group, whose Smart Green Supply Chain programme (Ben Zakour, 2024) pursues three aims: lean, attacking cost, delay and over-stock; green, attacking the carbon footprint of each transport leg to the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023); and predictive, anticipating disruption through probabilistic modelling in the manner of a digital supply chain twin (Ivanov; Dolgui, 2020). This work sits inside the third, against the Industry 4.0 premise that cyber-physical systems and digital twins can turn a reactive supply chain into a self-organising one (Kagermann et al., 2013).

The programme's existing platform, DISCO, is a serious game and digital twin simulating a complete supply chain for a fictitious satellite product across 22 component suppliers (Baxas, 2025). Players act as supply chain buyers and decide supplier selection, transport routes and factory allocation.

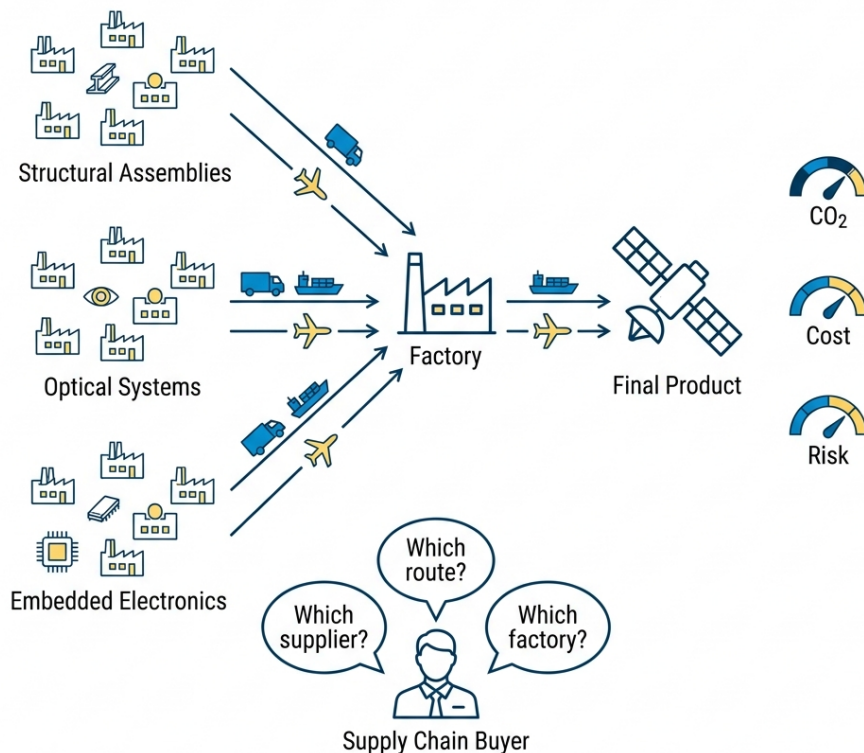


Figure 1: The DISCO serious game network

Its scoring engine applies Min-Max normalisation over the values of the current game session,

illustrated here on the carbon dimension:

$$\text{ScoreCO}_2 = 1 + 9 \times \frac{\text{realCO}_2 - \text{co}_{2,\min}}{\text{co}_{2,\max} - \text{co}_{2,\min}}, \quad (1)$$

with analogous formulae for cost and risk, on a scale from 1 for the best achievable performance to 10 for the worst.

This work sits within a body of prior DIN research. Baxas (2025) defines the DISCO software architecture including the Time To Service (TTS) and Time To Recovery (TTR) formalisation; Baulain et al. (2024) developed the Physical Internet routing module, with transit times through the HERE Maps API, Vincenty distances, mandatory Heavy Goods Vehicle (HGV) pause compliance and the Itinerary Performance Index (IPG); Kabouche (2025) the supplier risk scorer combining delay history, geographic vulnerability and financial fragility; Cordova (2024) the per-route CO₂ quantification module on the GLEC and VECTO methodology; and Mondelice (2025) the Transport Management System interface providing real-time transit-time extraction. The system delivered here, SupplyScore, is a standalone Python application that runs alongside these modules rather than inside them. Their coupling is a stated perspective of the programme, not a delivered capability, and this thesis makes no claim to the contrary.

1.2 Problem Statement

This thesis addresses the layer that comes *before* any planning decision: the urgency signal is systematically biased at the source, and must be measured before it can safely inform anything downstream (M. Zhu et al., 2018; Rusou et al., 2020). Where capacity is shared the bias does not stay local. Over-triage takes capacity from whoever needed it more; under-triage is worse, because a risk nobody has declared is a risk nobody is watching (Li et al., 2020; Ivanov; Dolgui, 2020).

Digital twins and real-time decision support are strong at operational monitoring, disruption detection and resource reallocation (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021; Karam et al., 2022; X. Zhu et al., 2025), and they do it by computing the physical state. To the best of the review in Section 2, none models the gap between a declared urgency and a computed one. Figure 2 shows that gap over the life of a task: where the declared step sits above the computed curve the operator claims more urgency than the state justifies, and where the curve sits above the step the state is worse than anyone has declared.

Formally, let $U_d \in [0, 1]$ denote the declared urgency and $U_r(t) \in [0, 1]$ the real urgency computed from operational data in the digital twin. The mismatch between the two signals is

$$\Delta U(t) = U_d - U_r(t) \in [-1, 1]. \quad (2)$$

A tolerance band $\varepsilon \in [0, 1]$ is introduced so that small deviations are not treated as operationally meaningful. Over-declaration then corresponds to $\Delta U(t) > \varepsilon$, under-declaration to $\Delta U(t) < -\varepsilon$, and approximate agreement to $|\Delta U(t)| \leq \varepsilon$.

The two conditions need not cost the same. Weighting them asymmetrically, with $\lambda_{\text{under}} \geq \lambda_{\text{over}}$, prices under-prioritisation above over-prioritisation. Two bodies of work support do-

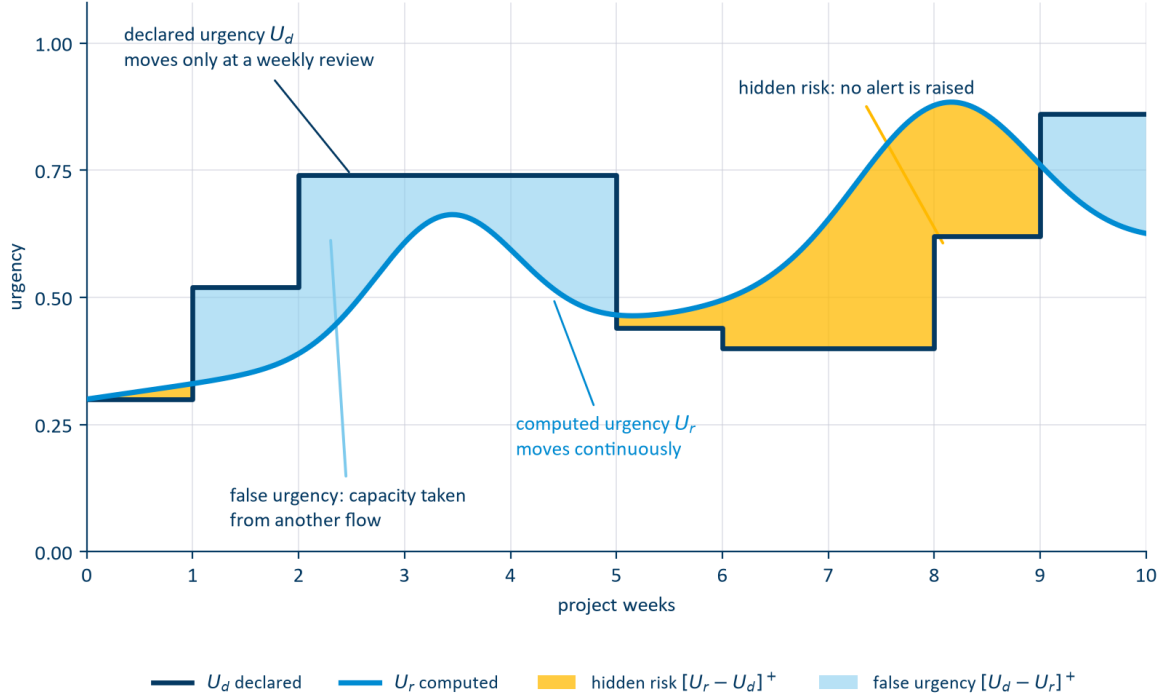


Figure 2: Declared and real urgency, and the two mismatch pathologies

ing so: operations research showing that decision errors carry unequal downstream consequences (Kim et al., 2020; DiwasSingh et al., 2020), and scheduling work that models differentiated tardiness penalties instead of a uniform delay cost (Baker et al., 1982; Chauhan et al., 2022). How much asymmetry is right is a calibration question, answerable only in a specific setting.

1.3 Aim, Objectives and Scope

The aim is to establish whether the divergence between declared and computed urgency is measurable to a standard that would justify acting on it. That wording commits to a falsifiable target rather than to building a system: a score that cannot beat a trivial baseline against recorded outcomes has not met the standard, however well engineered the system producing it. It also makes measurability the question rather than assuming it, because an architecture that lets the same person supply both signals is measuring its own input.

Three objectives follow. **O1**, define a multi-block, DAG-propagated real urgency $U_r(t)$ and a cognitive adequacy score $A(t)$ measuring its gap with the planner's declared urgency U_d . **O2**, implement network propagation and criticality-driven prioritisation to identify structurally critical nodes. **O3**, validate the resulting signal against recorded outcomes, and measure what happens to declaration behaviour when the system's own forecast is placed in front of the declarant.

They are pursued through four questions: how a physically grounded U_r can be computed from heterogeneous KPI blocks and propagated across a multi-tier graph; how A should be formulated to penalise under-triage asymmetrically and whether it has predictive validity; how pairwise elicitation and outranking identify structurally critical nodes; and how the framework behaves when operationalised as a weekly-cadence serious game, including what happens to

declared urgency once the forecast is shown. Six hypotheses with numeric thresholds were frozen before any data existed; Section 3.4.3 states them with the campaign they belong to, and Section 5.3 is the only place their verdicts appear.

Four boundaries bound what the results support. The declarants are language-model agents, not human consultants, and the human campaign for which the protocol was frozen has not been run. The evidence is two scenarios, an eight-node satellite chain and a fifteen-node airframer chain, with seven positive outcomes between them. SupplyScore is standalone: it does not integrate with DISCO, an ERP, or the routing and carbon modules above, and every KPI reaches it through a weekly review or a scripted event. And the satellite observation work of Appendix I is reported as an opening, not a second contribution, with no claim that the two systems have been connected.

1.4 Thesis Structure

Section 2 reviews the streams that bear on the problem and states the gap each leaves. Section 3 keeps three things apart that are easily confused: the mathematical model, the choices made in building it into software, and the experiment designed to test it. Section 4 reports what was measured, then diagnoses it. Section 5 states what the work contributes, what bounds it, and what should follow. The appendices carry the apparatus, the full scoring tables and the two workstreams reported as openings.

2 Literature Review

2.1 Introduction and Key Definitions

This review answers a single question: which existing methods can be assembled to measure, correct and propagate a biased urgency signal, and which part of that assembly is missing from the literature? The relevant work is broad but fragmented across operations research, behavioural science and digital twin engineering, and read individually each stream solves one piece. Read together they leave a specific slot empty: no retrieved work combines, within a single supply chain decision architecture, a multi-block operational urgency function, a structured client-side urgency input, an adequacy layer comparing stated and operational urgency, network propagation across a multi-tier DAG, and empirical validation of the adequacy signal against recorded outcomes (Sawik, 2013; Sawik, 2014; Sawik, 2016; Cohen et al., 2019; Hosseini; Ivanov, 2020). The review is organised by the role each stream plays in that argument rather than by chronology,

Six terms carry specific meanings throughout. **Urgency** is how far a task needs prompt action to avoid a material loss of operational performance; *importance* is about strategic value and *priority* about current execution order, and a task can be important and not urgent, or first in the queue for reasons connected to neither (J. Chen et al., 2025; M. Zhu et al., 2018). **Real-time scheduling** covers algorithms assigning execution order under temporal and resource constraints (Pinedo, 2012), drawn on here only as historical grounding for deadline-sensitive value functions, since no scheduling algorithm is delivered. A **digital twin** is a synchronised digital representation of a physical system supporting monitoring, simulation and decision-making from real-time and historical data (Ivanov; Dolgui, 2020). **Multi-Criteria Decision-Making (MCDM)** covers methods combining potentially conflicting performance dimensions into a single decision (Saaty, 1980; Brans; Vincke, et al., 1986). The **Physical Internet (PI)** is the model of open, mutualised logistics networks routing shipments through shared infrastructure in standardised modular containers (Montreuil, 2011). And a **serious game** is a simulation environment designed to train decision-makers through realistic scenarios with immediate feedback and scoring (Sterman, 1989; Prensky, 2001).

2.2 Risk Propagation and Deadline-Sensitive Value

Within real-time systems, deadline-sensitive scheduling theory models task value as a function of completion or response time, an approach known as time-value or utility-accrual scheduling (K. Chen et al., 1996; J. Chen et al., 2025). Chen and Mühlethaler (K. Chen et al., 1996) show that task utility is typically constant up to a deadline and then non-increasing with delay, and a broader survey confirms that deadline-sensitive literature models task value as a function of response time with bounded, structured degradation after critical thresholds (J. Chen et al., 2025). This establishes the precedent for treating urgency as time-varying, but it stops at a single task in isolation.

A complementary line of evidence comes from network risk propagation. Ivanov and colleagues formalise the ripple effect, whereby a local disruption at one supplier cascades through downstream tiers and amplifies as it travels (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). Probabilistic risk inference for supply chain systems (Zheng et al., 2020a) further shows that aggregating heterogeneous risk indicators into a single node-level score is itself an open modelling question, distinct from how that score should then be propagated.

Table 32 compares four candidate aggregation and growth models found across these two lines, and Figure 3 plots what each one does.

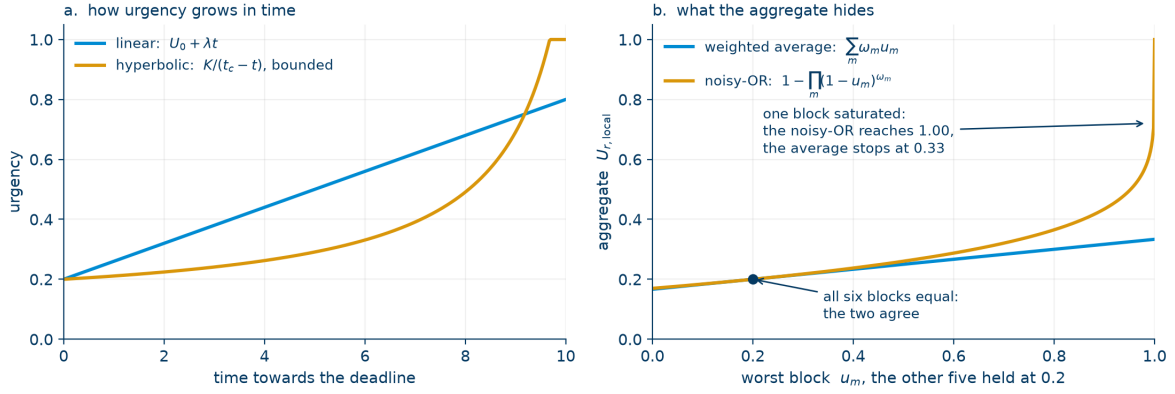


Figure 3: The four models of Table 32. Panel b is the choice this thesis has to make: a weighted average is fully compensatory, so a catastrophic reading on one block is offset by acceptable readings elsewhere and the aggregate stops at 0.33; a noisy-OR is not, and any factor approaching its maximum drives the aggregate towards its own maximum whatever the others read. Which behaviour is desirable depends on whether the aggregate describes average condition or exposure to the worst active cause, and the composite-indicator literature treats that as a modelling decision rather than a solved question (Greco et al., 2018; Becker et al., 2017).

The hyperbolic family carries a further caveat. A structurally similar acceleration appears in Log-Periodic Power Law (LPPL) models of financial crash precursors (Johansen et al., 2001; Johansen et al., 2000), but empirical analyses find LPPL-family models highly parameter-sensitive and unsuited to use as production risk models (Feigenbaum, 2001; Brée et al., 2010).

Gaps in risk propagation and deadline-sensitive value. Neither stream specifies how to aggregate heterogeneous, block-structured operational risk indicators at a single node before propagating them. Neither addresses declared urgency, or the mismatch between what is stated and what is computable. And the aggregation literature stops at the observation that the choice matters, without establishing whether equal values on structurally different indicators carry equal risk meaning, which is the question any multi-block aggregate has to answer before it can be interpreted.

2.3 Cognitive Bias in Urgency Declaration

Human decision-makers systematically misrepresent urgency. Two lines of behavioural operations research establish it from different angles, and both bear directly on any system that takes operator-declared urgency as an input.

M. Zhu et al. (2018) report the Mere-Urgency Effect in five controlled experiments: participants consistently over-prioritised imminent low-value tasks over distant high-value tasks, and the bias persisted even when participants were informed of its existence. Steel et al. (2006) provides a complementary formal model through the Temporal Motivational Theory (TMT):

$$V(t) = \frac{U \times E}{1 + \Gamma \times D(t)^k}, \quad (3)$$

where U is objective utility, E is the expectancy of success, $D(t)$ is the delay to deadline, and Γ and k are individual discounting parameters. As the delay to deadline shrinks towards zero, the motivational value $V(t)$ converges to the full product of utility and expectancy, so immediacy becomes more salient as the deadline approaches. Near-term tasks can therefore attract disproportionate attention even when their broader objective value is lower (Steel et al., 2006). Table 33 gives the operational role of each parameter.

These laboratory findings are supported by evidence from operational settings. Rusou et al. (2020) show that operators under workload preferentially complete small, easy tasks, the operational analogue of under-triage for large-scope supply chain activities. DiwasSingh et al. (2020) confirm in a large field study that completing easy tasks first significantly hurts overall performance. Hospital admission control under occupancy uncertainty (Kim et al., 2020) produces over- and under-triage patterns analogous to those seen in logistics operators facing uncertain lead times. Human-centred decision support systems such as that of Rosenthal et al. (2020) use visual feedback to show the mismatch between human plans and system temporal constraints, helping users calibrate their schedules.

Gaps in cognitive bias research. Behavioural research diagnoses the problem and quantifies it in controlled settings, but the magnitude of the bias in industrial supply chain contexts remains a matter of calibration, and this literature does not prescribe a real-time operational correction.

2.4 Indicators of Delivery Rupture Risk

If urgency is to be aggregated from several indicators, the literature has to supply the indicators. This section reviews the body of work that identifies which observable quantities precede a delivery rupture, meaning a failure to deliver at the agreed time, place, quantity and quality. It is the source from which Section 3.2.2 later selects.

Heckmann, Comes and Nickel establish rupture as the terminal event a supply risk measure should anticipate, and criticise the field for measuring exposure without connecting it to that event (Heckmann et al., 2015). Indicator-based supplier monitoring proceeds by identifying observable precursors rather than modelling the event directly (Mokhtar et al., 2019).

Table 34 gives the five families of precursor that recur across that work, with the sources establishing each and the reason it is a precursor rather than a symptom. Section 3.2.2 selects from this table, and Section 3.2.2 states what it kept.

Gap in the indicator literature. The families are named individually and no retrieved work supplies a criterion for deciding which of them belong in one composite, nor a rule for combining them once chosen. That is the question Section 3.2.1 answers with an explicit independence assumption rather than with a weighting convention.

2.5 Closest Antecedents: Network Risk Twins and MCDM Supplier Scoring

The works most structurally similar to this thesis combine multi-criteria supplier scoring with disruption-aware network monitoring. Sawik (2013) develops mixed-integer programming models for resilient supplier order assignment integrating cost, service level and disruption

risk, extended in later work to joint and integrated formulations (Sawik, 2014; Sawik, 2016). Cohen et al. (2019) and Afrasiabi et al. (2022) extend this to MCDM supplier prioritisation under disruption, combining criteria weighting with allocation optimisation. Chakraborty et al. (2020) confirm that rule-based heuristics remain competitive against exact methods in dynamic industrial environments. These works are cited for the closeness of their subject matter, not because this thesis reuses their optimisation machinery, which addresses allocation and sequencing rather than urgency measurement.

Gaps in the closest antecedents. Two gaps distinguish this stream from the present work. First, none introduces a network-propagated urgency signal computed independently at every node and combined across tiers; risk is typically evaluated once, at the point of decision. Second, none compares a client-stated urgency against a model-derived urgency: where the declared signal exists at all, it is used directly, without validation against live operational proxies.

2.6 Enabling Method: Bayesian and Probabilistic Risk Updating

Bayesian methods are a well-supported approach for revising risk estimates from new evidence in supply chain research (Pearl, 1988; Koller et al., 2009). Rather than a full Bayesian network with an explicit conditional-probability-table structure, this study draws on the narrower and more tractable case of Beta-Bernoulli conjugate updating: a failure probability p is treated as a Beta-distributed belief with an equivalent prior sample size n_0 , revised from a small number of newly observed equivalent trials (k, n) each time a calibrated operational event is declared.

Three convergent findings support this choice. Hosseini; Ivanov (2020) survey Bayesian methods applied to supply chain risk and resilience, finding that probabilistic belief-revision methods are often preferred to deterministic scoring when sparse new evidence must update an existing estimate. Huang et al. (2020) and Deng et al. (2020) develop Bayesian models for disruption and delay probability estimation. Lockamy et al. (2011) and Qazi et al. (2017) support Bayesian revision as a practical tool under sparse, event-driven evidence, consistent with the small-sample setting of a weekly-cadence digital twin. Plain frequency counts, by contrast, are unstable when only a handful of events are declared per node per project (Zheng et al., 2020b).

Gaps in Bayesian risk updating. Bayesian belief revision quantifies disruption exposure but does not, by itself, compare it against a client-declared urgency, and it says nothing about whether the resulting probabilities are well calibrated once they reach a human reader.

2.7 Enabling Method: Digital Twins and Real-Time Decision Support

The digital twin concept was formalised by Grieves (2014) as a virtualised representation of a physical process. Tao et al. (2018) extended it to manufacturing, and Fuller et al. (2020) surveyed the enabling technologies. In supply chain management, Ivanov; Dolgui, et al. (2021) define the Supply Chain (SC) Digital Twin (DT) as integrating simulation, optimisation and machine learning. Ivanov; Dolgui (2020) address digital supply chain twins for disruption risk and resilience management. Wang et al. (2020) demonstrate twin-driven planning in a manufacturing context analogous to DISCO's satellite component network. In the Industry 5.0

setting, these systems have evolved towards human-in-the-loop Cognitive Digital Twins, where semantic models and reasoning methods capture human feedback and synchronise mental models with simulated physical operations (Nilloofar et al., 2023).

Real-time freight planning work shows that operational urgency proxies, meaning slack, lateness risk and shortage exposure, are already present within these systems (Hrušovský et al., 2021; Perez et al., 2022). Model-predictive control for vehicle routing (Karam et al., 2022) and twin-based disruption response (Ding et al., 2023; Leung et al., 2022; X. Zhu et al., 2025) confirm the state of the art in live operational replanning.

Gaps in digital twin decision support. In the retrieved literature, no system combines a declared urgency input, a computed operational urgency, and their propagated mismatch within one architecture. A second gap matters for this thesis specifically: these systems are evaluated on the accuracy of their physical state estimation, and their inputs are assumed to arrive. Where an input depends on the same operator whose judgement the system exists to support, that circularity is not usually treated as a measurement problem.

2.8 Enabling Methods: Multi-Criteria Decision Methods

This research builds on two distinct domains within multi-criteria decision-making: pairwise comparison methods for eliciting weights, and outranking aggregation for turning those weights into a defensible ranking of nodes.

Table 35 compares the four methods on the three axes that decide which is usable here: what an elicitation costs the operator, what the method returns, and whether a good score on one criterion can hide a bad score on another.

The last column of that table is why both an elicitation method and an outranking method are used rather than one. Weighted averages are fully compensatory: an excellent score on one criterion masks a critical failure on another. Outranking builds pairwise preference flows instead, which prevents that masking, and a cross-industry survey confirms its suitability for supplier and node evaluation. Composite-index work nonetheless warns that nominal weights often diverge from effective importance through correlation, normalisation and aggregation dynamics, and that the choice of normalisation is itself non-neutral because it can produce rank reversal (Greco et al., 2018; Becker et al., 2017; Triantaphyllou, 2000).

Gaps in multi-criteria decision methods. These methods weight criteria and rank alternatives, and they are static elicitation and aggregation tools. None is routinely compared against a computed operational counterpart. And the composite-indicator critique is generally raised as a caveat at the end of a paper rather than turned into something a running system can check on its own data.

2.9 Forecast-Assisted Decision and Automation Bias

The streams above concern how to compute a signal. This one concerns what happens when the signal is shown to the person whose judgement it was built to support, and it is included because the campaign reported in Section 4.9 produced a result in exactly that register.

Parasuraman et al. (1997) set out the canonical taxonomy of misuse, disuse and abuse of automation, establishing that the failure modes of automated aids are as much about human reliance as about machine accuracy. Skitka et al. (1999) demonstrate automation bias directly: participants supported by an automated aid committed both omission errors, failing to react to a problem the aid did not flag, and commission errors, following the aid against contradicting evidence. Cummings (2004) extends the finding to time-critical decision support, the family to which a weekly supply chain review belongs. Goddard et al. (2012) provide a systematic review in clinical decision support, and report that automation bias is frequent, that it is mediated by the confidence with which advice is presented, and that it is only partially mitigated by training.

The direction of the effect is not uniform. Dietvorst et al. (2015) document algorithm aversion, where people abandon an algorithm after seeing it err even when it outperforms them, while Logg et al. (2019) document algorithm appreciation, where people weight algorithmic advice more heavily than human advice. Bansal et al. (2021) show that adding explanations to a model's output can increase acceptance of its recommendations whether or not those recommendations are correct, so that explanation improves persuasion more reliably than it improves joint accuracy.

Two consequences follow for the present work. First, an advisory signal changes the behaviour it measures, which means a system that both elicits declared urgency and displays a forecast to the declarant is not observing an independent signal. Second, a confidently presented but poorly calibrated probability is a specific hazard rather than a cosmetic defect, since the confidence with which advice is presented is itself a mediator of reliance (Goddard et al., 2012).

Gaps in forecast-assisted decision research. This literature is dominated by single-decision laboratory tasks and clinical decision support. It does not address a repeated, multi-turn setting in which the same operator is shown a forecast about their own node, week after week, while their declaration feeds back into the model's state. Nor does it address the case where the displayed forecast is systematically miscalibrated in a known direction, which is the situation measured in Section 4.6.

2.10 Validation Methodology: Serious Games

The Beer Game (Sterman, 1989) is the foundational supply chain serious game. Sterman shows that players systematically misperceive the feedback in their own supply line, underestimating orders already placed but not yet received, and over-order with a lag once a shortage appears, amplifying the resulting oscillation as it travels upstream, the mechanism known as the bullwhip effect (Lee et al., 1997). This decision bias is analogous to the false-urgency pathology, where an operator over-reacts to a signal after a delay rather than adjusting smoothly. Prensky (2001) establishes the theoretical basis for digital game-based learning, and Deterding et al. (2011) formalise gamification design principles.

Mohaddesi et al. (2020) introduce the Gamette methodology, a bite-sized approach to capturing decision-making patterns for behavioural modelling, directly applicable to a weekly-review protocol in which operators repeatedly submit urgency assessments. Bellotti et al. (2013) provide an evaluation framework for serious game learning outcomes. Lafkihi et al. (2025)

use a freight transportation game to study carrier behaviour on online spot-market platforms, a setting connected to the Physical Internet’s vision of mutualised capacity allocation.

Gaps in serious game validation. Serious games are established as instruments for observing decision bias, but they are usually scored on learning outcomes or on aggregate system performance. They are rarely used as a controlled comparison in which the same frozen scenario is replayed with and without a treatment applied to the declarant, which is the design adopted in Section 3.4.4.

2.11 Probabilistic Forecast Verification

Because this thesis scores a probabilistic forecast against recorded outcomes, it inherits the vocabulary of forecast verification. Brier (1950) introduced the mean squared error of probabilistic forecasts, and Murphy (1973) decomposed it into reliability, resolution and uncertainty, establishing the distinction this thesis relies on: a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures require different remedies. Gneiting et al. (2007) formalise strictly proper scoring rules, under which a forecaster minimises expected loss only by reporting their true belief.

The distinction has been rediscovered in machine learning. Niculescu-Mizil et al. (2005) show that several strong classifiers produce distorted probabilities that require post-hoc correction, and C. Guo et al. (2017) show that modern neural networks are systematically over-confident, with accuracy improving while calibration degrades. Both report that the correction is a separate, cheap, post-hoc layer rather than a change to the model.

Gaps in forecast verification practice. The verification literature assumes the outcome being predicted is unambiguously defined. In an operational supply chain setting it is not: whether a node-week counts as an adverse outcome depends on choices about which events qualify and over what horizon. Section 4.7 shows that this choice, and nothing else, moves the measured skill of the same forecasts by a factor of five.

2.12 Analogical and Contextual Literature

Three streams provide vocabulary and context rather than foundations, and are flagged as non-foundational wherever they are used.

Medical triage as a conceptual analogy. Medical triage provides an analogy for prioritising under severe capacity constraints and cognitive pressure. Protocols such as the Manchester Triage System (MTS) (Mackway-Jones et al., 2013) categorise patients into five urgency levels using flowcharts, whereas the Emergency Severity Index integrates predicted resource consumption. Comparative clinical studies show substantial classification discrepancies between the two systems, at a weighted $\kappa = 0.51$, so even trained professionals disagree on urgency assessment (Kuligowski et al., 2025). The clinical literature also establishes an asymmetry that logistics work rarely states explicitly: under-triage, meaning a patient classified as less urgent than they are, is more dangerous than over-triage, because the first costs a life and the second costs a queue (Kim et al., 2020; Mackway-Jones et al., 2013).

Physical Internet and synchromodality. The Physical Internet is a model of interconnected logistics networks in which routing, storage and transport assets are mutualised across actors (Montreuil, 2011). In such networks urgency stops being a local concern and becomes a network-level quality of service variable, because the capacity one actor claims is capacity another loses.

A related but separate concept is synchromodality. Where the Physical Internet describes who shares the infrastructure, synchromodality describes when the mode is chosen: a shipment can be moved between road, rail and waterway in real time as conditions change, instead of being committed to one mode when it is booked. The two can operate together and neither presupposes the other. Dynamic routing under synchromodality needs priority signals that stay valid across rolling planning windows (W. Guo et al., 2020), and disruptions travel quickly through interconnected networks, producing cascading ripple effects (Li et al., 2020).

Regulatory carbon accounting. Corporate sustainability reporting requirements have made transport emissions a reported quantity rather than an optional one (Cordova, 2024). The GLEC Framework v3.2 (Smart Freight Centre, 2023) and Vehicle Energy Consumption calculation TOol (VECTO) (European Commission, 2017) define emissions baselines for heavy goods vehicles, and emissions factors relate non-linearly to vehicle speed and cargo load, so a carbon figure cannot be recovered from cost or distance alone (ADEME, 2014).

Earth observation of built and logistics assets. The opening reported in Section I rests on a separate literature. Overhead object detection has a standard benchmark in DOTA (Xia et al., 2018), whose oriented bounding boxes and vehicle classes make it the reference for overhead vehicle detection, while building footprint extraction is benchmarked by Inria (Maggiori et al., 2017) and SpaceNet (Van Etten et al., 2018). Promptable segmentation foundation models (Kirillov et al., 2023) have made it possible to sample candidate sites from imagery alone rather than from a cadastral database. Separately, a body of economics work establishes that satellite observation can serve as an independent proxy for activity that official statistics record slowly or not at all, from night-time lights as a growth proxy (Henderson et al., 2012) to daytime imagery as a poverty predictor (Jean et al., 2016).

Gap in Earth observation for supply chain state. That economics literature measures aggregate activity at regional scale. It does not produce the per-facility quantities a supply chain twin needs, meaning storable volume and transfer capacity at a named site, and the detection literature stops at footprints and vehicles without converting them into an operational capacity figure.

2.13 Source Review Grid

Each source analysed above is synthesised in an evaluation grid recording its main argument, methodological approach, conclusions, thematic relevance and a critical appraisal of its limitations. The grid is given in Appendix B, and it makes explicit which claims in this thesis rest on foundational work and which rest on analogy.

2.14 Identified Gaps

Each stream above closes on the limit of what it settles, and collected those limits describe one absence rather than several. Figure 4 places each stream against the two things this thesis needs from it, and the empty column is the gap.

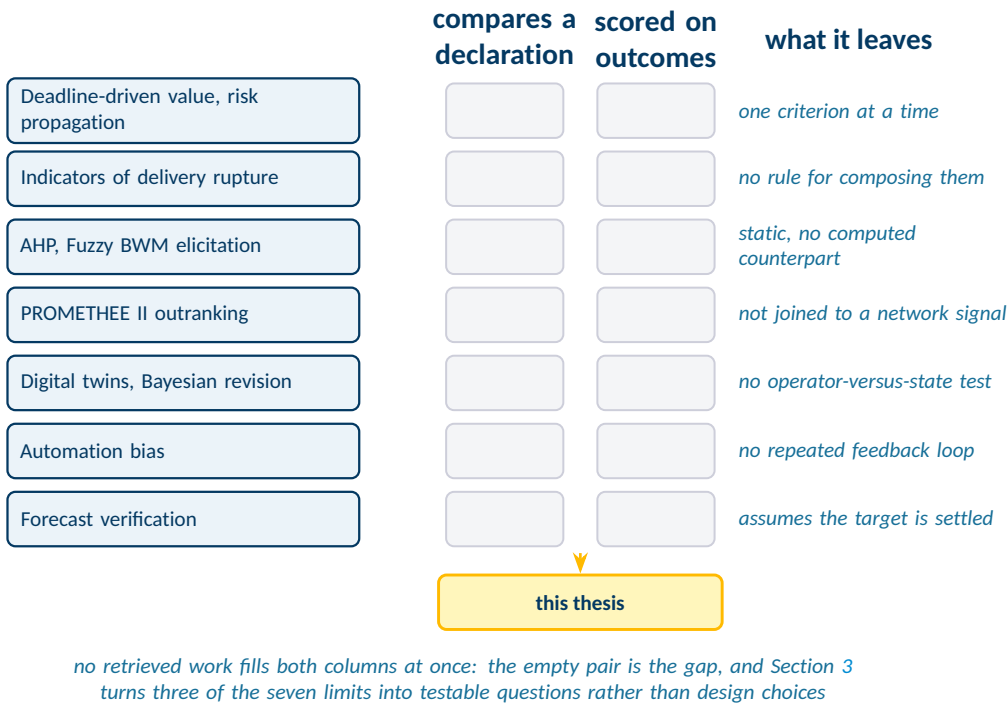


Figure 4: Each stream against the two properties this work requires: that it compare a human declaration with a computed state, and that it be scored against recorded outcomes. Every retrieved stream leaves at least one of the two empty, and none fills both.

No retrieved work combines those streams with a structured declared-urgency input and an empirically tested adequacy layer. That missing combination is the gap, and closing it is the contribution this thesis claims: a measurement of the divergence between declared and computed urgency, propagated across a supplier network, and scored against recorded outcomes rather than asserted. Three of the gaps become testable questions rather than design choices, and they are the ones carried into Section 3: whether the divergence can be measured at all once both signals come from independent sources, whether the resulting signal predicts anything, and what happens to the declared side when the computed side is shown to the person declaring.

3 Methods

3.1 Methodological Framework

This research is structured as an agile test-and-learn programme rather than a linear sequence, operating through repeated iterations of literature analysis, mathematical specification, software implementation and scenario-based simulation. That structure maps onto the IMRaD form of this thesis: the specification and implementation iterations are reported here as the method, the simulation and campaign iterations in Section 4 as the results.

The section keeps three things apart, because mixing them is what makes a framework of this kind hard to read. Section 3.2 states the mathematical model on its own terms, saying nothing about tooling. Section 3.3 gives the application choices, meaning how each object in that model is elicited, computed and exposed. Section 3.4 gives the use case built to test it. The method's own limitations, its reproducibility provisions and its ethical considerations close it.

3.1.1 The Proposed Structure

The response designed here is a two-layer loop, and stating it before the mathematics makes the equations that follow easier to place.

The first layer is a cognitive adequacy check. It captures the operator's declared urgency U_d through consistency-gated pairwise comparisons, computes a real urgency $U_r(t)$ from live operational state, and scores their mismatch through an asymmetric penalty. The second layer propagates and prioritises. Local urgency travels across the supply network, declared urgency descending towards suppliers and real risk ascending towards the final customer, and the resulting network state feeds an outranking and a shock-simulation criticality index that together indicate which nodes warrant attention. Figure 5 shows how the two layers sit relative to each other and where the declared and computed signals meet.

Two choices inside that design are worth naming immediately, since Section 2.2 left both open. The aggregation over blocks is a noisy-OR rather than a weighted average, because a complete capacity failure should not be offset by acceptable cost and carbon readings, and non-compensatory behaviour is the property that guarantees it. The penalty over the mismatch is asymmetric, pricing under-declaration above over-declaration, on the clinical analogy of Section 2.12, where under-triage costs more than a queue. Both are design commitments open to challenge, and Section 5.4 records what each one costs.

3.2 Mathematical Model

This section states what real urgency and declared urgency are, how each propagates across the network, and how their mismatch is scored. Appendix F carries a single node through every equation below, from its six blocks to the adequacy score its coordinator's declaration produces, with the arithmetic shown at each step. It deliberately leaves out tooling, elicitation mechanics and operational cadence, which are application decisions addressed in Section 3.3.

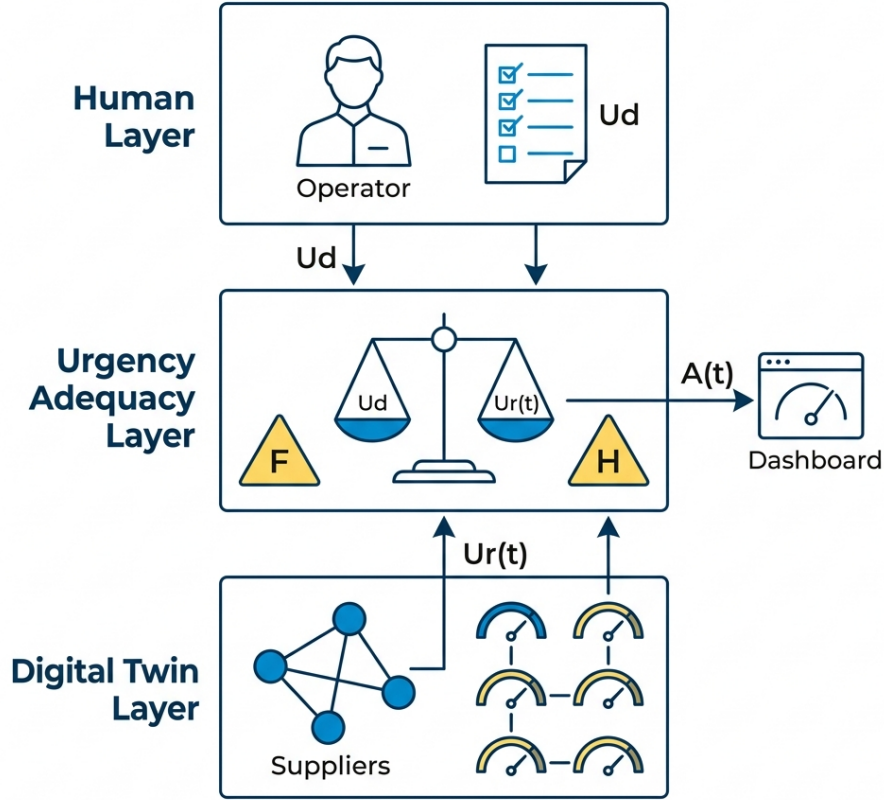


Figure 5: The layered adequacy architecture

3.2.1 Local Real Urgency

Real urgency is not read from a single deadline-proximity signal. It is aggregated at each node from six independent KPI blocks, namely time, capacity, performance, risk, cost and carbon, each producing a partial urgency $u_m \in [0, 1]$, or no value at all when the underlying KPI data is unavailable. The aggregation is a weighted probabilistic OR, so that a single saturated block is sufficient to drive local urgency towards its maximum regardless of how well the others score:

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}, \quad (4)$$

where $\omega_m \geq 0$ is the aggregation weight of block m and the product runs over blocks with available data. Table 1 gives each block's schematic formula and what it measures. The per-term weights a_k and c_k are renormalised over the terms whose data is actually available.

Where the form comes from. Equation 4 is not an arbitrary choice among aggregators. It is what follows from treating the six blocks as competing causes of the same event, and the derivation is short enough to give in full, because everything the model can and cannot claim descends from it.

Table 1: The six real urgency blocks

Block	Formula	Explanation
u_{time}	$P(L > d^* - t) + \kappa_{\text{retard}}[r]_+ - \kappa_{\text{avance}}[-r]_+$	Probability that the lead time L , modelled as Normal(μ , σ), exceeds the time left before the next active deadline d^* ; forced to 1 once the deadline has passed. The planning term r compares theoretical with reported progress.
u_{cap}	$a_1(1 - \frac{V_{\text{rem}}}{V_{\text{max}}}) + a_2(1 - \frac{W_{\text{rem}}}{W_{\text{max}}}) + a_3[\frac{D-\Phi}{D}]_+$	Saturation of the node's capacity: how little storage volume and weight remain, plus the fraction of demand D not covered by the current flow rate Φ .
u_{perf}	$1 - \text{OEE}$	Complement of Overall Equipment Effectiveness, itself availability $\times$ performance $\times$ quality.
u_{risk}	$1 - \exp(-p_{\text{fail}} \times t_{\text{recov}} \times \text{sev}/T_{\text{ref}})$	Expected unavailability: failure probability times recovery time times impact severity, normalised by a reference horizon, then combined by probabilistic OR with environmental and political exposure.
u_{cost}	$c_1 \frac{c_{\text{op}} - c_{\text{nom}}}{c_{\text{nom}}} + c_2 [\text{tariff} - 1]_+ + c_3 \frac{c_{\text{stock}}}{c_{\text{ref}}}$	Financial stress: operating cost drift above nominal, tariff multiplier above neutral, and storage cost relative to a reference.
u_{co2}	$[\frac{e - e_{\text{target}}}{e_{\text{max}} - e_{\text{target}}}]_+$	Normalised overshoot of measured emissions above target, saturating at the maximum threshold.

Assumption 3.1 (Independent competing causes). Delivery rupture at a node is an event R . Each block m describes a mechanism M_m that can produce R on its own, $u_m = P(M_m)$ is the probability that mechanism m fires during the period considered, and the mechanisms fire independently of one another.

Under Assumption 3.1 the node escapes rupture only if no mechanism fires. The mechanisms being independent, the probability of escaping all of them is the product of the individual escape probabilities, so

$$P(\text{no rupture}) = \prod_m (1 - u_m), \quad U_{r,\text{local}} = P(R) = 1 - \prod_m (1 - u_m), \quad (5)$$

which is Equation 4 with every $\omega_m = 1$. This is the noisy-OR of Bayesian network practice (Pearl, 1988), and the name is literal: it is the probabilistic form of the logical statement *rupture occurs if mechanism 1 fires, or mechanism 2 fires, or ...*

The exponent ω_m generalises that statement: $(1 - u_m)^{\omega_m}$ is the escape probability of ω_m independent repetitions of mechanism m . Setting $\omega_m = 0$ removes a block, which is how missing data is represented and is a removal rather than a claim of health; a weight above one makes a block more able to drive the aggregate alone, which is the model's only lever for saying that some mechanisms are more diagnostic than others. The campaign runs at $\omega_m = 1$ throughout, so every result in Section 4 concerns the plain noisy-OR of Equation 5.

Why not a weighted average. The alternative an engineer reaches for first is $U_{r,\text{local}} = \sum_m \omega_m u_m$ with weights summing to one, and it fails for a reason that is structural rather

than empirical. A weighted average is *compensatory*: a block at 1.0 can be pulled back below any threshold by enough blocks near zero. Figure 6 holds five blocks fixed and lets one rise, and the two aggregators separate as it does. At the right-hand edge, where the deadline has passed and $u_{\text{time}} = 1$, the noisy-OR reads 1.0 exactly, because the factor $(1 - 1) = 0$ collapses the product, while the average is still at 0.40 and would rank the node below one that is mildly stressed everywhere.

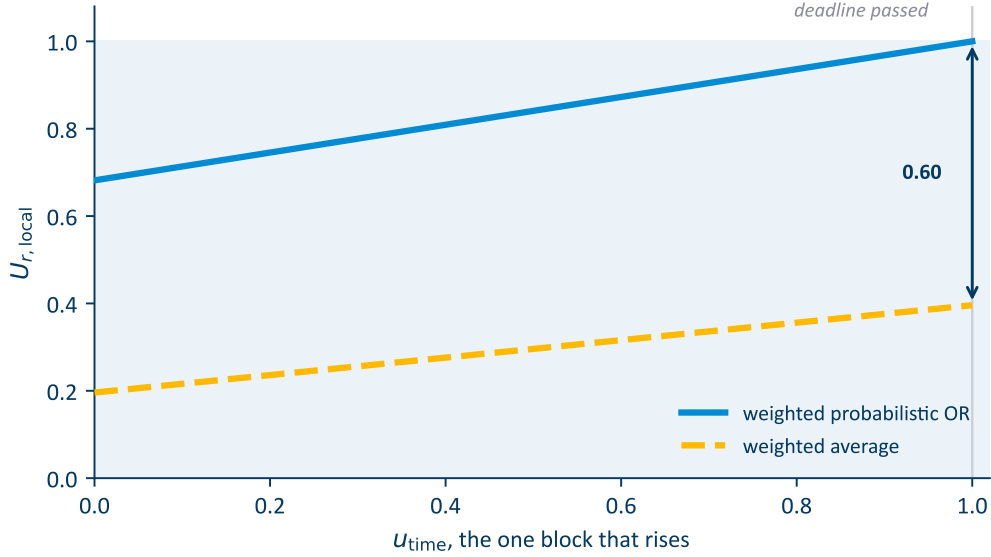


Figure 6: The two aggregators on the same node, five blocks held at their values of Table 22 and one allowed to rise. The gap at the right-hand edge is the non-compensatory property: a mechanism that has reached the state past which it cannot get worse takes the node to 1 on its own, and no healthy block can pull it back.

Physically the first answer is wrong: a missed deadline is not compensated by a comfortable carbon budget. This is the non-compensatory requirement that the composite-indicator literature raises as a caveat and that Section 2.8 traces through the outranking methods (Greco et al., 2018; Brans; Vincke, et al., 1986), and here it is met by the aggregator itself rather than by a downstream correction.

The price is stated with the benefit. Independence between the six mechanisms is a strong assumption and a false one in detail: a supplier under financial stress often has degraded performance, so u_{cost} and u_{perf} share information, and counting both double-counts the underlying cause. The noisy-OR then overstates urgency, and it does so in the direction of caution rather than of complacency. Section 4.4 reports the observed correlations between blocks, which is the check this assumption makes possible: two blocks carrying the same information are visibly correlated on the campaign data, and the assumption is falsifiable to that extent.

Three properties of Equation 4 matter for reading it, and they are also where the model is most open to challenge.

What the endpoints mean. Each raw KPI carries physical units, whether days, euros, kilograms or percent, and passes through a transfer function onto $[0, 1]$. At $u_m = 0$ the factor $(1 - 0)^{\omega_m}$ equals 1 and leaves the product unchanged. At $u_m = 1$ the factor is 0, so the product

collapses and $U_{r,\text{local}} = 1$ whatever the other blocks read. The upper endpoint therefore has a precise operational meaning: the block has reached the state past which it cannot make the node any more urgent, whether that state is a deadline already passed, a fully depleted buffer or a near-certain failure. The word used for it here, saturation, is a label adopted for this model rather than a concept imported from the literature.

How a missing block is handled. An unavailable KPI gives its block weight $\omega_m = 0$, so its factor is $(1 - u_m)^0 = 1$ and the aggregate is untouched. Absence is therefore not represented by $u_m = 0$, which would be a positive claim that the block is healthy, and the two cases are distinguished in the data model as well as in the mathematics. The remaining weights are renormalised, which keeps a node with three live blocks comparable with a node with six.

Whether the blocks are exchangeable. The product is symmetric in its factors, so it treats the blocks as interchangeable up to their weights. That is the strongest assumption in the model and it does not hold cleanly. A value of $u_{\text{time}} = 0.3$ means a 30% probability of missing the next deadline, a quantity with a frequentist reading; $u_{\text{cost}} = 0.3$ means operating cost has drifted three tenths of the way along a normalising scale, an ordinal statement dressed in a cardinal number. No transfer function makes them equally informative about rupture. Any reading of the aggregate as a probability is therefore approximate, and it is used for ranking and for comparison against U_d rather than as a calibrated one. The assumption is testable: two blocks carrying the same information are observably correlated, which is the check reported in Section 4.4.

Two lifecycle rules apply before aggregation. A node marked DONE has its effective local U_r forced to 0.0, since a completed task carries no further physical risk, and a node marked ABANDONED has it forced to 1.0. These apply identically whether the local urgency comes from the analytic blocks or from the Monte Carlo estimate below.

A stochastic variant of the time block. The analytic u_{time} block assumes a Normal lead time at a single node. On a multi-tier network that assumption breaks, since a node can only start when all its predecessors have delivered, and the maximum of several skewed lead times has no convenient closed form. The module `mc/lead_time.py` therefore replaces the analytic estimate with a stochastic one when enabled.

Each node draws L_i from a configurable distribution, lognormal by default. Completion times propagate in topological order, $S_i = \max_{j \in \text{Pred}(i)} C_j$ and $C_i = S_i + L_i$, and the delay probability is the fraction of draws missing the deadline, $u_{\text{time}} = P(C_i > d_i)$, over $N = 10\,000$ draws by default, with a memory guardrail capping the simulation at $N \times n \leq 5 \times 10^7$ across n nodes. The recurrence is implemented vectorised over draws rather than looped over them, so a node's ten thousand completion times are one array operation.

Two properties of the estimator decide how it is used. It is a sample mean of Bernoulli indicators, so it is unbiased for $P(C_i > d_i)$ and its standard error is at most $1/(2\sqrt{N})$, which at $N = 10\,000$ is 5×10^{-3} in the worst case at $\hat{p} = 0.5$. That is an order of magnitude finer than the differences the urgency signal is asked to resolve, so the simulation contributes negligible noise to $U_{r,\text{local}}$, and increasing N further buys nothing at the $\sqrt{N}$ rate. The cost is $\Theta(N(n + e))$ time and $\Theta(Nn)$ space, the space being what the guardrail bounds.

The default lognormal is parameterised by moments rather than by log-parameters, so that a node's configured mean m and standard deviation s are the ones actually realised: $\sigma_{\ln}^2 = \ln(1 + s^2/m^2)$ and $\mu_{\ln} = \ln m - \sigma_{\ln}^2/2$. The normal law is truncated by clipping at zero rather than by rejection, a documented choice: rejection is exact but of unbounded cost, whereas the clip moves the mass $P(X < 0)$ onto $\{0\}$, a bias that is negligible while $m \gg s$ and never optimistic, since it cannot produce a negative lead time.

3.2.2 Why These Six Blocks

The six blocks are not an arbitrary KPI shortlist. The selection started from the question that defines urgency in a supply chain: which mechanisms make the risk of a delivery rupture rise? A rupture, meaning the inability to deliver at the agreed time, place, quantity and quality, is the terminal event the urgency signal must anticipate (Heckmann et al., 2015). Working backwards from that event, and consistently with indicator-based supply risk monitoring (Mokhtar et al., 2019), four direct causal mechanisms and two arbitration variables were identified.

- **Time erosion** (u_{time}): the remaining slack between deadline and realistic lead time. Time is the only non-compressible resource, so a task whose slack reaches zero can no longer be saved (Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013).
- **System saturation** (u_{cap}): queues, transport capacity, workshop load and bottlenecks. A saturated system introduces hidden waiting delays on top of the nominal lead time (Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024).
- **Critical-resource performance** (u_{perf}): drift in the performance of the resource that produces the part. OEE is used as a composite proxy because precise performance data is rarely accessible on the supplier side (Mokhtar et al., 2019; Hlioui et al., 2017).
- **Adverse-event exposure** (u_{risk}): the expectation of unavailability caused by a disruption. The default incident probability of 0.02 is revised weekly from declared events through the Bayesian operator of Section 3.3.2, building on the DIN supplier risk scorer (Kabouche, 2025) and on probabilistic ripple-effect models (Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020).
- **Financial flexibility** (u_{cost}): expediting surcharges, tariff rises and storage costs. Cost drift is not a rupture cause in itself but a stress indicator, and it bounds which corrective actions remain economically viable (Shen et al., 2019; Tomlin, 2006).
- **Carbon overshoot** (u_{co2}): emissions above target. Like cost, it is an arbitration criterion rather than a rupture mechanism, becoming decisive only when a regulatory or ESG constraint binds and restricts the admissible responses (Hoen et al., 2014; Smart Freight Centre, 2023; International Organization for Standardization, 2023).

Each candidate indicator was screened against the four filter questions of Table 36. Only indicators passing all four were retained, which yields four direct causes plus two arbitration variables and explains why the list has six entries rather than four or eight.

The selection remains a conceptual construction until the underlying data is shown to exist and each block is shown to contribute information to the aggregate. Section 4.11 returns to this, because the campaign measured one block failing that test.

3.2.3 Local Declared Urgency

Symmetrically, declared urgency is not read as a single free-text estimate. It is reconstructed from a structured elicitation so that it can be scored, propagated and compared with $U_{r,\text{local}}$ on the same scale. At each node the operator compares four criteria pairwise through the Analytic Hierarchy Process, which yields a consistency-checked weight w_j per criterion, then rates each criterion on a severity scale $s_j \in \{1, \dots, 9\}$. Local declared urgency is the weight-averaged severity rescaled to $[0, 1]$:

$$U_{d,\text{local}} = \sum_{j=1}^4 w_j \frac{s_j - 1}{8}, \quad \sum_{j=1}^4 w_j = 1. \quad (6)$$

The elicitation tool, its four criteria and the consistency gate are application choices described in Section H.2. What matters here is that $U_{d,\text{local}}$ and $U_{r,\text{local}}$ are both bounded, node-local quantities in $[0, 1]$ before either is propagated.

3.2.4 Network Propagation

Declared urgency propagates downstream, from the final customer at rank 0 towards raw-material suppliers, while real urgency propagates upstream, from suppliers towards the final customer. Both use a leaky product rule that keeps the signals bounded in $[0, 1]$ and monotonically increasing as more neighbours become urgent:

$$U_{d,i} = 1 - (1 - U_{d,\text{local},i}) \prod_{k \in \text{Succ}(i)} (1 - \gamma_{ik} U_{d,k}), \quad (7)$$

$$U_{r,i} = 1 - (1 - U_{r,\text{local},i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} U_{r,j}), \quad (8)$$

where $\gamma_{ik} \in [0, 1]$ is the downstream attenuation coefficient on arc (i, k) and $\beta_{ji} \in [0, 1]$ the upstream amplification coefficient on arc (j, i) . Table 37 defines each term, and Figure 7 shows the two signals travelling in opposite directions across the same graph. Backup arcs are documentary and inert with respect to both propagations, a modelling choice that the campaign scenario exploits deliberately.

Both recurrences are evaluated in a single pass each. The order of traversal is what makes them computable at all: taken in topological order, every predecessor of a node is resolved before the node itself, so neither pass ever needs a value it has not yet produced, and no iteration to a fixed point is required.

What the recurrence guarantees. Four properties follow directly and are used throughout. Taken in topological order the pass terminates and returns the unique solution of Equations 7 and 8, since every predecessor is resolved before the node that reads it. The result always lies in $[0, 1]$ without clipping, because each factor $1 - \beta_{ji} U_{r,j}$ lies in $[0, 1]$ and a finite product of such factors does too; the implementation still clips, but only against floating-point drift. Propagated urgency is non-decreasing in the local urgency of every ancestor, the partial

derivative being a product of non-negative terms. And the pass costs $\Theta(n + e)$ in time and $\Theta(n)$ in space, since a topological order is $\Theta(n + e)$ and each arc is examined once.

Monotonicity is what licenses the criticality index of Section 3.2.5: forcing a node's local urgency upward can never lower the urgency read at the final customer, so the worst-case shock is well defined. The upper bound is what makes that index degenerate under saturation.

Recomputing only what changed. A full pass is cheap but not free, and the weekly cadence writes one node at a time. The engine therefore keeps two dirty sets and, on a write, restricts each pass to the cone that can have changed: the descendant cone for real urgency and the ancestor cone for declared urgency. By Proposition 3.2.4(iii) a node outside both cones has no path along which the changed value could reach it, so reading its cached value is exact rather than approximate. The restricted pass costs $\Theta(|C| + |E(C)|)$ on the cone C , which on the campaign graphs is a small fraction of the whole.

When each quantity is computed. The weekly cadence governs when new data *arrives*, not when the model runs, and the two are easy to conflate. The sequence is as follows. A review submitted for one node writes that node's KPI values and questionnaire answers. That write immediately triggers recomputation of the node's own u_m blocks and its two local quantities, $U_{r,\text{local}}$ and $U_{d,\text{local}}$. The new local values then propagate, but only along the directions that can be affected: a change in local real urgency is pushed through the node's downstream cone towards the final customer, following Equation 8, and a change in local declared urgency is pushed through its upstream cone towards the suppliers, following Equation 7. Nodes outside those two cones are not recomputed, because their values cannot have changed.

Two consequences matter for reading the results. Nothing is computed on a global weekly timer, so the network is never in a state where all eight nodes were evaluated at the same instant; a node reviewed on Monday and a node reviewed on Thursday hold values one review apart, and the propagated figures reflect whatever the neighbours had most recently submitted. And the campaign turn, not the calendar week, is the unit that makes the state comparable across nodes, which is why every result in Section 4 is indexed by turn.

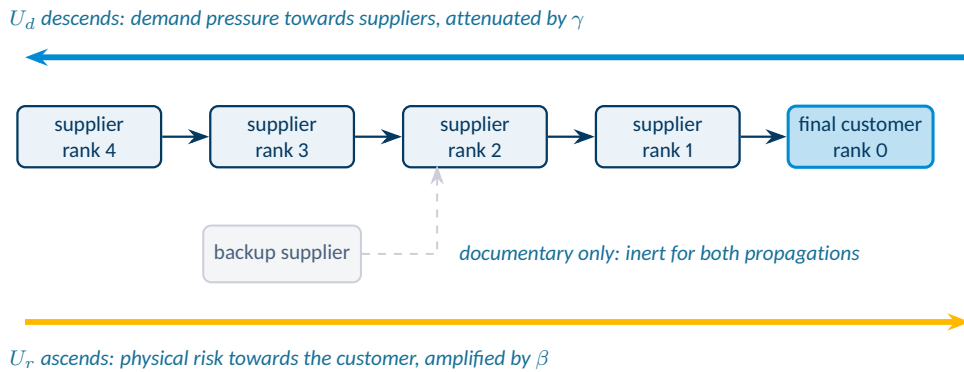


Figure 7: Bidirectional propagation. Declared urgency descends from the customer towards suppliers and is attenuated on each arc; computed risk ascends the other way and is amplified. A backup arc appears in the graph and carries neither.

3.2.5 Network Criticality Index

The criticality service answers one question systematically: which node, if it failed completely, would cause the most damage to final-customer delivery? For every active node in turn, the service forces the node's local real urgency to 1.0, re-runs the ascending propagation, and measures two deltas, the rise of U_r at the final-customer nodes and the largest rise anywhere in the network. Every shock is run twice, once on the standard pipeline and once on an ε -regularised twin that carries the same propagation in log-survival, $\ell(p) = -\ln(1 - p)$, with urgency clipped to $1 - \varepsilon$ so that a survival product is never exactly zero. Nodes are ranked by the rise at the final customer first, then by the corresponding log-survival delta $\Delta\ell_{\text{final}}$, then by name.

One behaviour of this index matters for interpretation, and it is what the second sort key exists for. A node already saturated at $U_r \geq 1.0$ yields a delta of zero and sinks to the bottom of the ranking, since its risk is already realised in the baseline. In a network-wide crisis, where many nodes saturate at once, ΔU_r collapses to zero throughout and the ranking degenerates exactly where it would be most useful; nine of the nineteen campaign turns are in that state, which is what the exclusion rule used by hypothesis H3 in Section 3.4 was written for.

That degeneracy is not an accident of the data. It is forced by the model, and Proposition 3.2.5 derives it from the range and monotonicity established above. Figure 8 shows the two regimes side by side.

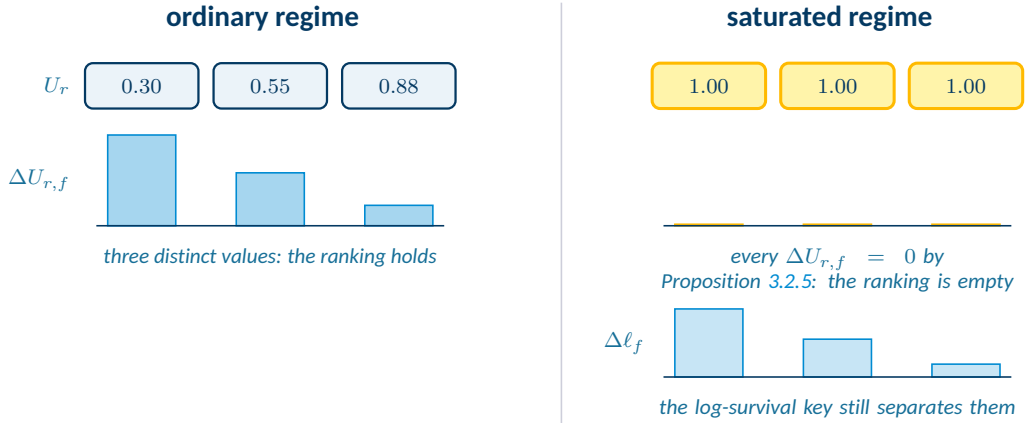


Figure 8: Left, the ordinary regime: the worst-shock deltas differ and the criticality ranking is informative. Right, a network already at $U_r = 1$ at the customer, the state nine of nineteen campaign turns reach: every delta is exactly zero and the ranking carries nothing, while the log-survival delta beneath it still orders the nodes.

Saturation collapses the ranking, and the model forces it. Suppose the baseline already reads $U_{r,f} = 1$ at the final customer. Raising any node's local urgency cannot lower a downstream value and cannot push $U_{r,f}$ above its ceiling, so the shocked value is again exactly 1 and $\Delta U_{r,f} = 0$. The node was arbitrary: every node scores zero and the ranking carries nothing. This is a consequence of the range and monotonicity above rather than an accident of the campaign data.

The response is to measure the same shock in a coordinate that the ceiling at 1 does not compress. Writing $s_i = 1 - U_{r,i}$ for the survival at node i , Equation 8 is an identity in survival space, and taking logarithms turns its product into a sum.

The twin is the same recurrence in another coordinate. Writing $s_i = 1 - U_{r,i}$ and substituting into the propagation factor gives $1 - \beta_{ji}U_{r,j} = (1 - \beta_{ji}) + \beta_{ji}s_j$, so Equation 8 is an identity in survival space,

$$s_i = (1 - U_{r,\text{local},i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} + \beta_{ji}s_j), \quad (9)$$

and taking $-\ln$ turns the product into a sum,

$$\ell_i = -\ln(1 - U_{r,\text{local},i}) - \sum_{j \in \text{Pred}(i)} \ln(1 - \beta_{ji} + \beta_{ji}e^{-\ell_j}). \quad (10)$$

Each factor is a convex combination of 1 and s_j . Clipping local urgency to $1 - \varepsilon$ therefore keeps every factor at or above ε , so no survival product vanishes and ℓ_i is finite at every node. The twin changes nothing about the model and everything about what survives the ceiling at 1.

The twin therefore changes nothing about the model and everything about what survives the clip. Two consequences follow, and the second is the reason the index has a second sort key at all.

Off saturation the two keys agree; on it only one still separates. The baseline is common to every shocked node, so ranking by either delta is ranking by the corresponding shocked value at the customer. On $[0, 1)$ the map $U \mapsto -\ln(1 - U)$ is strictly increasing, so the two rankings coincide exactly. Under saturation the first is constant at zero while the second stays strictly increasing in the shocked local urgency, because the clip acts at $1 - \varepsilon$ on the local values and never on the accumulated sum. The first claim is registered as a property-based invariant and tested on random DAGs with random coefficients.

Part (i) is registered as a property-based invariant and tested as one, on random DAGs with random coefficients. Section 4.10 reports what part (ii) recovers on the campaign, where nine turns of nineteen are in the saturated regime.

Each shocked propagation is a fresh pass, so by Proposition 3.2.4(iv) the index costs $\Theta(n_{\text{act}}(n + e))$, doubled by the twin. No incremental shortcut applies here: the shocks are independent what-if queries rather than a sequence of updates, and each starts from the same unmodified baseline.

A second entry point answers the distributional question rather than the worst case. Instead of forcing local urgency to 1.0, it draws shock severities from the graduated flat-rate severities of the calibrated event engine, minor, major and critical failures at 0.4, 0.7 and 1.0 and financial alerts at 0.3, 0.6 and 0.9, the two families weighted equally and, within a family, by decreasing field frequency. Each draw is applied as a ratchet, $\max(U_{r,\text{local}}, \text{severity})$, on the same principle as the event engine, since a failure never improves a node's state. The batch is propagated in one vectorised pass, in duplicate for both measures, and each node receives $P(\Delta U_{r,\text{final}} > 0.2)$ together with the median and ninth decile of its $\Delta \ell_{\text{final}}$ distribution. The computation is read-only, seeded and deterministic.

3.2.6 Asymmetric Cognitive Adequacy

The mismatch between declared and real urgency is split into two named pathologies, false urgency F for over-declaration and hidden risk H for under-declaration:

$$F = \max(U_d - U_r, 0), \quad H = \max(U_r - U_d, 0). \quad (11)$$

Naming H a *hidden risk* rather than a discrepancy commits the model to a reading of where the gap comes from, and that reading is an assumption about the declarant rather than a property of the subtraction.

Assumption 3.2 (Sincerity of the declaration). The declared urgency U_d recorded by the weekly questionnaire is a sincere report of what its author perceives at the time of writing. A gap between U_d and U_r is therefore a difference in perception, not a difference between what the author believes and what the author is willing to write down.

Assumption 3.2 is what makes the two signals independent in source, which is the precondition for their difference to mean anything at all: a declaration derived from the computed state, whether by the software or by a declarant reading the dashboard before answering, would make H an artefact of the wiring rather than a measurement. Section C.2 describes the controls that enforce the software half of that independence, and Section 4.3 reports an early run in which it was violated and the run discarded.

The human half cannot be enforced, only assumed. A coordinator who reads their supplier's state correctly and still reports it low, because a recovery plan is not ready to announce, produces exactly the same positive H as one who has misread it. The two are indistinguishable in the data this instrument collects, and they call for opposite interventions: better information in the first case, and a different reporting incentive in the second. Every behavioural reading in Section 4 presumes the first, and Section 5.4 records that this is the boundary of what the design can support. The free-text field beside each declaration is the only evidence that bears on the question, and Section 4.5.2 reads three of them where the distinction matters.

Drawing on Prospect Theory (Kahneman; Tversky, 1979), the two are penalised asymmetrically and with diminishing marginal sensitivity to large errors:

$$\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha, \quad (12)$$

and the penalty is rescaled into an adequacy score $A \in [0, 100]$ through an exponential mapping bounded by the penalty of a maximal one-sided error, with each term defined in Table 38:

$$A = 100 \times \frac{\exp(-\text{penalty}) - \exp(-\text{penalty}_{\text{max}})}{1 - \exp(-\text{penalty}_{\text{max}})}, \quad \text{penalty}_{\text{max}} = \max(\lambda_{\text{under}}, \lambda_{\text{over}}). \quad (13)$$

What the score guarantees. At most one of F and H is non-zero, so with $\delta = |U_d - U_r|$ the penalty is $\lambda\delta^\alpha$ for the λ of the side in question. Four properties follow. The score lies in $[0, 100]$ and reaches 100 only at $U_d = U_r$. It decreases strictly in δ on each side. For a common δ , under-declaration scores below over-declaration exactly when $\lambda_{\text{under}} > \lambda_{\text{over}}$, which is the governance constraint enforced at construction rather than left to configuration. And for

$\alpha < 1$ the score is not differentiable at $\delta = 0$: both one-sided derivatives are infinite, because $\alpha\delta^{\alpha-1} \rightarrow \infty$.

That last property is why the curvature is there and is not cosmetic. The score is most sensitive where the gap is small, so a coordinator drifting slightly from the computed state loses points fast while the difference between a large error and a very large one barely registers. The signal is meant to catch a misreading while it is still cheap to act on, not to grade catastrophes against each other.

Part (iv) is the reason the curvature is there and is not cosmetic. The score is at its most sensitive exactly where the gap is small, so a coordinator drifting a little away from the computed state loses points fast, while the difference between a large error and a very large one barely registers. That ordering matches what the signal is for: it is meant to catch a misreading while it is still small enough to act on, not to grade catastrophes against each other. Part (iii) is the governance constraint stated as an inequality, and it is enforced at construction time rather than left to configuration.

Figure 9 draws the resulting score. The governing constraint is $\lambda_{\text{under}} > \lambda_{\text{over}}$, and the default ratio of 2.25 reprises the loss-aversion coefficient while $\alpha = 0.88$ reprises the curvature of the value function, both measured by Kahneman; Tversky (1979) and restated later (Kahneman, 2011). These are literature-derived starting points, not measured properties of supply chain coordinators, and both they and the aggregation weights ω_m are configurable per project rather than hard-coded.

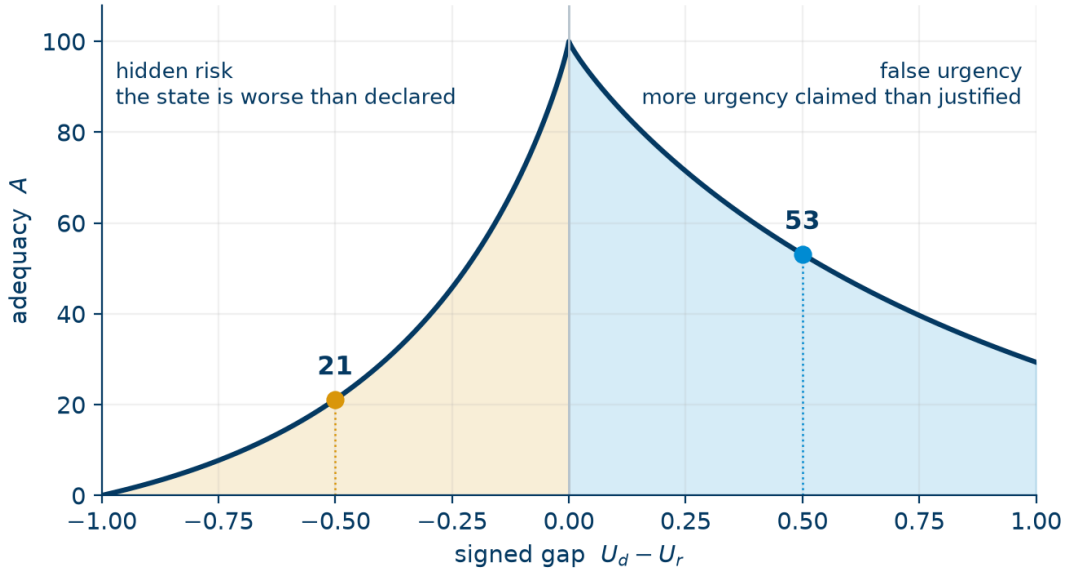


Figure 9: The adequacy score against the signed gap. The two branches meet at $A = 100$ when declared and computed urgency agree, and diverge on either side: the same absolute gap of 0.5 scores 21 when the state is worse than declared and 53 when more urgency is claimed than justified. A coordinator who panics wastes premium capacity; one who under-reports misses the window to prevent a line stoppage, and the curve prices the second above the first.

3.3 System Implementation

The model above is deliberately silent on tooling. This section describes how each object it defines is elicited from an operator, computed inside the running system, exposed back to a coordinator and kept safe. The delivered system, SupplyScore, comprises 73 Python modules across 10 packages, covered by 1831 automated test functions in 106 test files, including property-based tests.

3.3.1 Architecture

Five decisions shape it, and Figure 10 shows how they relate. All operations go through one orchestrating facade (`services/orchestrator.py`); the web interface, the command-line entry point and the test suite drive the system exclusively through it, and there is no second write path. Data is split between one global registry database and one SQLite database per node holding its private operational history, which prevents write contention between operators and enforces the boundary that one supplier's raw history is not readable through another's, and which is what makes each node of the campaign auditable after the fact. Every business write crosses a single mutation service that whitelists the touched fields, validates ranges, diffs against current state so a no-op leaves no trace, records author and source in an immutable audit log, and synchronises the in-memory graph. The network itself lives in a networkx-backed repository whose propagation is incremental, recomputing only the cones of nodes whose local urgency changed. And each project carries its own clock, a system clock for live use, a game clock advancing week by week and a fixed clock for deterministic tests, which is what makes a compressed campaign possible without touching the model.

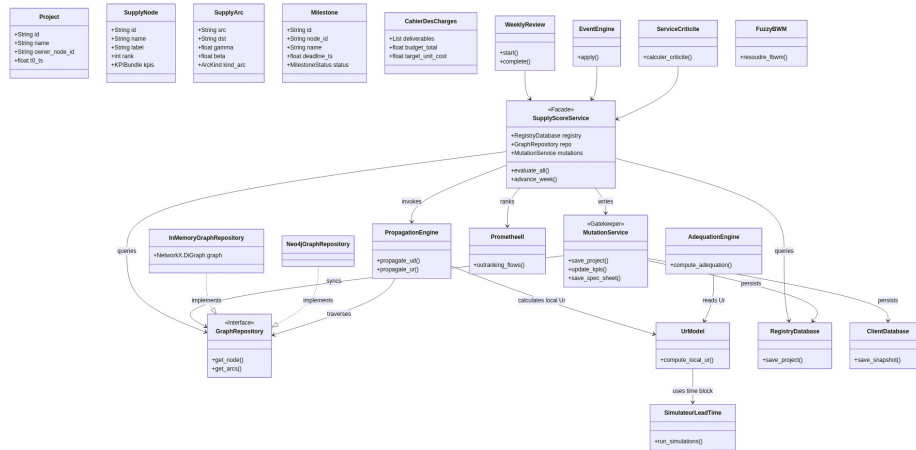


Figure 10: The SupplyScore architecture

3.3.2 Weekly Review Cycle and Calibrated Event Engine

The weekly review cycle keeps the twin's KPI state current without free-form manual data entry: every write is structured, signed and auditable. At each new week every active node is flagged as due for review, and the operator completes four sequential sections: the AHP questionnaire, pre-filled with the previous week's values; the week's KPI values, shown against the previous week; progress on active milestones, compared against theoretical planning progress; and any supply chain events that occurred during the week. All four are written as

a single signed, atomic transaction, so no partial or anonymous review can corrupt a node's history.

Rather than asking operators to recompute the KPI impact of a disruption by hand, the system exposes a taxonomy of fourteen calibrated event types, each mapped to one or more KPI updates through one of three calibration operators: a Beta-Bernoulli Bayesian update for failure probabilities,

$$p_{\text{post}} = \text{clip}\left(\frac{p \times n_0 + k}{n_0 + n}, 10^{-4}, 0.99\right), \quad (14)$$

with a prior weight $n_0 = 26$, roughly six months of weekly observations; an exponential moving average for continuous variables such as OEE, $x_{\text{new}} = (1 - \lambda)x_{\text{old}} + \lambda x_{\text{obs}}$ with $\lambda \in [0.3, 0.5]$; and a ratchet update for severity, $s_{\text{new}} = \max(s_{\text{old}}, s_{\text{event}})$. Figure 11 follows one declared event through to the KPI fields it revises, showing which of the three operators applies where.

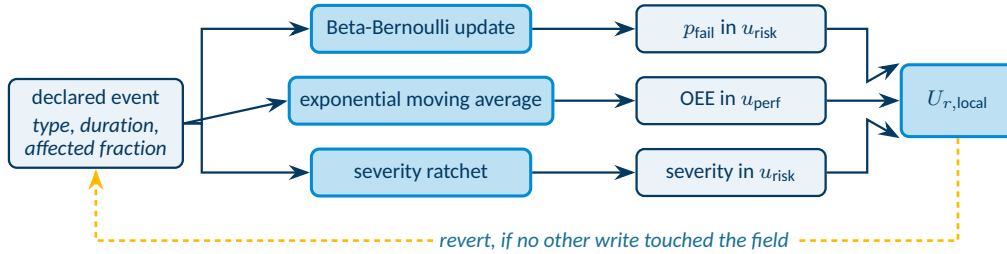


Figure 11: One declared event through to the KPI fields it revises. Each of the fourteen event types selects one or more of the three calibration operators, so no operator recomputes a KPI impact by hand.

As a worked example, a node declaring a strike lasting $d = 24$ hours and affecting a fraction $p_f = 0.50$ of the workforce implies $H_{\text{lost}} = d \times p_f = 12$ equivalent lost hours over a 168-hour week, so $f_{\text{lost}} \approx 0.0714$ and the observed availability is $A_{\text{obs}} \approx 0.9286$. Updating a baseline availability of 0.95 with $\lambda = 0.5$ gives $A_{\text{new}} \approx 0.9393$, while risk severity updates in ratchet mode to $\max(0.30, 0.50) = 0.50$.

3.3.3 Explainability, Export and Operational Hardening

When the twin flags a node, the coordinator's first question is why. The per-node explanation page answers it with exact decompositions rather than approximations. Declared urgency splits into one additive contribution per AHP criterion, $\kappa_j = w_j(s_j - 1)/8$, summing exactly to $U_{d,\text{local}}$. Local real urgency splits the same way through the logarithm of Equation 4: each block contributes $\ell_m = -\omega_m \ln(1 - u_m)$, the shares $c_m = \ell_m / \sum_k \ell_k$ sum to one, and $U_{r,\text{local}} = 1 - \exp(-\sum_m \ell_m)$ reconstructs the score exactly. Figure 12 shows both decompositions side by side for a single node. This is what allows a score to be traced back to the block that produced it without any manual computation, and it is how the diagnosis of Section 4.6.1 was made. Separately, one action compiles the registry and per-node history into a twelve-worksheet Excel workbook so that analyses can run outside the tool on frozen data, and the persistence layer enables write-ahead logging, checks integrity before every backup and copies each database through the native `sqlite3.backup` API without locking the application.

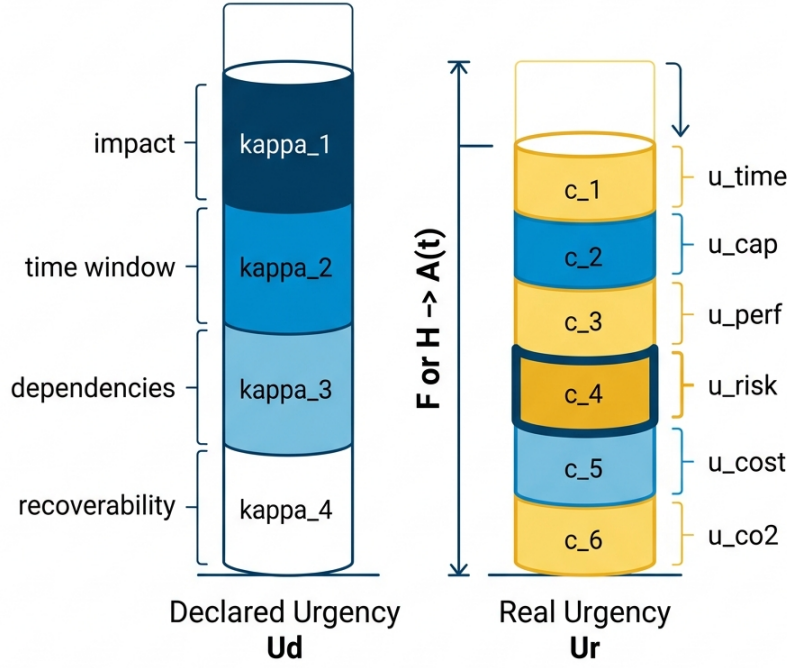


Figure 12: Per-node explainability decomposition

3.4 Experimental Design: The HÉLIOS Campaign

Everything above establishes that the model is implemented, internally consistent and tested. It does not establish whether the adequacy signal says anything true about an operator facing a crisis. This section presents the experiment built for that purpose.

3.4.1 Scenario and Network

The first chain is HÉLIOS, an eight-node satellite supply chain carrying a semiconductor shortage over nineteen weekly turns, with a foundry freeze as the scenario's one critical event. The second is AIRB, an airframer's supply base of fifteen nodes and thirty arcs over five ranks, with no exogenous shock at any point and three structurally central suppliers set to degrade endogenously. Both were frozen and hashed before the first turn. Appendix C gives the network, the calibrated event list and the turn-by-turn injection schedule.

3.4.2 Data Provenance

The two urgency signals come from different sources, which is the point of the design.

Real urgency is fed from public time series of the actual crisis, namely INSEE production and price indices, WSTS worldwide semiconductor billings and a semiconductor-shortage dataset, frozen in a data pack whose SHA-256 hashes were recorded before the first turn. Declared urgency comes from the participants through the weekly AHP questionnaire, with the consistency gate active.

Every third-party dataset is passed through an authenticity check before use. A logistics dataset initially considered for the freight node was rejected at this check for the signatures of synthetic

training data, namely pre-computed target columns, GPS coordinates scattered outside the region the dataset claimed to cover, and uniformly flat distributions. The semiconductor-shortage dataset passed, showing a secular price decline, a genuine 2021 upturn and sectoral employment consistent with official series. The rejected dataset is retained in the raw-data folder for audit but is never loaded by the analysis pipeline, and the freight node's cost block is driven entirely by calibrated events instead.

Two provenance rules protect the real-urgency series. A single-pilot rule forbids double counting: each KPI field of each node is driven either by a time series or by a calibrated event, never by both, so when an event writes a field on a given turn the series skips that turn. Second, one honest artefact of the real data is kept rather than smoothed away. The INSEE business-failures series falls through 2020 and 2021 under public support schemes, so a failure-rate indicator taken alone would lower external risk exactly while the sector crisis was worsening. Keeping it makes the multi-block argument concrete: a single macroeconomic indicator can tell the opposite of the ground truth.

3.4.3 Pre-registered Hypotheses

The six hypotheses below were written with their numeric thresholds and frozen before the campaign produced any data. They are reproduced unchanged: where the campaign later showed a threshold to be badly chosen, or a hypothesis untestable on the scenario as scripted, that is reported as a finding in Section 5.3 rather than repaired by rewriting the hypothesis.

H1 (Cognitive latency): Declared urgency follows computed real urgency with a delay of at least one weekly cycle on the majority of nodes. Test: the lagged correlation between U_d and U_r peaks at a lag of one week or more for at least five of the eight campaign nodes.

H2 (Early detection of hidden risk): The hidden-risk signal $H = \max(U_r - U_d, 0)$ exceeds 0.3 at the crisis-origin node several weeks before the rupture becomes publicly visible.

H3 (Structural criticality): The systematic shock simulation ranks the true crisis origin first, by impact on the final customer, on at least 80% of the informative, non-saturated weekly rounds.

H4 (Predictive validity): At threshold $H \geq 0.5$ with a four-week horizon, the hidden-risk signal predicts recorded negative outcomes, meaning a missed milestone, a critical event or a node abandonment, with precision and recall both above 0.5.

H5 (Urgency contagion): On weeks when crisis news becomes public, operators of nodes that are not physically affected still raise their declared urgency, reproducing the mere-urgency effect (M. Zhu et al., 2018) at network scale.

H6 (Perception hysteresis): Once the physical situation improves, declared urgency decays at less than half the rate of real urgency.

Table 2 maps each hypothesis to the question it serves and states what its failure would mean, so a negative result carries a consequence rather than a verdict.

One omission from the register is worth flagging in advance, because Section 4.7 measures its cost: the thresholds were pre-registered and the definition of an adverse outcome was not, and that definition moves the measured result by more than the thresholds do.

Table 2: Each hypothesis, its research question, and the cost of failure

Hypothesis	Serves	If it fails
H1 cognitive latency	RQ2	Declared urgency tracks the physical signal without delay, so there is no lag to correct and the adequacy layer has nothing to add.
H2 early detection	RQ2	The hidden-risk signal arrives no earlier than the rupture it is meant to anticipate, which makes it a description rather than a warning.
H3 structural criticality	RQ3	The network ranking cannot be trusted to identify the bottleneck, so criticality-driven prioritisation is unsafe to act on.
H4 predictive validity	RQ2	The signal does not predict recorded outcomes, and the score is a diagnostic curiosity rather than a decision aid. This is the hypothesis the aim of Section 1.3 stands or falls on.
H5 urgency contagion	RQ1, RQ4	Operators do not import urgency from context, so declaration bias is individual rather than systemic and does not propagate.
H6 perception hysteresis	RQ2	Over-declaration corrects itself once the situation eases, so the asymmetric penalty is mis-specified and λ_{under} needs recalibrating.

Distinct from these testable hypotheses, the framework rests on stated modelling assumptions, each verifiable in the delivered code: the network is a directed acyclic graph, so a linear topological sort is valid; reviews follow a strict ISO weekly cycle; declared urgency is damped as it descends towards suppliers while real risk is amplified as it ascends, each scaled by a per-arc coefficient; the six KPI blocks are treated as independent so a probabilistic aggregation is admissible; task statuses drive explicit override rules; and the cognitive loss is asymmetric, with under-declaration priced at roughly twice the cost of panic. These are design commitments rather than empirical claims, and the limitations they induce are catalogued in Section 5.4.

3.4.4 The Three Arms

The experiment compares three arms on the same frozen scenario, which is what allows the effect of a treatment to be separated from the behaviour of the scenario. Figure 13 gives the three loops.

Arm A is the control: eight isolated agents declare their urgency each turn seeing only their own role card and turn briefing, and it is the source of the prospective forecasts scored in Section 4.6. Arm B changes one thing, showing each agent the predictive analysis of its own node from turn 4 onward, and adds one field to the declaration, `influence_prediction`, taking one of four values: no influence, confirms my assessment, revised upward, or revised downward. That field is the direct measure of the treatment effect. Arm C lets the agents act on the terrain through a catalogue of interventions applied by the mutation service and recorded in an intervention journal, and was executed in its agent version after the repair of Section 4.10.

A rupture avoided thanks to an alert makes a good prediction look wrong, which is why the

comparison of arm C with arm A in Section 4.10 is made on final state, and why the headline count of unfinished milestones there is not the figure to read.

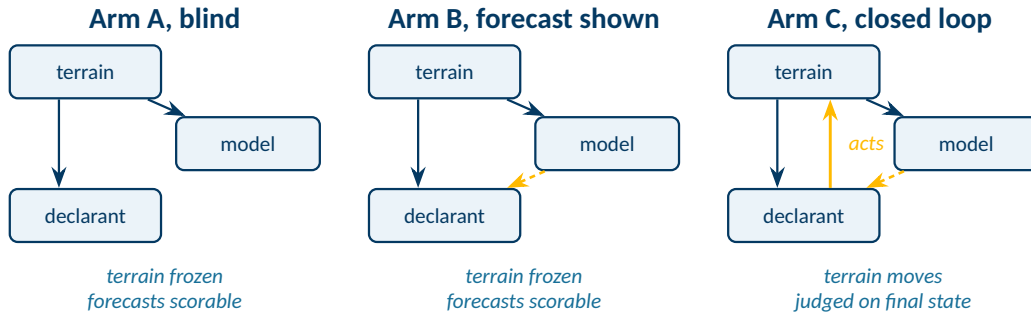


Figure 13: The three arms. In A and B the terrain is frozen, so the forecasts remain scorable and the only difference between the two is whether the model’s output reaches the declarant, the dashed ochre arrow. In C that arrow is joined by a second: the declarant acts on the terrain, the loop closes, and the reference truth moves with it, which is why C is judged by its final state rather than by the accuracy of its predictions.

Two properties of the design carry the results. The agents are language models, one per role, each confined to its own role card and turn briefings with no access to the other roles, the dashboard or future turns, an anti-lookahead discipline enforced by the harness rather than by instruction; a single model instance per role has no statistical standing as a substitute for eight independent human judges, which Section 4.11 treats as the principal threat to external validity. And the declaration format in arm B is identical to arm A but for the influence field, which is what makes the two comparable.

3.4.5 Scoring Protocol

The forecast layer emits, for each node and each turn from turn 4 onward, a probability p_h that an adverse outcome occurs at horizons $h \in \{1, 2, 3, 4\}$ weeks. Scoring those forecasts requires an outcome definition, and this thesis uses the one the production calibration service itself applies, so that the model is judged against the target it was built for. A node-week is a positive observation when at least one of three conditions holds within the window $[S + 1, S + h]$: a milestone of the node whose deadline falls in the window is abandoned or is still active with its deadline passed; the node is abandoned; or an event of gravity critical or defect is recorded and not subsequently reverted.

Four quantities are computed, following standard forecast verification practice (Brier, 1950; Murphy, 1973; Gneiting et al., 2007):

- **Discrimination**, by the area under the ROC curve, computed by mean ranks so that tied forecasts are handled correctly, with a percentile bootstrap 95% interval over 2 000 resamples. AUC answers whether the forecast orders node-weeks correctly.
- **Calibration**, by the Brier score and its skill against the climatological forecast that always announces the base rate. A negative skill means the forecast is worse calibrated than announcing the base rate.
- **Reliability**, by a banded table comparing the mean announced probability with the observed frequency in each band. The bands are deliberately uneven, because a uniform ten-bin split hides the failure mode that matters here.

- **Operating point**, by the confusion matrix at the pre-registered threshold of 0.5 used by hypothesis H4.

How those quantities are to be read. Eight nodes over nineteen turns is a large number of node-weeks and a very small number of events. Every table in Section 4 rests on a positive class of between one and six observations, and at that size a point estimate on its own is close to meaningless. Three rules are therefore fixed here, before any result is seen, and applied throughout.

Proportions carry a Wilson score interval rather than a normal approximation (Wilson, 1927). The reason is specific to this data rather than stylistic: the counts of interest sit at the boundary, three of three and zero of three, where the normal interval has zero width and is simply wrong. The Wilson interval stays inside $[0, 1]$ and keeps a defensible width at both endpoints.

Discrimination is tested rather than only estimated. The AUC equals the Mann-Whitney U statistic divided by the product of the class sizes, so the null hypothesis $AUC = 0.5$ has an exact conditional test, and it is that test which decides whether a reported AUC is distinguishable from chance. The percentile bootstrap interval is still reported, because it conveys the spread, but it is not what any claim rests on: resampling a positive class of size four produces an interval whose coverage is not the nominal one, and Section 4.8 shows the two disagreeing on this data at one horizon.

No claim is made from a comparison the design cannot support. Comparing two AUCs computed against different label sets is not a paired comparison, so the three outcome definitions of Section 4.7 are reported side by side and never differenced. Where two scorers are applied to one label set, as in the baseline comparison of Section 4.9.2, the comparison is paired and a permutation test on the observed difference is the appropriate one.

The consequence is stated plainly rather than buried. With this many events the campaign can establish the direction of an effect and the existence of a mechanism; it cannot establish a rate, and no rate is claimed from it.

Separating discrimination from calibration is essential rather than decorative (Murphy, 1973): a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures call for different remedies. Because the outcome definition is itself a modelling choice, the same forecasts are additionally scored against a deliberately naive target, any non-reverted event on the node regardless of gravity and ignoring milestones. The difference between the two scorings is reported in Section 4.7 as a result in its own right.

3.5 Limitations and Simplifications of the Method

Four simplifications are built into the method itself, as distinct from the limitations of the delivered framework catalogued in Section 5.4.

The six KPI blocks are treated as probabilistically independent so that the noisy-OR aggregation is admissible. They are not independent in reality: a capacity loss raises cost, and a performance drop lengthens lead time. The consequence is that correlated blocks double-count evidence, which the composite-indicator literature identifies as the standard failure mode of weighted aggregation (Greco et al., 2018; Becker et al., 2017). The explainability

decomposition of Section 3.3.3 is the mitigation: the coordinator sees per-block contributions rather than only the aggregate, so a correlation is visible rather than hidden.

The network is assumed acyclic, which permits a linear topological sort and makes propagation deterministic and cheap. Real supply networks contain feedback, most obviously in returns and rework loops.

The scenario is scripted, which is what makes the ground truth knowable and the forecasts scorable, but it also means the events are the only source of adverse outcomes. The consequence, quantified in Section 4.6, is that the positive class is extremely small, and any conclusion about discrimination is correspondingly weak.

The declarants are language models. This buys perfect protocol compliance, exact reproducibility and no scheduling cost, and it loses the property that matters most for hypothesis H4, namely an imperfect human perception of a crisis. This is discussed at length in Section 4.11 rather than treated as a detail.

3.6 Reproducibility

Every numerical result in Section 4 is regenerated from raw artefacts by a single script, `Utilities/figures/derive.py`, which writes a JSON file that both the figures and the text of this thesis read. No number in this document is transcribed by hand from an intermediate report.

Execution environment. Everything reported here was developed, run and measured on a single laptop: an AMD Ryzen 7 PRO 8840HS, eight cores and sixteen threads, with 11.7 GB of usable memory, running Windows 11 and CPython 3.12.1 in a project-local virtual environment with a pinned lock file. No GPU is used at any stage, including the segmentation work of Section I, and no computation is distributed. Persistence is a single file-based SQLite database per project, with no server process of any kind.

That is worth stating for two reasons beyond reproducibility. It bounds what the timings of Section H.4 mean, since they are consumer-hardware figures rather than server figures and would only improve on a deployment machine. And it settles the deployability question the industrial supervisor asked at the outset: the whole system, model and interface and database together, runs inside the resources of a standard engineering workstation, so a pilot needs no infrastructure request.

The inputs to that script are the two campaign databases, one per arm, and the raw per-agent declaration and forecast logs. The outcome labels are not reimplemented: the script calls the production `CalibrationService` directly, so the thesis scores the model against the same target definition the running system applies. Random procedures are seeded, specifically the bootstrap interval on the AUC and the Monte Carlo lead-time simulation.

Two provisions guard against silent drift. The script recomputes the arm A against arm B forecast comparison on every run and reports whether the two arms carry identical prediction-layer output, which they must since the replay is deterministic. It also reports the outcome base rate at every horizon, so a change in the underlying database is visible immediately rather than absorbed into a summary statistic.

One earlier analysis file in the campaign folder, `analysis/rapport_experience.md` with its companion JSON, holds a superseded run whose participant-influence counts disagree with the raw logs. It is used nowhere in this thesis, and the derivation script names it in an exclusion list so the choice is recorded in the output.

The satellite observation system of Section I is a separate codebase with its own verification script. Its network calls are cached on disk, so a re-run does not depend on the availability of the public services it draws from, and its calibrated constants are stated in the source with the sample size behind each.

3.7 Ethical Considerations

The research was conducted under the Cranfield University Research Ethics System.

The study as executed involves no human participants, since the declarants of both campaigns are language-model agents and no personal data was collected, processed or stored at any point. It was submitted as a Category 1, low-risk study, and the approval letter is filed with this thesis; Appendix A records what the human version of the protocol would require.

The protocol was originally designed for eight human consultants, and that version would require participant consent for two features of incomplete disclosure, the cover story and the two placebo sheets, together with the right to withdraw and a full debrief. Appendix A records both and what each would require, so the obligation is not lost if the human campaign is run.

Two further considerations apply to the work as it stands. The data behind the scenario is public, namely INSEE and WSTS series and an open dataset, and the ALTEN material is covered by the confidentiality statement at the front of this thesis. The satellite work draws only on public French open-data services, retains no imagery of private individuals and operates on industrial buildings; Appendix A states the constraints in full.

3.8 Conclusion

The method combines a probabilistic multi-block urgency model, a structured elicitation of declared urgency, a bidirectional propagation over the supply network and an asymmetric scoring of the mismatch between the two, as a tested, auditable and persistent system. Around it, a pre-registered three-arm experiment on a frozen scenario allows the forecast to be scored with proper scoring rules, and the effect of showing it to the declarant to be isolated from the behaviour of the scenario. What the method establishes is bounded by its own design, and those bounds have been stated rather than left implicit.

4 Results and Discussion

4.1 Introduction

Two claims are worth separating before any figure is given, because conflating them is the easiest way to misread what follows. That the gap between declared and computed urgency carries information about which supplier will fail is supported by the campaign. That the gap can be converted into a well-calibrated probability attached to a named week is not, yet. The distance between those two statements is the subject of Sections 4.6 and 4.10, and the sample that bounds both is stated wherever a figure is given.

Every number below is regenerated from raw artefacts by the derivation script described in Section 3.6.

4.2 The Delivered System

SupplyScore is delivered as a standalone Python application: 73 modules across 10 packages, covered by 1831 automated test functions in 106 test files, driving a multi-page Dash web interface over a multi-tenant SQLite persistence layer. The suite expands to 2 044 collected cases, of which 2 028 pass and 15 are skipped for want of an optional dependency; one fails on two missing image files referenced by a presentation script outside the library. All of the mathematical objects of Section 3.2 are implemented and exercised: the six-block noisy-OR aggregation, the AHP elicitation with its consistency gate, the Fuzzy BWM weighting, the bidirectional DAG propagation, the asymmetric adequacy score, the Monte Carlo lead-time estimate, the PROMETHEE II network outranking and the systematic shock-simulation criticality index.

Three operational capabilities carry what follows. The per-node explainability page reconstructs any score term by term, which is what allowed the diagnosis in Section 4.6.1 to be made from the application rather than by hand. The Excel export produces frozen data for analysis outside the tool. And the hardened persistence layer is what makes a campaign database an auditable record weeks after the run. Two pages carry most of the operator's interaction with the system: the weekly review (Figure 14) and the criticality ranking (Figure 15).

4.3 Mechanical Validation: The Synthetic Dry Run

Before any declaration data existed, the full campaign mechanics were exercised end to end on a disposable database. Five iterations calibrated the scenario and eliminated engine artefacts, and the final iteration replayed all nineteen turns with synthetic respondents standing in for the eight declarants.

This is an engineering check, not a scientific one, and the reason is in how the synthetic declarations were made: each is anchored on the node's own local real urgency, following $1 + 8 \times U_{r,\text{local}}$ with a fixed panic, under-report or neutral bias per role. A declaration built that way cannot lag the signal it is derived from, cannot under-report it for long, and cannot react at all to a press event it never reads, so it returns not confirmed or not applicable on all six hypotheses by construction. Figure 16 shows that circularity: real urgency solid and synthetic declared urgency dashed, tracking each other closely at every node rather than one lagging or decoupling from the other.

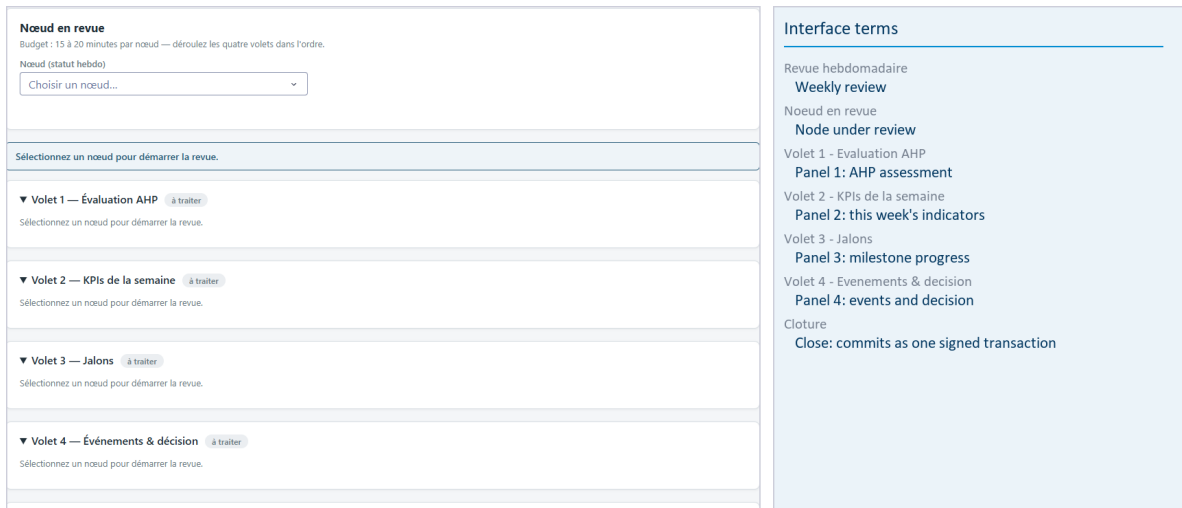


Figure 14: The weekly review page, with its interface terms keyed into English. The four panels commit as one signed transaction.

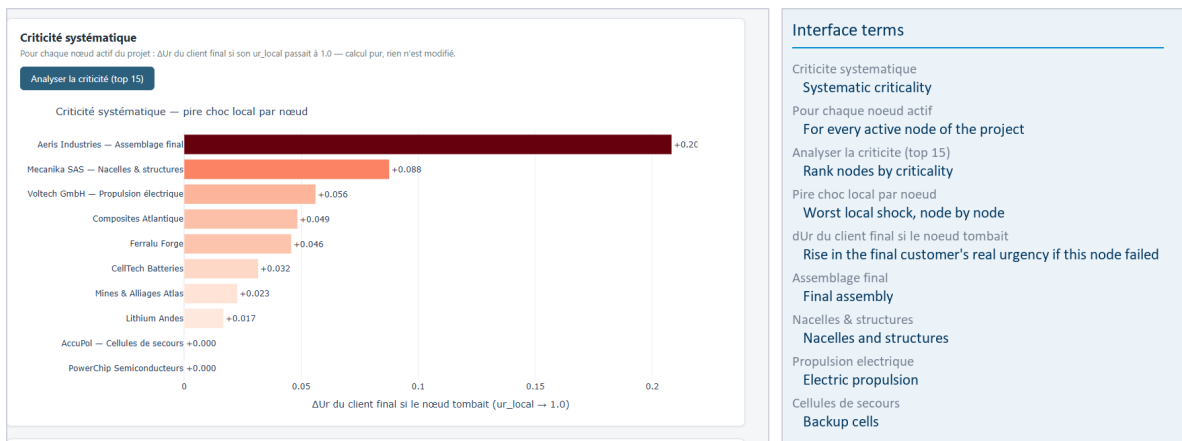


Figure 15: The criticality ranking a coordinator would use to decide where to look first: for each active node, the rise in the final customer's real urgency were that node to fail completely.

What the run does establish is that injection, weekly decay, milestone drift, event calibration, propagation, criticality, snapshotting and backup all run correctly under the two-turns-per-day cadence the campaign requires. It also calibrated the null baseline detector, a raw threshold $U_{r,local} \geq 0.5$, at precision 0.16 and recall 0.27: the bar the adequacy signal must clear to demonstrate value beyond a raw KPI reading.

4.4 Arm A: Blind Declaration by Eight Isolated Agents

Arm A replaced the synthetic generator with the eight isolated agents of Section 3.4.4. What matters here is that nothing was simulated around them: each answered the real AHP questionnaire in character, pairwise comparisons and severity scores passed through the production consistency-ratio code untouched, and campaign setup, turn injection and submission went through the same weekly-cycle service a browser-based participant would reach. Table 3 gives the verdicts on the six pre-registered tests.

Two results stand out. H5 is confirmed: declared urgency at nodes that are not physically

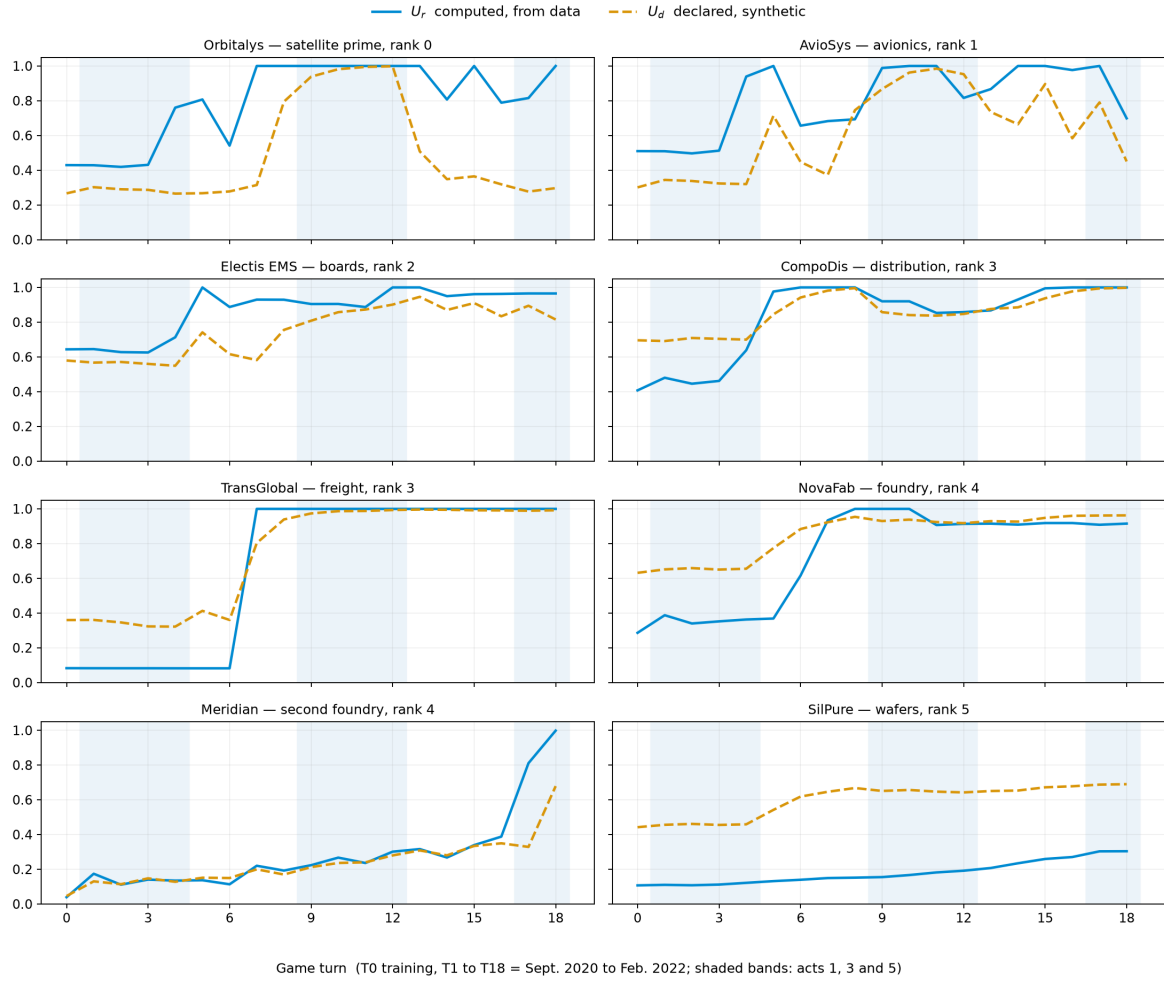


Figure 16: Real against synthetic declared urgency, dry run

Table 3: Arm A verdicts on the six pre-registered hypotheses

Test	Verdict	Measured value
H1 latency	Not confirmed	3 of 8 nodes at lag ≥ 1 turn (threshold: 5 of 8)
H2 early detection	Not confirmed	$H(\text{NovaFab})$ peaks at 0.017 on turns 6 to 12 (threshold: 0.3)
H3 criticality	Not confirmed	NovaFab top-1 on 7 of 10 informative turns, 70% (threshold: 80%; saturated turns excluded per Section 3.2.5)
H4 predictive validity	Not confirmed	precision 0.00, recall 0.00 at $H \geq 0.5$
H5 contagion	Confirmed	non-affected nodes: $\Delta U_d = +0.079$ on press turns against $+0.005$ on calm turns
H6 hysteresis	Not applicable	real urgency shows no measurable decline over the recovery turns

affected rises by 0.079 on press-event turns against 0.005 on calm turns, matching the pre-registered direction and ordering, which is the mere-urgency effect (M. Zhu et al., 2018) reproduced at network scale. And H4 stays not confirmed for a specific and informative reason: these agents read their briefing carefully and declared urgency close to what it described, so U_d tracks U_r too closely to leave the gap that H is built to catch, more tightly even than the fixed-bias dry run, with $H(\text{NovaFab})$ peaking at 0.017 against 0.070.

Read from the resulting project after the eighteenth turn, the network-average adequacy is 65.5, four of the eight nodes carry a hidden-risk reading above 0.1, one carries a false-urgency reading above 0.1, and the worst-scoring node by a wide margin is Orbitalys at $A = 15.3$ (Table 41).

Orbitalys' own explanation page traces the chain from a single scripted event to its score without manual computation. Its next milestone had slipped past its deadline by turn 18, which forces u_{time} to 1 under the rule of Table 1; with the node's real urgency already saturated by this cause alone, the page attributes the whole of it to the local block and none to the upstream network. The agent playing Orbitalys declared $U_d = 0.41$, so the hidden risk is $H = 0.59$, penalised through Equation 12 at $2.25 \times 0.59^{0.88} = 1.42$, giving $A = 100 \times (e^{-1.42} - e^{-2.25}) / (1 - e^{-2.25}) = 15.3$, the same number the dashboard reports. The network-wide PROMETHEE II ranking places NovaFab and Orbitalys at the top of the priority list, at $\phi = +0.220$ and $+0.215$, which are the expected bottleneck and the node its own missed milestone makes most urgent. The same ranking carries a caveat the tool raises on its own: on this run the time and performance blocks are perfectly correlated, $\rho = 1.00$, which is exactly the composite-indicator risk flagged in Section 2.8.

4.5 What the Adequacy Signal Gets Right: Node-Level Targeting

The scoring that follows in Section 4.6 asks whether the forecast layer attaches correct probabilities to node-weeks. That is the right question for a forecast and the wrong question for the instrument. A coordinator does not act on the proposition that node X has a 0.14 chance of an adverse outcome between weeks $t + 1$ and $t + 4$; they act on a shortlist of suppliers to look at this week. This subsection scores that, because it is the decision the adequacy signal was designed to support, and because the answer differs from the one below.

At each turn the eight nodes are ranked by hidden risk $H = [U_r - U_d]_+$, and the set flagged at $H > 0.10$ is compared with the set of nodes that go on to miss at least one milestone at its originally committed deadline, the definition of Section 4.7. Four of the eight nodes are positive under that definition: the satellite prime, the avionics integrator, the electronics manufacturer and the foundry. The comparison is at node level and prospective, since the flag at turn t is confronted with misses occurring later in the campaign. Nothing in it uses the forecast layer.

Table 4 gives the two regimes the campaign separates, and the boundary between them is the turn on which the scenario's foundry freeze becomes public.

For six consecutive turns the signal flags exactly three nodes, all three of which later miss a commitment, with no false alarm on any of the six. The fourth positive node is never flagged early, so recall stops at three of four. On every one of those turns the worst-scoring node of the eight is the satellite prime, which misses its turn-8 delivery and delivers at turn 14: the flag precedes the miss by eight turns, and it is raised while that node's own manager is

Table 4: Nodes flagged by hidden risk against nodes that later miss a commitment, arm A

Regime	Turns	Precision	Recall	False alarms per turn
Before the crisis is public	T0 to T5	1.00	0.75	0
After	T6 to T18	≤ 0.50	0.25	1 to 3

declaring an urgency of 0.001.

That last clause is the construction working as specified rather than a coincidence. The flag does not come from a pessimistic declaration; it comes from an optimistic declaration standing against a computed state that is not. A system aggregating indicators alone would have ranked that node on its indicators. A system collecting declarations alone would have believed it. Only the difference between the two carries this information, which is the proposition the thesis set out to test and the one the literature review of Section 2.14 found unaddressed.

4.5.1 Why the Signal Degrades, and What the Sample Supports

From turn 6 the precision falls to 0.50 and the recall to 0.25, for a structural reason. Once the shortage is public every declarant raises their urgency, U_d converges towards U_r , and a quantity defined as $[U_r - U_d]_+$ shrinks by construction: a perception-gap measure cannot survive the disappearance of the perception gap. The consequence is a scope statement rather than a limitation to be repaired. The instrument is an early-warning device and not a crisis-management one, and its useful window is the interval in which acting is still cheap. The repair of Section 4.10 sharpens the flag without altering which nodes it selects: at turn 3 the satellite prime scores 26.6 out of 100 against 46.0 for the next worst node, and after repair the same turn gives 2.2 against 44.6, so the ordering is identical and the separation widens by an order of magnitude.

The standing of the result is bounded and stated plainly. It rests on eight nodes, four of them positive, over one scripted scenario. A precision of 1.00 sustained over six turns on three flagged nodes is an encouraging observation and not an estimated rate: its Wilson interval runs from 0.44 to 1.00, which Section 4.8 reads against the interval of the second chain. What would settle it is more chains rather than a different model.

4.5.2 The Second Chain, and the Boundary It Draws

A second chain was then built and run: AIRB, an airframer's supply base of fifteen nodes and thirty arcs over five ranks, with no exogenous shock at any point and three structurally central suppliers set to degrade endogenously. Its milestone truth is derived from the indicator trajectory rather than scripted, so a supplier that slows down misses its milestone because it slowed down. Fifteen declarants, one per node, each seeing only its own monthly sheet and its own past declarations, played all nineteen turns.

On that chain the adequacy signal does not select the failing nodes at all: across all nineteen turns, precision at rank three is 0.00 for both $[U_r - U_d]_+$ and the adequacy score, against a base rate of 0.20, while the forecast layer scored on the same runs holds at 0.56.

The mechanism is legible in the snapshots and is not a failure of aggregation. The three

degrading suppliers declare the highest urgencies in the chain from turn 0 onwards, before their indicators have moved: $U_d = 0.94, 0.88$ and 0.98 at turn 0, still 0.97 and 0.96 at turn 16 against a computed U_r of 1.00 . A quantity defined as $[U_r - U_d]_+$ is then bounded by 0.03 . Nothing is concealed, so nothing is detected. The nodes that top the gap ranking at every turn are the downstream assemblers, whose local urgency is zero and whose computed U_r is inherited entirely through the graph from suppliers they cannot observe.

This is a boundary rather than a refutation, and it is sharper than the scope statement above. The role cards of those three suppliers described concealers, and the declarants behaved as such in prose, turn after turn, writing that they would put off calling their customers and keep overtime in reserve. At the same turns they marked near-maximal severity on the questionnaire. They minimised in *action* and maximised in *measurement*. The questionnaire asks what an actor perceives; withholding is something an actor does. A gap defined between a computed state and a declared perception therefore detects blindness, not discretion. The synthetic declarants of the semiconductor campaign carried a bias applied to the number itself, so the gap opened and precision reached 1.00 ; a declarant that reasons does not lower the number.

The claim that survives both chains is therefore narrower: the perception gap is informative where declarants misjudge their own state, and silent where they judge it correctly and simply do not escalate. Distinguishing those two failure modes requires an instrument that observes escalation behaviour and not only declared severity, which the present design does not provide. Section 5.5 states it first rather than as a caveat, because it bears on what the instrument measures and not merely on how well.

4.6 Arm A Rescored: Discrimination Without Calibration

The hypothesis verdicts above test the adequacy signal at a single threshold, and a single operating point cannot distinguish a forecast that orders outcomes badly from one that orders them well but announces the wrong numbers. This subsection therefore scores all 120 prospective forecasts of arm A, covering turns 4 to 18 across eight nodes, against the production outcome definition of Section 3.4.5. Table 5 gives the results at each horizon.

Table 5: The arm A forecast layer, scored by horizon

Horizon	Positives	Base rate	AUC [95% CI]	Brier	Brier skill
1 week	2	0.017	0.555 [0.470, 0.639]	0.130	−6.93
2 weeks	4	0.033	0.372 [0.258, 0.492]	0.166	−4.14
3 weeks	5	0.042	0.428 [0.315, 0.546]	0.191	−3.77
4 weeks	6	0.050	0.485 [0.389, 0.583]	0.207	−3.36

There is no discrimination to speak of. The AUC ranges from 0.372 to 0.555 and every 95% interval contains 0.5 , so on this campaign the forecast does not order node-weeks better than chance. That statement carries its sample size: the positive class contains between 2

and 6 observations depending on horizon, far too few to support a confident claim in either direction, so the campaign provides no evidence of discrimination rather than refuting it.

The calibration result is much stronger, because it does not depend on the small positive class in the same way. The Brier skill against climatology is between -3.36 and -6.93 , meaning the forecast is between four and eight times worse, in mean squared error, than a trivial forecaster that announces the base rate every time. Figure 17 bands the four-week forecasts by announced probability and compares the mean announced value with the observed frequency in each band.

The pattern is unambiguous, each marker's area proportional to the number of forecasts in its band. Where the model announces a low probability it is right: 74 forecasts announcing a mean of 0.070 were followed by an adverse outcome 8.1% of the time. Where it announces a high probability it is wrong every single time: twenty-one forecasts announced a mean of 0.948 and none was followed by the outcome it predicted. The confusion matrix at the threshold of 0.5 records 0 true positives, 21 false positives, 6 false negatives and 93 true negatives.

Hypothesis H4 is stated on a different quantity and should not be read off that matrix: it tests the hidden risk H , not the four-week forecast, at its own threshold of $H \geq 0.5$. Measured on the campaign database through the production calibration service, that second matrix reads 0 true positives, 3 false positives, 8 false negatives and 141 true negatives over 152 node-weeks at four weeks. The two signals fail at the same threshold and for related reasons, and both return precision and recall of exactly zero, but they are separate measurements and the H4 verdict rests on the second.

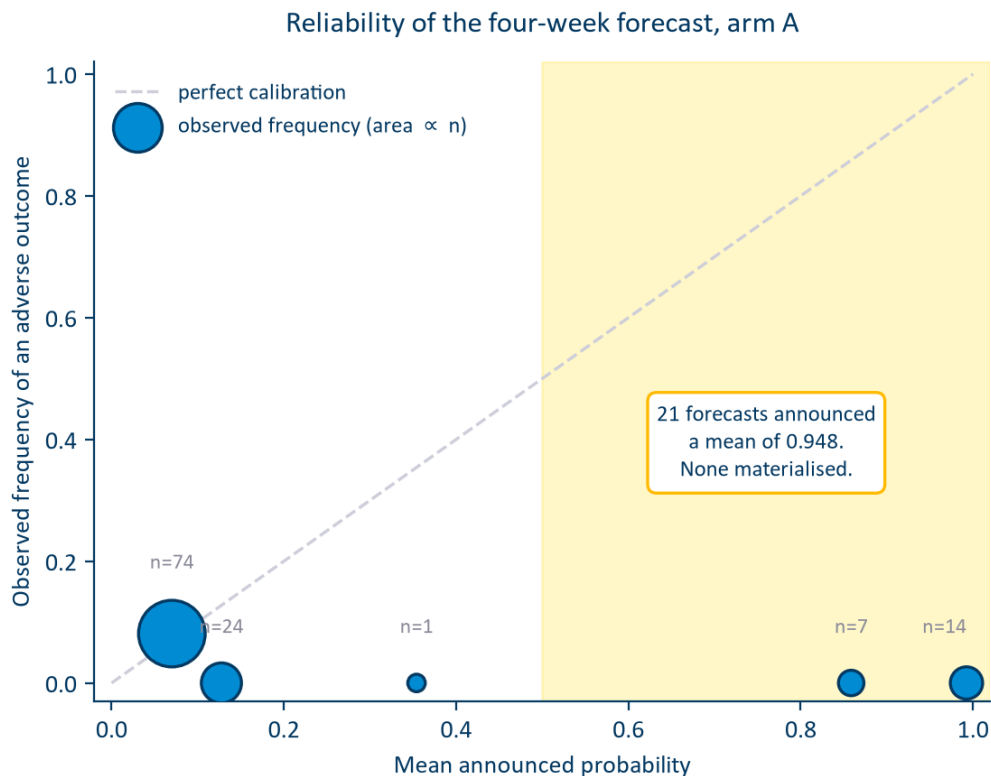


Figure 17: Reliability of the four-week forecast, arm A

The two failures are therefore of different kinds, and this is the dissociation that Murphy's decomposition anticipates (Murphy, 1973): the ordering is uninformative, and the level is badly wrong at the top of the range while being close to right at the bottom.

An obvious objection is right-censoring: forecasts made near the end of the campaign have observation windows extending past the last turn, so if the confident ones were concentrated there, censoring rather than miscalibration would explain the result. They are not. The twenty-one confident forecasts are spread over turns 5 to 18 and across seven of the eight nodes, and repeating the scoring with every forecast whose window runs past the final turn removed leaves $n = 88$, still 0 true positives against 16 false positives, with the Brier skill moving from -3.36 to -7.82 : worse rather than better.

4.6.1 The Mechanism

Figure 18 states the defect before the evidence for it.

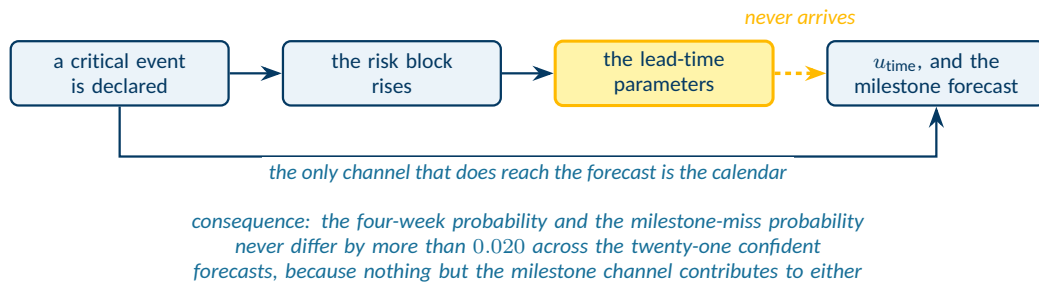


Figure 18: The defect located in this section. An event raises the risk block but never reaches the lead-time parameters that feed u_{time} , so the milestone forecast is driven by the calendar alone and is blind to the shocks the scenario injects.

The confident tail has a single source, and the explainability layer of Section 3.3.3 makes it visible without instrumentation: across all twenty-one confident forecasts, the four-week probability and the milestone-miss probability never differ by more than 0.020. Nothing but the milestone channel contributes to it.

Set against the registry, this is decisive. Of the eighteen milestones in the campaign, ten completed and eight remained active, none was abandoned, and exactly one was still open past a deadline that had fallen inside the campaign. The model announced near-certain misses for AvioSys, CompoDis, Électis, NovaFab, TransGlobal and Meridian, and all of them delivered. That reading is more forgiving than it appears, because the registry stores each milestone's *current* deadline and a replanning overwrites it, so a milestone re-dated and subsequently delivered is recorded as completed. Read against the deadline each node originally committed to, six of the eleven milestones whose original deadline fell inside the campaign were missed, and five of the twenty-one confident forecasts were followed by a missed original commitment within four weeks rather than none. The confident tail remains wrong far more often than right, but the distance between zero of twenty-one and five of twenty-one is produced entirely by which deadline the truth is read against, which is the subject of Section 4.7.

The same defect is visible directly in the state snapshots, from the opposite side. Reading the u_{time} block at turn 6, the turn on which the scenario's one critical event lands, gives the values of Table 42.

Six of the eight nodes read as numerically zero in the middle of a crisis and the seventh reads as saturated: the block behaves as a step function rather than a risk gradient, either off or fully on. The cause is structural. The block compares a lead-time distribution against the time remaining to the next deadline, while the scripted events act on capacity, cost and failure probability without touching any quantity on the time axis, so a shock never moves what the milestone probability is computed from. The strongest form of the evidence depends on no scoring choice at all: across the whole campaign, eight nodes and 41 indicator snapshots each, the accumulated-delay field the block would need was written a non-zero value exactly zero times.

This is a defect in the model rather than a property of the scenario, and it is the one finding of the campaign with a concrete repair. It also invalidates, for this run, the assumption of Section 3.2.2 that each block contributes useful information to the aggregate. One block did not.

4.7 The Target-Definition Trap

The scores above depend on a choice that is easy to overlook: what counts as an adverse outcome. Section 3.4.5 uses the production definition, requiring a missed milestone, an abandoned node or an event of gravity critical or defect. Two alternatives are worth scoring the identical forecasts against: the one an evaluator without access to the model's own definition would reach for, any event recorded on the node; and one that keeps the production definition intact but reads a missed milestone against the deadline the node originally committed to, dropping the forecasts whose window runs past the last campaign turn. Table 6 gives the three.

Table 6: The same forecasts under three outcome definitions

Outcome definition (4-week horizon)	<i>n</i>	Base rate	AUC	Brier skill
Production definition: missed milestone, abandonment, or critical event	120	0.050	0.485	−3.36
Naive definition: any recorded event	75	0.387	0.484	−0.64
Original-commitment definition: production outcome, deadline as first committed, censored points excluded	90	0.289	0.654	−0.55

The model is identical, the forecasts are identical, and the horizon is identical. Only the definition of truth changes. Between the first two rows the measured base rate moves from 5.0% to 38.7% and the Brier skill improves by a factor of more than five, from −3.36 to −0.64. An analyst choosing the easier target would report a system that appears five times better calibrated without a single line of the model having changed.

The third row moves the answer in a different direction and for a defensible reason rather than a convenient one. Holding the definition of an adverse outcome fixed and only changing which deadline a milestone is judged against carries the Brier skill from −3.36 to −0.55 and the AUC from 0.485, indistinguishable from chance, to 0.654. The argument for preferring it is about what a forecast is for rather than about statistics: a replanning is itself the formal admission of a slip, so a truth in which re-dating erases the miss scores the forecast against an outcome it was built to warn about before the re-dating happened. It is the definition used

for the re-measurement of Section 4.10, and stating which of the three any figure refers to is not pedantry: the three disagree by more than any modelling decision taken in this work.

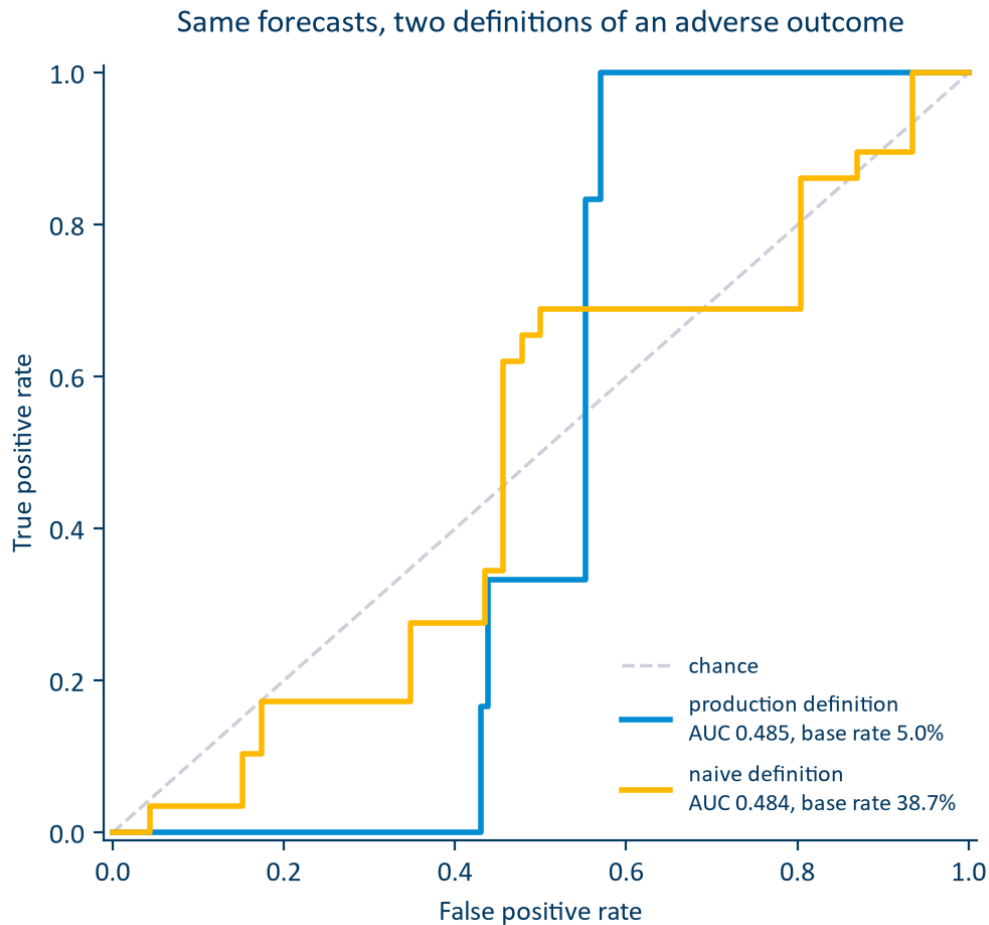


Figure 19: ROC under both outcome definitions

Discrimination, by contrast, is almost untouched, moving from 0.485 to 0.484 (Figure 19). That contrast is itself informative: the ordering of node-weeks is a property of the forecast, while the apparent quality of the probabilities is substantially a property of how often the target fires, since a frequent target flatters a forecaster who announces high probabilities. A calibration figure reported without its outcome definition is therefore close to meaningless, and the definition belongs in the pre-registration alongside the thresholds, where the H1 to H6 register does not currently carry it.

4.8 How Much of This Is Signal

Every score reported so far is a point estimate resting on a positive class of between one and six observations. The protocol of Section 3.4.5 fixed in advance how such estimates are to be read; this section applies it to the headline claims and states what survives. Nothing new is measured here. The forecasts, the labels and the scores are the ones already reported, and only their uncertainty is added.

Table 7 tests every arm A discrimination figure against chance, and Figure 20 draws the same eight rows against the 0.5 line.

Table 7: Arm A discrimination against chance, under two outcome definitions. The interval is the percentile bootstrap; the p -value is the exact conditional test of $AUC = 0.5$

Outcome definition	h	n	pos.	AUC [95% boot.]	p
Production	1w	120	2	0.555 [0.470, 0.639]	0.797
	2w	120	4	0.372 [0.258, 0.492]	0.388
	3w	120	5	0.428 [0.315, 0.546]	0.590
	4w	120	6	0.485 [0.389, 0.583]	0.904
Naive, any event	1w	75	11	0.603 [0.404, 0.786]	0.279
	2w	75	19	0.601 [0.456, 0.741]	0.192
	3w	75	25	0.545 [0.409, 0.675]	0.529
	4w	75	29	0.484 [0.358, 0.621]	0.823

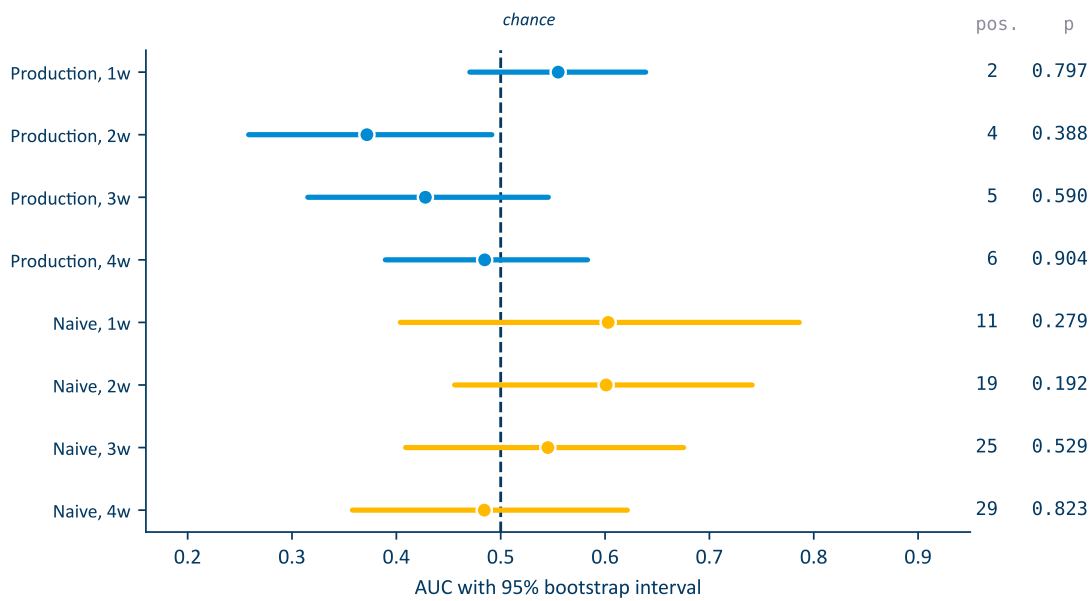


Figure 20: Every arm A AUC with its bootstrap interval, read against the dashed chance line. An interval crossing that line is a result the campaign cannot separate from chance. The positive-class size and the exact p -value are given at the right of each row.

Not one of the eight is distinguishable from chance. The smallest p -value in the table is 0.192, and it belongs to the naive definition, the one this thesis argues against using. The claim of Section 4.6 that arm A discriminates without calibrating is therefore weaker than the AUC figures alone suggest: what the campaign establishes is that calibration is bad, which the Brier skills settle at any sample size, and that discrimination is unproven in either direction rather than demonstrated.

One row deserves separate comment, because it is where the two methods disagree. At two weeks the bootstrap interval $[0.258, 0.492]$ excludes 0.5 while the exact test returns $p = 0.388$. The exact test is the one to follow, and the disagreement has a cause rather than being a coincidence: with four positives, a bootstrap resample redraws that class with replacement, so a large share of resamples contains three or fewer distinct positives and the resulting interval is narrower than its nominal coverage. This is precisely the failure the protocol anticipated, which is why the interval was recorded as descriptive and the test as

decisive before any of these numbers were seen.

Table 8 does the same for the claims that are counts rather than rankings.

Table 8: Wilson 95% score intervals for the proportions reported on the two chains

Claim	Count	Estimate	Wilson 95%
Chain 1, precision of the adequacy flag	3/3	1.00	[0.44, 1.00]
Chain 1, recall over the failing nodes	3/4	0.75	[0.30, 0.95]
Chain 1, failure base rate	4/8	0.50	[0.22, 0.79]
Chain 2, precision of the adequacy flag	0/3	0.00	[0.00, 0.56]
Chain 2, failure base rate	3/15	0.20	[0.07, 0.45]

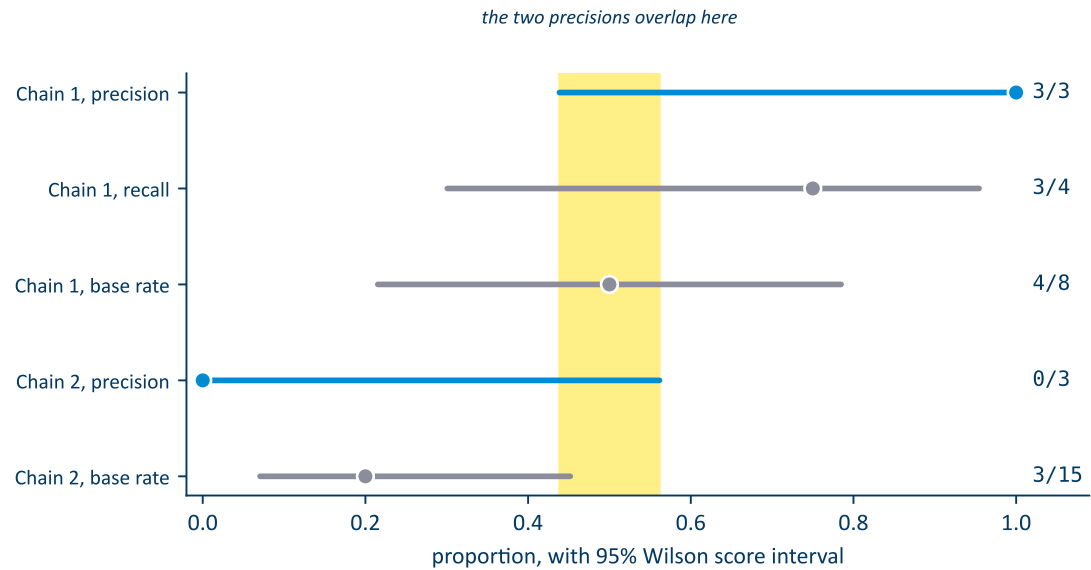


Figure 21: The five proportions with their Wilson intervals. The shaded band is the range the two precision intervals share: a precision of 1.00 on three flags and one of 0.00 on three flags are not, on this evidence, distinguishable.

The two precision rows are the ones that matter, and Figure 21 shows why: their intervals share the shaded band. The contrast between the two chains, which Section 4.5.2 reads as a boundary, is therefore a difference between two point estimates and not a demonstrated difference between two populations. That does not make the boundary wrong, and the mechanism argued for it stands on its own; it means the boundary is a hypothesis the campaign generated rather than a result the campaign established, and it is reported as one.

What survives this, and why. Three findings do not depend on the size of the positive class, and they are the ones this thesis rests on.

The first is structural. The defect located in Section 4.6.1 is a property of a code path, not of a sample: the block in question cannot respond to the events the scenario injects, which is verifiable by reading the propagation of a single event on a single node and reproducible with one observation. A defect that is present in every run is not made more or less true by how many runs there are.

The second is a theorem. The collapse of the criticality ranking under saturation is Proposition 3.2.5, proved from the range and monotonicity of the propagation, and the campaign merely provides an instance of it in nine turns of nineteen. No inference is involved.

The third is a within-forecast comparison. The three outcome definitions of Section 4.7 score the identical forecasts, so the spread between them is a fact about the definitions and not an estimate of anything in a population. It needs no interval because nothing is being generalised.

What does not survive is any statement of a rate. No number in this thesis should be read as an estimate of how often the instrument would flag correctly in service, and the campaign was never sized to produce one. Section 5.5 gives the sample the question would need.

The same standard applied to the one confirmed hypothesis. H5 is the single hypothesis this campaign confirms, and it is reported as a difference between two means, $\Delta U_d = +0.079$ on press turns against $+0.005$ on calm turns at nodes that are not physically affected. It carries no interval and no test, and applying the protocol to the five negative verdicts while exempting the positive one would be the exact asymmetry the protocol exists to prevent. The comparison is a paired one, the same nodes observed under two conditions, so a permutation test on the turn labels is available and costs nothing; it was not run because the per-node-turn declarations behind the two means are not among the artefacts the derivation script of Section 3.6 currently exports. That is a reproducibility gap rather than a statistical one, and it belongs on the list of Section 5.5: until it is closed, H5 should be read as a confirmed direction of effect and not as a measured effect size.

4.9 Arm B: What Happens When the Forecast Is Shown

Arm B replays the identical frozen scenario with the identical eight agents, changing one thing: from turn 4 onward each agent is shown the predictive analysis of its own node before declaring. The loop stays open, so the terrain does not change and the forecasts remain scorable. One property of the design must be established before any interpretation: the prediction layer is deterministic, so the two arms should carry identical forecasts on the turns they share, and they do, all 72 shared node-turn forecasts being numerically identical. Any behavioural difference therefore comes from the act of displaying the forecast, not from a different forecast being displayed.

The dataset comprises 97 declaration records, reduced to 96 after removing one duplicate submission at turn 1, each carrying an influence field recording whether the displayed forecast had no influence, confirmed the agent's assessment, or caused an upward or downward revision. Figure 22 gives the distribution turn by turn.

The first exposure is the result. At turn 4, when the forecast becomes visible for the first time, all eight agents record an influence and the direction is entirely one-sided: five revised downward and three had their existing assessment confirmed. Not one revised upward. The first upward revision anywhere appears at turn 5, and the scenario's single critical event, the foundry accident, lands at turn 6, two turns after the moment every declarant was either reassured or confirmed.

Over the full campaign the distribution is close to balanced, 15 upward against 16 downward, so what the data shows is a strong first-exposure effect rather than a permanent suppression of declared urgency. Two factors aggravate it. The forecast displayed at turn 4 is drawn from

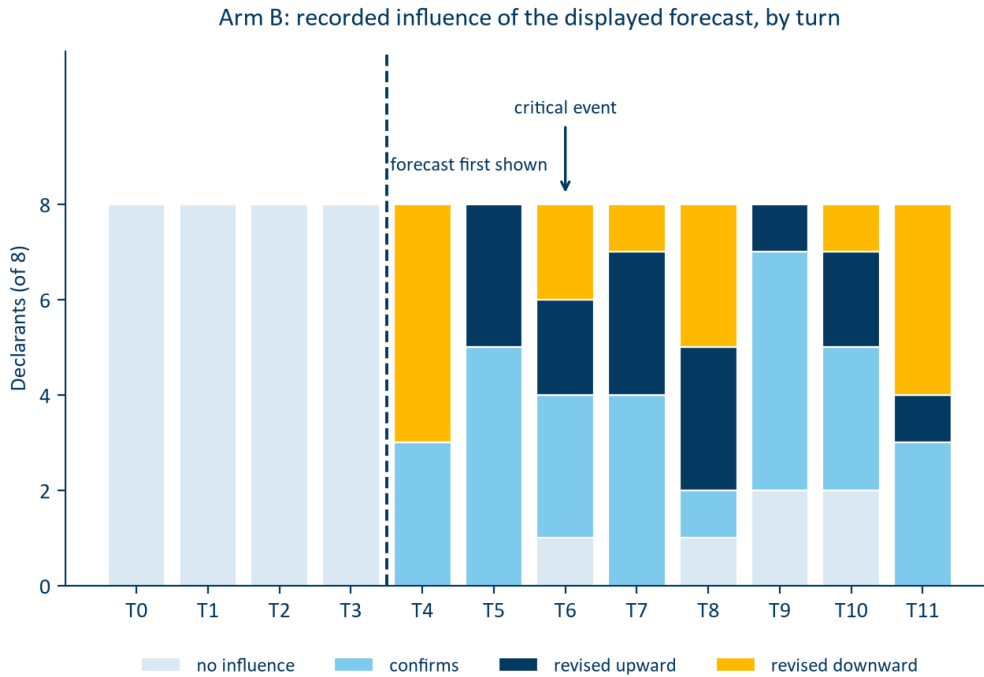


Figure 22: Arm B: influence of the displayed forecast, by turn

the same layer whose confident predictions never materialise and whose u_{time} block is blind to the very shock about to arrive. And it is displayed as a numeric probability with a confidence interval, a format the automation-bias literature identifies as increasing reliance (Goddard et al., 2012; Bansal et al., 2021). The system was presenting an unfounded reassurance in the most persuasive available form, immediately before the event it had failed to anticipate.

One qualification cuts against the strongest reading. The influence field records the effect on the declared urgency level, not on the decision taken, and the free-text notes show agents continuing to negotiate allocation, alert the customer and order ahead of schedule at the same turn. The measured effect is on what was declared rather than on what was done. Since the declared value is precisely the input the adequacy layer consumes, that is still a consequential effect on the measurement system.

4.9.1 The Same Test on the Second Chain

The airframer chain of Section 4.5.2 was then run a second time with the forecast displayed, on the same indicator pack and the same derived truth as its control run, so that the act of display is the only difference between the two. Fifteen declarants, nineteen turns, 225 forecasts per arm, and 225 declarations made with a forecast on screen. Figure 23 carries the three measurements that follow.

Four results follow, and the first is a negative one that simplifies the method. The forecast scores identically in both arms: a Brier skill of -1.601 against -1.592 at one week, an AUC of 0.826 in both, differences confined to the third decimal, and an unchanged node ranking. The forecast is computed from indicators the declaration channel does not reach, so the loop is too weak to close and a treatment arm can be scored exactly as a control arm can. The caution stated when this campaign was designed, that showing the forecast would destroy

the reference, is refuted by measurement.

The second replicates first contact at five times the scale. On the turn the forecast first appears, eleven of fifteen declarants report it confirms what they already thought, two revise downward, and *none* revises upward: the only turn in the campaign without a single upward revision. One node states the mechanism in its own note, its second lot having sat at zero for four consecutive turns, which should have raised its time-window urgency further, but the tool's reading of eleven per cent tempered that concern. The forecast suppressed an escalation the indicators warranted.

The third qualifies the first two and is the one that generalises. Across all fifteen turns with a forecast visible, upward revisions outnumber downward ones 26 to 9: the one-sided suppression is a first-exposure effect, and the persistent effect is different and larger. A declarant follows the number in whichever direction it moves, one node raising its declared urgency at turn 12 while its own indicators were improving because the tool had risen. The display does not sedate the operator; it transfers authority from the shop floor to the model, reassuring when the model is low and the ground is bad, alarming when the model is high and the ground is good. On a layer whose Brier skill is -1.59 at one week, that transfer is the harm, and it is the composition of two defects documented separately: miscalibration, and display.

The fourth was not sought, and it is the one a designer can act on. The share of declarants reporting that the forecast changed nothing rises from two at turn 4 to twelve at turn 18, and over the last four turns seventy-three per cent report no effect. Their reasons are measurable: between consecutive turns, on the same node and with the terrain unchanged, the median move in the four-week probability is 0.024, but fourteen of 139 moves exceed 0.20, eight exceed 0.50, and the largest is 0.93. A forecast that falls from thirty per cent to eight in one turn teaches its reader to stop looking. Disengagement is a rational response to instability rather than negligence, and it is more costly than the first-contact effect: by the closing turns, when the layer finally does place a high probability on the supplier that goes on to miss, no one is reading it.

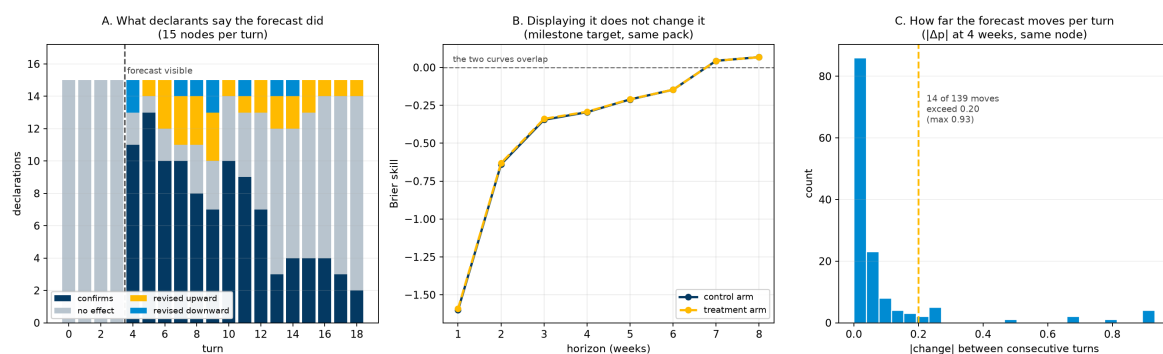


Figure 23: The second chain, run twice on one indicator pack. **A:** what declarants report the forecast did to their judgement, by turn: first contact at turn 4 produces no upward revision at all, and reported influence decays as the campaign proceeds. **B:** the forecast scores identically whether or not it is displayed; the two curves overlap. **C:** the turn-to-turn movement of the four-week probability on an unchanged node, which is what declarants cite when they stop trusting it.

4.9.2 What the Model Being Followed Actually Knows

The preceding result rests on a premise worth testing: that the layer declarants defer to is doing work a declarant could not do unaided. Two observations put that in doubt. Enriching two dead indicator blocks on this chain moved computed urgency on 174 of 285 node-turns and left the forecast identical to nine decimal places on 151 of 154 comparable points, and reading the simulation loop explains why. The autoregressive state fitted to local urgency enters only the client-impact channel; the scored outcome is the disjunction of a milestone term and an event term, the milestone term depends only on a lead-time draw against the fraction of the milestone still outstanding, and the event term is empty on a chain carrying no exogenous shock. Computed urgency never reaches the quantity being scored.

That is a claim about code, so it was tested against a baseline built to be as weak as possible. For a node whose next unfinished milestone falls due at turn D with fraction r of its work outstanding and nominal lead time L in turns, the baseline announces $p(k) = \text{clip}(1 - (D - t - k)/rL, 0, 1)$: no simulation, no fitted parameter, no calibration and none of the six urgency blocks. The service declines to forecast on 71 of the 225 node-turns, where no milestone is in flight, and the baseline is silenced on exactly those, so both are scored on an identical sample at every horizon.

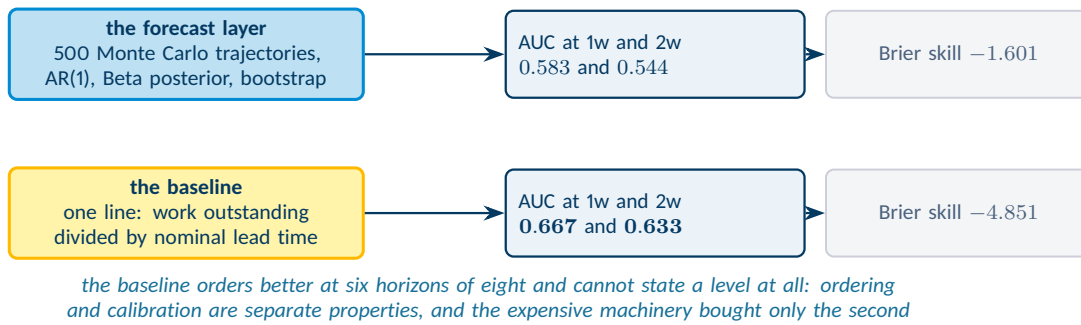


Figure 24: The comparison that decides whether the forecast layer earns its complexity. The baseline is silenced on the 225 node-turns where no milestone is in flight, and on the rest it ranks the misses better while producing no usable probability.

The baseline ranks the misses *better* than five hundred Monte-Carlo trajectories at six horizons of eight. At one week its AUC is 0.908 against 0.826 and its PR-AUC 0.396 against 0.153; the Monte-Carlo layer regains the advantage only at seven and eight weeks (0.667 and 0.633 against 0.583 and 0.544). Since the baseline reads the remaining fraction from the same progress variable the truth is derived from, it was re-run in a form that reads nothing at all, remaining work taken as the calendar fraction a perfectly healthy milestone would show, using only the due date and the nominal lead time. Its AUC and PR-AUC are identical to three decimal places at every horizon. The discrimination is entirely the ratio of weeks remaining to weeks required.

What the simulation layer does contribute is visible in the same table and is not discrimination. Its Brier skill is -1.601 at one week against the baseline's -4.851 : the baseline produces an ordering and has no means of turning it into a probability, and converting an ordering into a scale is what the five hundred trajectories are doing. It also holds its ranking at the long horizons where a calendar ratio collapses, at nearly double the PR-AUC at seven and eight weeks. The layer is therefore a calibration wrapper around a lead-time ratio, which is a

defensible thing to be, but it is not what the interface presents and not what the declarants in Section 4.9.1 believed they were deferring to.

This sharpens rather than softens the display result. Twenty-six upward revisions and nine downward ones were made in deference to a short-horizon signal that a two-number ratio reproduces and slightly exceeds: the transfer of authority documented above was not a transfer to a model that knew more but, at the horizons displayed, to one that knew less. The defect is nameable and cheap to repair, since the autoregressive state must enter the milestone term and not only the client-impact term, and the repair would be measurable on these same frozen forecasts. It was not attempted here, and the sample bounds it: five positives at one week, one chain, one outcome definition.

4.10 The Repair, in Brief

The defect of Section 4.6.1 was repaired and the identical frozen forecasts were rescored against it. Two results belong in the body and the working is in Appendix J.

The repair changed the recommendation as well as the model. Routing the calibrated events into the lead time, which is the obvious reading of the diagnosis, does not work: a quantity routed that way is scaled by the fraction of work remaining, so it attenuates to nothing on exactly the milestones nearest their deadline. Lost time has to be added to the completion date instead. The diagnosis was right and the instruction it implied was wrong, which was only visible because the repair was re-measured rather than assumed to work.

What it bought is bounded. The structural repair lifts both calibration and discrimination at every horizon, the Brier deficit roughly halving from -0.553 to -0.216 at four weeks. The system moves from badly wrong to less wrong, not to useful, and the Brier skill remains negative throughout. Two further defects surfaced with it: the forecast treated declared progress as exact, which made the milestone probability binary in 85% of cases, and the campaign ran at 25.6% indicator coverage with three of six blocks inert.

4.11 Discussion

Six threads run through this discussion and they are not claims of the same kind. Four survive the sample size because they concern how a measurement is built; two do not, because they concern how often something happened. Reading them as though they were equivalent is how a campaign of this size gets over-read.

The circularity problem is the central methodological finding. Across the dry run, arm A and arm B, the same difficulty recurs in three different forms. In the dry run, synthetic declared urgency was anchored on real urgency by construction, so the gap the system exists to measure could not appear. In arm A, agents read their briefings carefully and declared close to what the briefing described, so the gap again did not appear, with H peaking at 0.017 where the hypothesis required 0.3. In arm B, showing the model's own output to the declarant closed the loop explicitly, so the declaration is no longer independent of the model at all. A system that measures the divergence between a human judgement and a computed state is only measuring anything when those two quantities have independent sources. I expected that to be a property of the design. It turned out to be a property that has to be defended in three separate places, and I missed it in all three before the data showed me.

Ordering and level are separate problems needing separate fixes. The scores of Section 4.6 do not support a single verdict on the forecast layer. On this campaign it neither orders well nor announces well, but the evidence for those two claims is of very different strength: discrimination rests on between two and six positive observations and is therefore inconclusive, while the calibration failure rests on 21 confident forecasts none of which materialised and is not plausibly a sampling artefact. The remedies differ accordingly. Miscalibration of this shape is routinely corrected by a post-hoc layer without touching the underlying model (Niculescu-Mizil et al., 2005; C. Guo et al., 2017). Poor discrimination is not, and would require the model to change.

The specific defect is repairable and general, but not in the obvious way. The mechanism identified in Section 4.6.1 is not a scenario artefact. The u_{time} block computes milestone risk from a lead-time distribution against a deadline, while the event engine writes to capacity, cost and failure probability. No path connects a shock to any quantity on the time axis, so the block cannot respond to a disruption, and any deployment of this framework would inherit that gap. The natural repair is to make the calibrated events write a lead-time impact, and Section 4.10 reports that this was implemented, measured and found to be wrong. The general form of the lesson is that a parameter describing a node's capacity to recover and a state accumulating what it has already lost carry the same units and are easy to confuse, and the confusion stays invisible while the database is incomplete.

Communicating ignorance is a design requirement, not a nicety. The arm B result reframes what an advisory layer owes its reader. Faced with an exogenous shock, a weather event, a fire, a geopolitical decision, the model has no basis for prediction at all. Displaying a low probability in a confident format is then worse than displaying nothing, because it converts an absence of information into an apparent reassurance. Two mitigations follow and neither requires new science: make the horizon salient, so that a low four-week probability cannot be read as a general statement of safety, and state explicitly which classes of risk the model does not cover. I would now treat both as acceptance criteria for an advisory layer rather than as refinements to add later.

Threats to validity. Four limit what the above can support. The declarants are language models, so the external validity of any claim about human declaration behaviour is unestablished; this is the principal threat, and it is why arm B is reported as a measured effect on this population rather than as a finding about operators. The campaign is a single scenario, so the outcome base rate, the event mix and the milestone structure are all properties of one script. The positive class is very small, which bounds every discrimination claim. Finally, arm B is not blinded, since the agents are told they are being shown a predictive analysis; a design in which some turns show a deliberately uninformative forecast would separate the effect of the content from the effect of the display.

Against the pre-registered thresholds. Five of the six hypotheses are not confirmed and one, H5 on urgency contagion, is confirmed. That is a largely negative result against the register, and it is reported as such. The value of the campaign lies less in those verdicts than in the diagnosis they made possible: the register asked whether the signal works, and the answer, with its mechanism, is that one identifiable block prevents it from working and that

showing the resulting forecast to the declarant is actively counterproductive until that block is fixed.

5 Conclusion

5.1 Answers to the Research Questions

Table 43 answers each research question in the terms it was asked, with the section carrying the measurement.

5.2 Contributions

This research set out to determine whether the gap between declared and computed urgency can be measured reliably enough, inside a working digital twin, to be used as a decision aid rather than as a descriptive statistic. The answer has two parts that must not be run together. As a device for deciding where to look, it works: the nodes the adequacy signal flags in the six turns before the shortage becomes public are, without a single false alarm, nodes that go on to miss a delivery commitment, and the worst-scoring of them is flagged eight turns ahead while its own manager reports that all is well. As a device for attaching a calibrated probability to a named week, it does not work yet, the Brier skill remaining negative after a repair that halved it. The first is the decision a coordinator actually takes; the second is a harder problem the campaign turned from an unknown into a measured distance.

What bounds both statements is the evidence base rather than the design: two chains, seven positives and thirty-three observable milestones cannot establish a rate, however clean the pattern. The five contributions below are therefore claimed as a working instrument with a measured behaviour and a method for interrogating it, not as validated performance figures.

Table 44 states each of the five with the measurement behind it and the bound on it. Two of them turned on a reversal rather than a number, and those are argued after the table.

The repair changed the recommendation, not only the model. Making the events write a lead-time impact is the obvious reading of the C3 diagnosis, and it does not work: a quantity routed into the lead time is scaled by the fraction of work remaining, so it attenuates to nothing on exactly the milestones nearest their deadline. Lost time has to be added to the completion date instead. The diagnosis was right and the instruction it implied was wrong, which is only visible because the repair was re-measured on the identical frozen forecasts rather than assumed to work. Two further defects surfaced with it: the forecast treated declared progress as exact, which made the milestone probability binary in 85% of cases, and the campaign ran at 25.6% indicator coverage with three of six blocks inert, a finding about the data rather than the mathematics.

The same trap was met from the other side. A post-hoc calibration layer fitted against a different target improved calibration while degrading ranking, and was deliberately not connected. Read with C5, the pair states the rule: a calibration figure reported without its outcome definition carries little information, and a calibration layer fitted without checking its target can degrade a system that works.

A sixth result is reported as an opening rather than a contribution. Individual loading docks are not observable from directly overhead, because of canopy occlusion rather than insufficient resolution, and the reframed problem of delimiting an equipped facade is both learnable and sufficient for capacity estimation.

5.3 Status of the Research Hypotheses

Table 9 gives the status of each pre-registered hypothesis across the three runs. It is the single authoritative statement of those verdicts in this thesis.

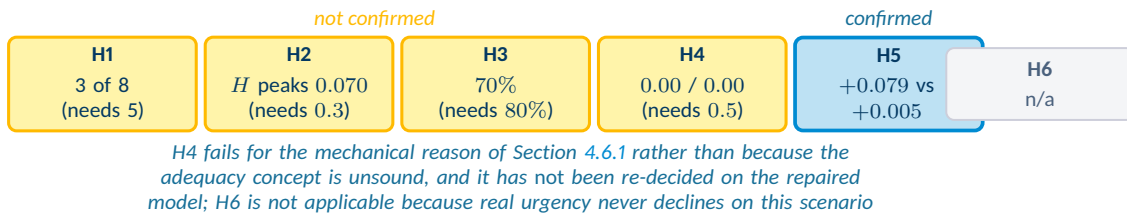


Figure 25: The pre-registered verdicts at a glance, with the measured value against the frozen threshold. Four fail on the number, one is confirmed, and one is inapplicable to the scenario as scripted.

Table 9: Status of the six hypotheses across the executed runs

Hypothesis	Dry run	Arm A	Reading
H1 latency	Not confirmed	Not confirmed	1 of 8 nodes on the dry run, 3 of 8 in arm A against a threshold of 5. The direction is right and the level is short.
H2 early detection	Not confirmed	Not confirmed	<i>H</i> peaks at 0.070 then 0.017 against a threshold of 0.3. The declared signal tracks the computed one too closely for a gap to open.
H3 criticality	Not confirmed	Not confirmed	Top-1 on 7 of 10 informative turns, 70% against a threshold of 80%. Identical across runs, since the test does not depend on U_d . Nine turns were excluded as saturated; Section 4.10 reports what a log-survival sort key recovers from them, as an exploratory re-measurement rather than a revised verdict.
H4 predictive validity	Not confirmed	Not confirmed	Precision and recall both 0.00 at the threshold. Section 4.6.1 attributes this to the time block rather than to the adequacy formulation.
H5 contagion	Not confirmed	Confirmed	$\Delta U_d = +0.079$ on press turns against +0.005 on calm turns. The mere-urgency effect reproduced at network scale.
H6 hysteresis	Not applicable	Not applicable	Real urgency shows no measurable decline over the recovery turns, so the comparison is not meaningful on this scenario.

One hypothesis of six is confirmed. Three qualifications belong with that count. H4 fails for an identified mechanical reason rather than because the adequacy concept is unsound. That mechanism has since been repaired (Section 4.10), but H4 has *not* been re-decided here: the verdict is a precision and recall measured on the arm A declarations at a pre-registered threshold, and re-running it on a model that has changed would not answer the question the register asked. It should be re-decided on the human campaign, against the repaired model, with the outcome definition written into the register. H6 was never testable on this scenario,

because the script does not de-escalate far enough for a decay rate to be compared, which is a design fault in the scenario rather than a result.

5.4 Limitations

Two of the limitations below go away if someone runs the instrument on more chains. The other two do not, and I want to be clear about which is which, because a reader who assumes all four are matters of sample size will over-read every result in Section 4.

Table 10 gives each one with the operational consequence I attach to it.

Table 10: Documented limitations and their implications

Limitation	Implication
The time block was blind to shocks, and the repair does not make it good	Measured, not suspected: on the campaign the calibrated events wrote nothing on the time axis, so milestone risk could not respond to a disruption and read as numerically zero mid-crisis at six of eight nodes. This is the cause of the H4 failure. The path has since been built and measured; the Brier deficit halves and remains negative, and the milestone probability is still binary in 71.7% of cases.
The campaign ran on a quarter of its indicators, and one predictive class is missing entirely	Coverage was 25.6%, with 24 of 42 indicators empty on every node and only three of the six urgency blocks contributing; a node with no lead time received a risk floor of zero through a documented fallback, so an absence of data presented itself as a reassuring prediction. Separately, the model reads current flow but not the forward booking horizon, which in the real shortage preceded the degradation of every flow indicator by months.
Validation rests on agent declarants	Both campaigns were executed with language-model agents, not human consultants, so every statement about declaration behaviour is a statement about that population. The human campaign, for which the protocol is frozen, has not been run.
Two scenarios, with a very small positive class	Seven positives in total and thirty-three observable milestones. Discrimination claims are correspondingly weak in both directions, and two scenarios authored by the same hand cannot separate a property of supply chains from a property of the author.
The framework produces probabilities, not certainties	Scores and intervals are model outputs, and the campaign showed the probabilities to be poorly calibrated at the top of their range.
Score freshness follows input cadence	Without an ERP connector, a node's score is never more current than its last submitted weekly review. Manual entry is an accepted trade-off, and Section I reports one route out of it for a single block.
Blocks are assumed independent, and elicitation is limited to four criteria	The noisy-OR aggregation requires independence, and the campaign produced a counter-example: two blocks were perfectly correlated on the arm A run, which the tool flagged on its own. Saaty's method degrades past roughly seven pairwise-compared criteria, which is why declared urgency stays on four while the larger set of KPI blocks is weighted through Fuzzy BWM instead.

Table 10, continued

Limitation	Implication
Event handling carries three known conditions	The Beta-Bernoulli prior weight and the EMA reactivity coefficients follow standard conjugate revision but are not empirically fitted. Reverting a declared event is guaranteed correct only if no other write has touched the same KPI meanwhile, and otherwise the system raises an explicit conflict rather than silently rebasing. And an operator can both declare an event and manually correct the same KPI in one week: the interface warns but does not forbid it, since the operator remains the final authority.
The satellite capacity model is not connected, and one of its constants is not reproducible	It reaches one block of six, covers metropolitan France only, produces a structural capacity rather than a live occupancy, and its trained segmentation model is not yet wired into the production capacity path. The fill factor of 0.80 comes from a 122-site calibration whose detailed output was not kept, and recomputing over the 68 sites still on disk gives a median of 0.97, so volumetric outputs carry an unquantified bias of order twenty per cent until the corpus is recollected. Dock capacity, which depends on the pitch constant instead, is unaffected.

5.5 Future Work

The work divides into what must happen before the framework can be evaluated fairly, and what becomes worth doing afterwards. Two items dominate both lists and are stated first because every other entry is bounded by them.

Table 45 gives what each item takes and whether it must precede a fair evaluation. Two carry an argument a table cannot hold, and follow it.

Why F1 is the sharpest question and not merely the first. The second chain established that $[U_r - U_d]_+$ selects failing nodes when declarants misjudge their own state, and selects nothing when they judge it correctly and choose not to escalate. Both are concealment operationally, and only the first is visible to a questionnaire, because a questionnaire records what an actor perceives while withholding is something an actor does. That makes F1 a question about *what* is being measured rather than about how well, which is why no improvement to the model reaches it. The free-text route carries one caveat: a measure taken on generated text has to be validated against human declarants before it can carry weight.

Why more data does not settle F2 on its own. Two scenarios authored by the same hand cannot separate a property of supply chains from a property of the author, and running a third by the same route would not either. A campaign an order of magnitude larger in positives would let the node-level result be stated as a rate with an interval, and would let the calibration layer be trained on the product's own target, which is the single step now blocking a positive skill. Both gains are real and neither touches the authorship problem, which only an instrumented live chain reaches.

One governance question belongs with F7 rather than after it: the detection library is distributed under a copyleft licence, immaterial for internal use and determinative the moment

the capability is distributed or exposed over a network.

5.6 Personal Conclusion

5.6.1 Technical Skills Developed

The project drew on five distinct areas, and the difficulty of the subject lay mostly in making them work together rather than in any one of them: multi-criteria decision analysis, through AHP and Fuzzy BWM elicitation, consistency validation and PROMETHEE II outranking; probabilistic modelling, through noisy-OR aggregation, Monte Carlo simulation, conjugate Bayesian revision in a sparse-data regime and, added during the campaign analysis, probabilistic forecast verification through proper scoring rules and reliability decomposition; graph algorithms, through propagation over a multi-rank directed acyclic graph, topological ordering, shock simulation and network criticality; software engineering, through the architecture of a multi-tenant Python system, hardened SQLite persistence, a Dash web interface and a milestone-based quality process with blocking test coverage; and geospatial data engineering, through public web services, dataset construction and annotation, and the training of a segmentation model under a strict CPU budget.

5.6.2 What I Learned

The hardest problem was not mathematical. It was recognising, three separate times, that the system was measuring itself. The dry run anchored declared urgency on real urgency by construction; the agents in arm A read their briefings so carefully that their declaration reproduced the computed state; and arm B closed the loop deliberately by showing the model's own output to the declarant. Each time the symptom looked like a modelling problem and each time the cause was circularity in the measurement. I would now design the independence of the two signals first, before designing either of them.

The second lesson concerns what counts as a result. For a long stretch of the project I treated a negative outcome as a failure to be fixed before reporting. The most useful findings in this thesis are negative: the forecast layer is badly calibrated, the confident tail is entirely false, one block is structurally blind, and showing that forecast to a declarant is counterproductive. None of those would have been found by a project optimising for a positive result, and each of them points at a specific repair, which a marginal improvement would not have. The related discipline, which the annotation episode of Section I taught bluntly, is that a confident self-report is not evidence, whether it comes from an automated process or from oneself, and that the cheap verification step should always come before the expensive one.

With hindsight I would change two things. I would have scored the forecast layer against recorded outcomes far earlier, since doing so took a few hundred lines and revealed the time block defect immediately, where months of implementation had not. And I would have written the outcome definition into the pre-registration alongside the thresholds, having since measured how much it moves the answer.

The skill that progressed most was not a technical one. It was the willingness to state a result that does not flatter the work, in enough detail that someone else can act on it. The subject of this thesis is, after all, the gap between what people declare and what is actually true, and it would be a poor outcome to have measured that gap in a supply chain while reproducing it in the reporting.

Professionally, the project confirmed a direction. I want to work where a model meets the person who has to act on it, because that boundary is where the interesting failures live: not in the mathematics, which can be checked, but in what a number does to the judgement of the person reading it.

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A Ethics Documentation

The study as executed involves no human participants. The declarants of both campaigns are language-model agents, and no personal data was collected, processed or stored at any stage. On that basis the project was submitted to the Cranfield University Research Ethics System as a Category 1, low-risk study, and the approval letter is filed with this thesis.

Two features of the protocol would require participant consent if the campaign were rerun with the eight human consultants for whom it was originally designed, and both are recorded here so that the requirement is not lost.

Incomplete disclosure through the cover story. Participants are told they are playing a fictional satellite programme. They are not told that the scenario replays the documented 2020 to 2022 semiconductor shortage, because a participant who recognised the historical chronology could anticipate events rather than react to them, which would destroy the latency measurement that hypothesis H1 depends on. The debrief reveals the real scenario and asks each participant at which turn they recognised it, which converts the principal threat to the measurement into a recorded quantity.

Deliberate misinformation through the placebo sheets. Two participants receive a false de-escalation notice, one at turn 9 and one at turn 15. These sheets control for suggestibility: a declared urgency that follows the placebo is reflecting the sheet rather than a reading of the situation. Participants are not told in advance which sheets are genuine.

Both are standard incomplete-disclosure designs in behavioural operations research, and both are resolved by the full debrief. Were the human campaign to run, the following would need to be in place before the first turn: written informed consent covering the use of incomplete disclosure; the unconditional right to withdraw at any point, with the participant's data removed on request; the debrief protocol, delivered to every participant regardless of whether they completed the campaign; and confirmation that no performance data is passed to line management, since the participants would be colleagues and the campaign records their judgement quality week by week.

Data ethics for the satellite observation work. All imagery and building data come from public French open-data services under their published licences. No imagery of private individuals is collected or retained, and the analysis operates on industrial buildings rather than on people. Vehicle detections are used only as an intermediate geometric cue for locating dock aprons; no vehicle is identified, tracked or associated with an operator, and the detections are not retained beyond the derivation of a facade extent. The regulatory database of classified installations is used solely to calibrate one constant across an aggregate of 122 sites, and never to infer or publish a result about a named site.

B Source Review Grid

Following the structured review method required by the programme's writing guidelines, the grid below records for each stream reviewed in Section 2 its principal sources, the methodological approach those sources take, what this thesis takes from them, and a critical appraisal

of the limitation that prevents the stream from answering the research question on its own. The final column is the one that matters for the argument: it is the accumulation of those limitations that defines the gap stated in Section 2.14.

Table 11: Source review grid

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Deadline-sensitive value and risk propagation <i>Foundation</i>	(K. Chen et al., 1996; J. Chen et al., 2025): analytical utility-accrual scheduling. (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020): conceptual and simulation models of the ripple effect	Urgency treated as time-varying and network-propagated rather than node-local; the noisy-OR choice over weighted averaging	Time-value work stops at a single isolated task. Ripple-effect work models propagation but not the node-level aggregation of heterogeneous blocks that must precede it. Neither addresses a declared signal at all
Cognitive bias in urgency <i>Foundation</i>	(M. Zhu et al., 2018): five controlled experiments. (Steel et al., 2006): formal discounting model. (Rusou et al., 2020; DiwasSingh et al., 2020): field studies	The justification for treating declared urgency as a biased signal; the asymmetric penalty structure; hypothesis H5	Diagnoses the bias but prescribes no operational correction. Effect sizes come from laboratory and clinical settings; the magnitude in industrial supply chains is uncalibrated, which this thesis inherits as an open parameter rather than resolves
Closest antecedents <i>Antecedent</i>	(Sawik, 2013; Sawik, 2014; Sawik, 2016): mixed-integer programming. (Cohen et al., 2019; Afrasiabi et al., 2022): MCDM supplier prioritisation	Confirmation that multi-criteria, disruption-aware supplier evaluation is established, and precise identification of what it omits	Risk is evaluated once, at the point of decision, not maintained per node and propagated. Where a client-declared signal exists it is consumed directly, never validated against an operational counterpart. The optimisation machinery targets allocation, not measurement

Table 11, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Bayesian risk updating <i>Enabling</i>	(Pearl, 1988; Koller et al., 2009): theory. (Hosseini; Ivanov, 2020): survey. (Huang et al., 2020; Deng et al., 2020; Lockamy et al., 2011; Qazi et al., 2017): applied models	Beta-Bernoulli conjugate revision of the failure probability in the risk block, at $n_0 = 26$	Suited to the sparse-evidence regime, but says nothing about whether the resulting probabilities are well calibrated once a human reads them. This thesis measured that gap directly and found it large
Digital twins and decision support <i>Enabling, primary gap source</i>	(Grieves, 2014; Tao et al., 2018; Fuller et al., 2020): concept and survey. (Ivanov; Dolgui, et al., 2021; Ivanov; Dolgui, 2020; Niloofar et al., 2023): supply chain twins and cognitive twins	The architectural family the delivered system belongs to; the human-in-the-loop framing	No retrieved system combines a declared input, a computed operational urgency and their propagated mismatch. A second gap matters here: these systems are evaluated on state estimation accuracy and assume their inputs simply arrive, so input circularity is not treated as a measurement problem

Table 11, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Multi-criteria decision methods <i>Enabling</i>	(Saaty, 1980): eigenvector method. (Rezaei, 2015; S. Guo et al., 2017): best-worst and fuzzy extensions. (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005): outranking. (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021): composite-index critique	The elicitation of U_d with its consistency gate; the Fuzzy BWM weighting of the six blocks; the PROMETHEE II network ranking	Static tools, rarely compared against a computed counterpart. The composite-index critique is usually offered as a caveat rather than instrumented; this thesis instruments it through the explainability decomposition, and the campaign then produced exactly the predicted failure, two perfectly correlated criteria
Forecast-assisted decision and automation bias <i>Foundation for the treatment arm</i>	(Parasuraman et al., 1997): taxonomy. (Skitka et al., 1999; Cummings, 2004): controlled studies. (Goddard et al., 2012): systematic review. (Dietvorst et al., 2015; Logg et al., 2019; Bansal et al., 2021): direction and explanation effects	The interpretive frame for the arm B result; the design requirement that an advisory layer communicate its own ignorance	Dominated by single-decision laboratory and clinical tasks. Does not cover a repeated multi-turn loop in which the declaration feeds back into the model, nor the case where the displayed forecast is known to be miscalibrated in a specific direction, which is the situation measured here

Table 11, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Probabilistic forecast verification <i>Enabling</i>	(Brier, 1950): score. (Murphy, 1973): decomposition. (Gneiting et al., 2007): proper scoring rules. (Niculescu-Mizil et al., 2005; C. Guo et al., 2017): post-hoc calibration	The scoring protocol of Section 3.4.5; the separation of discrimination from calibration that carries the central result	Assumes the predicted outcome is unambiguously defined. In an operational supply chain it is a modelling choice, and Section 4.7 measures that choice moving the reported skill by a factor of five
Serious games <i>Validation rationale</i>	(Stermann, 1989): the Beer Game. (Prensky, 2001; Deterding et al., 2011): design theory. (Mohaddesi et al., 2020; Bellotti et al., 2013; Lafkihi et al., 2025): methodology and applications	Justification of a weekly-cadence game as an empirical instrument for observing declaration bias	Usually scored on learning outcomes or aggregate performance, rarely used as a controlled comparison replaying one frozen scenario with and without a treatment on the declarant, which is the design adopted here
Medical triage <i>Analogy only</i>	(Mackway-Jones et al., 2013): protocol. (Kuligowski et al., 2025): comparative study, weighted $\kappa = 0.51$. (Kim et al., 2020): admission control	Vocabulary of over- and under-triage; the principle of asymmetric risk behind $\lambda_{\text{under}} > \lambda_{\text{over}}$	Analogy only, and flagged as such wherever used. Clinical triage has an outcome observable within hours and a trained, certified assessor; neither holds in a supply chain, so no numeric parameter is transferred from this stream

Table 11, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Physical Internet and carbon accounting <i>Context</i>	(Montreuil, 2011; W. Guo et al., 2020; Li et al., 2020): network context. (Smart Freight Centre, 2023; European Commission, 2017; ADEME, 2014; International Organization for Standardization, 2023): emissions frameworks	The motivation for treating urgency as a network-level quality of service variable; the carbon block as a separate arbitration criterion	Provides context and regulatory grounding, not method. The carbon block was the least exercised of the six on the campaign, so its inclusion remains justified by design intent rather than by measured contribution
Earth observation of built assets <i>Foundation for the opening</i>	(Xia et al., 2018): oriented detection benchmark. (Maggiori et al., 2017; Van Etten et al., 2018): footprint benchmarks. (Kirillov et al., 2023): promptable segmentation. (Henderson et al., 2012; Jean et al., 2016): activity proxies from imagery	The pre-trained detector and its vehicle classes; the sampling of candidate sites from imagery alone; the precedent that imagery can substitute for slow official statistics	The economics work measures aggregate activity at regional scale and produces nothing per-facility. The detection work stops at footprints and vehicles without converting them into an operational capacity. The conversion, and its calibration, is what Section I had to construct

C HÉLIOS Campaign Protocol Extract

This appendix records the elements of the frozen protocol that a reader would need in order to reproduce or audit the campaign, beyond the design already given in Section 3.4.

Cadence and Sequence

Nineteen turns are indexed T0 to T18. Figure 26 places what happens inside one of them.

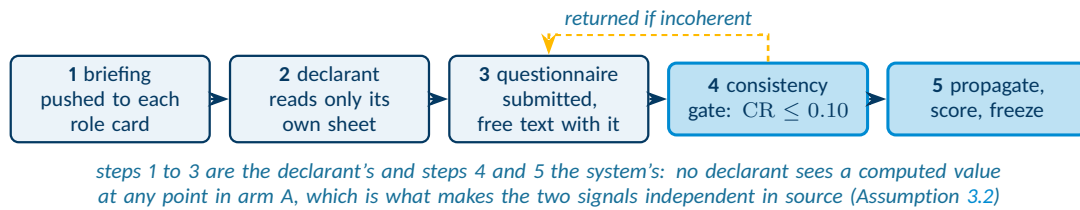


Figure 26: What happens inside a single turn. The dashed arc is the consistency gate returning an incoherent questionnaire; it is the only feedback a declarant receives, and it carries no information about the computed state.

T0 is a training turn used to verify that every declarant can complete the questionnaire inside ten minutes with the consistency gate active; its declarations are retained in the record but excluded from every analysis. The eighteen campaign turns run at two per day, morning and late afternoon, over nine working days. One game turn represents one real month, from September 2020 to February 2022, with real durations converted into engine hours at a fixed rate so that documented figures such as twelve to twenty-five week supplier lead times or a six-day canal blockage retain their true magnitude inside an eighteen-turn game.

Frozen Artefacts

Four artefacts were frozen before T0 and are the reference against which any later analysis is checked: the scenario definition with its twelve calibrated events and their turn assignments; the data pack of public time series, with a SHA-256 hash recorded per file; the role cards, one per node, each stating only what that declarant may know; and the hypothesis register H1 to H6 with its numeric thresholds. Any analysis not derivable from the register is labelled exploratory in the reporting.

Controls

- *Isolation.* Each declarant sees only its own role sheet, its own turn briefing and its own questionnaire. Dashboards, scores, the network view and other declarants' material stay hidden until the debrief. In the executed campaign this is enforced by the harness, which gives each agent a private working directory, rather than by instruction.
- *Placebo sheets.* A false de-escalation notice is handed to the avionics integrator at T9 and to the wafer supplier at T15. A declared urgency that follows the placebo is reflecting the sheet rather than a reading of the situation.
- *Weak-signal turn.* At T4 the press briefing mentions a tension with no engine event behind it, which measures who reacts to media noise alone. This is also the turn on which the arm B treatment first becomes visible, so the two effects are deliberately not separated on that turn and the arm A control is what isolates them.
- *Role assignment.* The two most information-heavy nodes, the foundry and the distributor, are assigned away from the most supply-chain-experienced participants, so that a role effect cannot be confounded with an expertise effect.
- *Journal.* Every deviation from protocol is logged the same day.

Contingencies Arbitrated in Advance

Three contingencies were decided before the campaign rather than improvised under time pressure. The campaign truncates at T16 if it overruns by more than two turns, rather than rushing the remaining responses. It drops to one turn per day if the questionnaire exceeds ten minutes at the training turn. Any absence, or any unresolved consistency failure, is carried forward unchanged and excluded from the reactivity analyses.

Declaration Record Format

Each declaration is one JSON record per node per turn, carrying the turn index, the six bipolar pairwise comparisons, the four severity scores, the derived criteria weights, the principal eigenvalue with its consistency index and consistency ratio, a flag for whether the consistency gate passed, the number of attempts, the raw and smoothed declared urgency, and a free-text note. Arm B adds one field, `influence_prediction`, taking one of four values: no influence, confirms, revised upward, or revised downward. The format is otherwise identical between arms, which is what makes them comparable.

Deviations Recorded in the Executed Campaign

Two are on record. One node submitted turn 1 twice; the first submission is retained and the duplicate discarded, which is why the arm B analysis reports 96 declarations from 97 raw records. Arm B ran to T11 rather than T18, so its forecast log covers turns 4 to 12 against turns 4 to 18 in arm A; all arm B comparisons in this thesis are therefore made over the turns actually recorded, and the arm A scoring uses its own full range.

C.1 Scenario and Network

The campaign, presented to participants under the cover story *Programme HÉLIOS*, replays the 2020 to 2022 semiconductor shortage over eighteen weekly game turns plus a training turn. The supply network has eight nodes, one per participant, anchored on the real actor types of the crisis (Table 39).

Figure 28 gives the network with its per-arc coefficients, γ attenuating demand as it descends and β amplifying risk as it ascends. NovaFab at rank 4 is the expected bottleneck, and the dashed arc into Meridian is the inert backup arc of Section 3.2.4: it appears in the diagram and carries nothing, which is the point of including it.

One game turn represents one real month, with real durations converted into engine hours at a fixed rate so that documented figures keep their true magnitude inside an eighteen-turn game. Figure 27 lays the campaign out: five acts drawn from the documented chronology, and the turn on which each calibrated event lands.

The twelve calibrated engine events driving the crisis are listed in Table 40, with the turn numbers as recorded in the campaign databases. One encoding choice shapes what the scenario can test: the Texas freeze is modelled as a production accident, which triggers the engine's Bayesian revision of the failure probability, a recovery clock and a severity ratchet, rather than as a capacity loss, which would only reduce storage volume. A hot shutdown of a fabrication line is mechanically closer to the former.

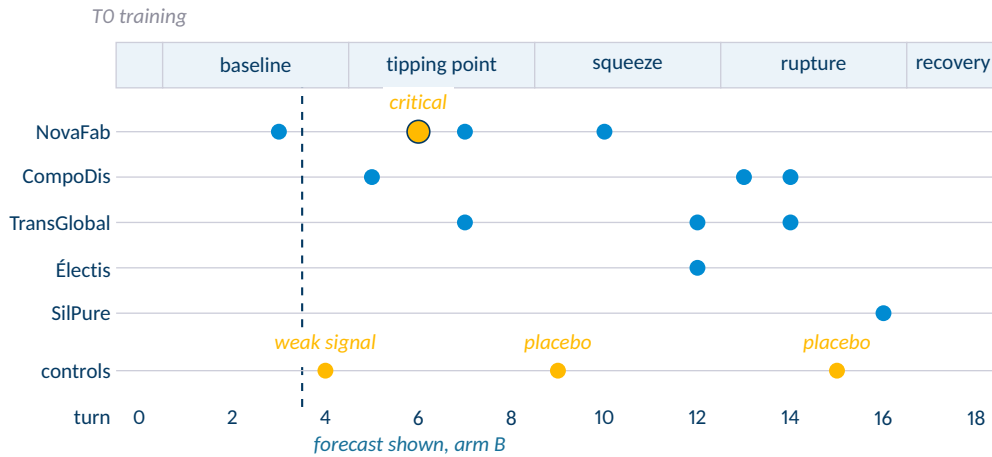


Figure 27: The campaign over its nineteen turns: five acts drawn from the documented chronology, and the twelve calibrated engine events on the node each strikes, with the protocol’s own controls on the lane beneath. Spot prices reach ten to forty times catalogue during the squeeze; the rupture is a manufacturer cutting production by roughly 40% while the announced backup foundry turns out to have no spare capacity; the recovery measures whether declared urgency lags the physical de-escalation. One event is of gravity critical, the fab freeze at turn 6, and Table 40 names each event with its real-world anchor.

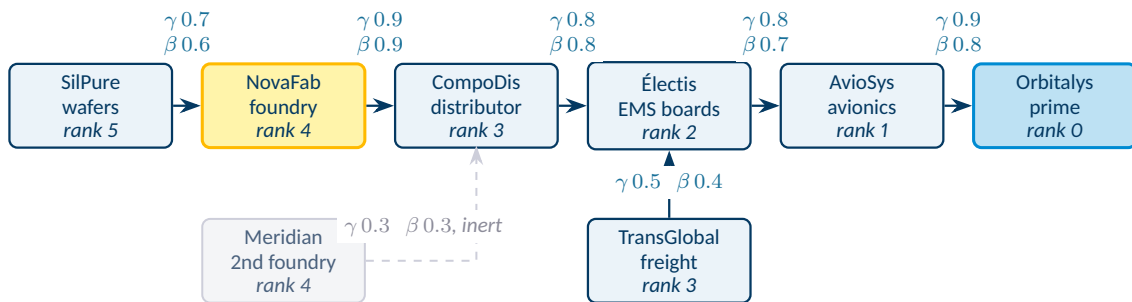


Figure 28: The eight campaign nodes, γ attenuating demand as it descends towards suppliers and β amplifying risk as it ascends towards Orbitalys, the final customer. NovaFab, in ochre, is the expected bottleneck. The dashed arc into Meridian is the announced plan B: it appears in the graph and carries nothing.

C.2 Protocol Integrity and Controls

The protocol was frozen before the first turn. Participants see only their own role sheet and questionnaire; dashboards and scores stay hidden until the debrief, and no discussion between roles is allowed. Three controls sit on the campaign lane of Figure 27. The two placebo sheets each hand one node a false de-escalation, so a declared urgency that follows the sheet is reflecting the sheet rather than a reading of the situation. The weak-signal turn has the press mention a tension with no engine event behind it, which measures who reacts to media noise alone. Beyond those, roles are assigned so that the two most information-heavy nodes do not go to the most experienced participants, keeping a role effect from being confounded with an expertise effect, every deviation is logged the same day in a campaign journal, and any analysis outside the pre-registered register of Section 3.4.3 is labelled exploratory.

D Full Scoring Tables

Every value in this appendix is written by the derivation script of Section 3.6 and is reproduced here in full, so that the summary tables of Section 4 can be checked against the complete output.

Turn to ISO Week Mapping

The campaign runs from ISO week 2026-S31 at T0 to 2026-S49 at T18, one turn per week. The outcome window of a forecast issued at turn t with horizon h therefore spans weeks $t + 1$ to $t + h$ inclusive.

Arm A, Production Outcome Definition

Table 12 gives every quantity behind the summary of Section 4.6, at all four horizons, including the confusion matrix at the pre-registered threshold and the climatological Brier score the skill is measured against.

Table 12: Complete arm A scoring, production outcome definition

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	120	120	120	120
Positive outcomes	2	4	5	6
Base rate	0.017	0.033	0.042	0.050
AUC	0.555	0.372	0.428	0.485
AUC 95% CI, lower	0.470	0.258	0.315	0.389
AUC 95% CI, upper	0.639	0.492	0.546	0.583
PR-AUC	0.027	0.031	0.041	0.054
Brier score	0.130	0.166	0.191	0.207
Brier, climatology	0.016	0.032	0.040	0.047
Brier skill	−6.93	−4.14	−3.77	−3.36
True positives at 0.5	0	0	0	0
False positives at 0.5	15	18	20	21
False negatives at 0.5	2	4	5	6
True negatives at 0.5	103	98	95	93

Arm A, Naive Outcome Definition

The naive definition counts any non-reverted event recorded on the node, regardless of gravity, and ignores milestones entirely. It is scorable on the 75 node-turns for which the node has at least one event in its history. Table 13 repeats the scoring under that definition, and the contrast with Table 12 is the measurement reported in Section 4.7.

Right-Censoring Sensitivity

Dropping every forecast whose observation window would extend past the final campaign turn leaves the four-week scoring on 88 observations with 2 positives, at 0 true positives

Table 13: Complete arm A scoring, naive outcome definition

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	75	75	75	75
Positive outcomes	11	19	25	29
Base rate	0.147	0.253	0.333	0.387
AUC	0.603	0.601	0.545	0.484
Brier score	0.155	0.248	0.337	0.389
Brier skill	−0.24	−0.31	−0.52	−0.64

against 16 false positives, and moves the Brier skill from −3.36 to −7.82.

The Confident Tail, Forecast by Forecast

Table 14 lists every arm A forecast announcing a four-week probability of at least 0.5. The rightmost column is the milestone-miss probability that the model reports alongside it. The two never differ by more than 0.020, which is the evidence that the confident tail is the milestone channel and nothing else. None of the twenty-one was followed by the predicted outcome under the production definition, which is what the Outcome column reports; five of them were under the original-commitment definition of Table 16.

Table 14: Every arm A forecast at or above 0.5 at four weeks

Turn	Node	p (4 weeks)	p (milestone miss)	Outcome
T5	AvioSys Intégration	0.898	0.880	none
T6	CompoDis Europe	0.996	0.996	none
T7	CompoDis Europe	1.000	1.000	none
T8	CompoDis Europe	1.000	1.000	none
T8	NovaFab	0.866	0.846	none
	Semiconductors			
T8	Orbitalys	1.000	1.000	none
T9	NovaFab	1.000	1.000	none
	Semiconductors			
T9	Orbitalys	1.000	1.000	none
T10	AvioSys Intégration	0.900	0.884	none
T10	NovaFab	1.000	1.000	none
	Semiconductors			
T10	Orbitalys	1.000	1.000	none
T11	AvioSys Intégration	0.898	0.884	none
T11	Orbitalys	1.000	1.000	none
T11	TransGlobal Fret	0.854	0.838	none
T12	Orbitalys	1.000	1.000	none
T13	Électis EMS	1.000	1.000	none
T15	AvioSys Intégration	0.896	0.884	none
T17	AvioSys Intégration	0.894	0.884	none
T17	CompoDis Europe	0.998	0.998	none
T18	CompoDis Europe	1.000	1.000	none
T18	Meridian Semi	0.704	0.690	none

Set against the registry, ten of the eighteen campaign milestones completed, eight remained active, and none was abandoned. Exactly one was still open past a deadline that had fallen inside the campaign. Every other node for which a near-certain miss was announced delivered, though five of those deliveries arrived after the deadline originally committed to, which the registry does not record because a replanning overwrites the deadline it stores. Table 16 gives that reading milestone by milestone.

Arm B, Per-Node Influence Record

Table 15 records the first-exposure turn node by node, which is the observation behind the result of Section 4.9.

Table 15: Arm B at turn 4, node by node

Node	Recorded influence of the displayed forecast
AvioSys Intégration	Confirms existing assessment
CompoDis Europe	Revised downward
Électis EMS	Confirms existing assessment
Meridian Semi	Confirms existing assessment
NovaFab Semiconductors	Revised downward
Orbitalys	Revised downward
SilPure Materials	Revised downward
TransGlobal Fret	Revised downward
Total	5 downward, 3 confirmed, 0 upward, 8 of 8 responding

NovaFab is the node that carries the scenario's critical accident event two turns later, at turn 6.

Milestones Against Their Original Commitment

Table 16 is the counterpart of the registry counts above. It lists every milestone whose *original* deadline falls inside the campaign, together with the turn at which it was actually delivered in arm A and in the closed-loop arm of Section 4.10. Seven further milestones carry original deadlines at turns 20 to 22 and are therefore unobservable within a nineteen-turn campaign; they are excluded here rather than counted as met.

The two rows set in bold are the only ones on which the arms differ. The closed loop saved the Électis series B, which slipped in arm A, and replanned the CompoDis S1 cover, which arm A met on its original date. Every other difference between the arms is a replanning that re-dated a milestone arm A also delivered late: an announced slip rather than a discovered one, which is not nothing and is not a save.

Under this definition of truth, five of the twenty-one confident arm A forecasts of Table 14 are followed by a missed original commitment within four weeks, at T8 and T9 on NovaFab, T10 and T15 on AvioSys, and T12 on Orbitalys, against none under the registry's current-deadline reading.

Table 16: The eleven milestones observable within the campaign, judged against the deadline first committed to

Milestone	Original deadline	Arm A delivery	Closed-loop deadline	Closed-loop verdict
Orbitalys, HÉLIOS-1 delivery	T8	T14	T14	Missed
Orbitalys, HÉLIOS-2 delivery	T16	not delivered	T19	Missed
AvioSys, avionics batch 1	T6	T6	T6	Met
AvioSys, avionics batch 2	T11	T12	T12	Missed
AvioSys, avionics batch 3	T16	T18	T18	Missed
Électis, HÉLIOS series A	T6	T6	T6	Met
Électis, HÉLIOS series B	T13	T14	T13	Met
CompoDis, component cover S1	T9	T9	T11	Missed
TransGlobal, annual flow contract	T12	T12	T12	Met
NovaFab, HÉLIOS wafer allocation	T10	T11	T12	Missed
SilPure, annual wafer contract	T15	T14	T15	Met
Met on original commitment		5 of 11		5 of 11

Arm A Rescored After Repair

Table 17 gives the complete scoring behind Table 46. The three models score the same 120 forecasts against the original-commitment target, censored points excluded, so the observation counts are identical and only the model differs: the frozen campaign as run, the structural repair of Equation 29 alone, and that repair with the progress-uncertainty term added.

Table 17: Arm A rescored after repair, original-commitment target

Model	Quantity	h = 1	h = 2	h = 3	h = 4
Observations		112	104	96	90
Positive outcomes		7	14	20	26
Base rate		0.062	0.135	0.208	0.289
Before repair	Brier score	0.1335	0.1969	0.2728	0.3190
	Brier skill	−1.278	−0.690	−0.654	−0.553
	AUC	0.736	0.636	0.641	0.654
	PR-AUC	0.163	0.205	0.266	0.353
Structural repair	Brier score	0.0965	0.1633	0.2243	0.2613
	Brier skill	−0.646	−0.401	−0.360	−0.272
	AUC	0.793	0.671	0.657	0.721
	PR-AUC	0.355	0.305	0.355	0.484
With progress uncertainty	Brier score	0.0988	0.1593	0.2120	0.2499
	Brier skill	−0.687	−0.367	−0.286	−0.216
	AUC	0.729	0.634	0.602	0.584
	PR-AUC	0.195	0.228	0.299	0.385

The cluster-bootstrap 95% intervals on the AUC are $[0.49, 0.93]$, $[0.60, 0.95]$ and $[0.56, 0.95]$ at one week for the three models respectively, and $[0.42, 0.85]$, $[0.54, 0.92]$ and $[0.21, 0.87]$ at four weeks.

The confidence intervals are cluster bootstraps resampling whole nodes rather than individual node-weeks, because forecasts on one node are strongly correlated and an i.i.d. bootstrap would understate the width. They are wide enough at every horizon to contain the AUC of the other two models, which is why the AUC movements are reported as uninterpretable rather than as gains or losses, while the Brier skill movements, which do not depend on the size of the positive class in the same way, are reported as measured.

Distribution of the Milestone Probability

Table 18 counts how many of the 120 milestone-miss probabilities fall at the extremes rather than in a usable band, which is the measurement behind the progress-uncertainty term of Section 4.10.

Table 18: Degeneracy of the milestone-miss probability, 120 forecasts

Model	Exactly 0 or 1	In $(0.05, 0.95)$	Forecasts ≥ 0.5 at 4 weeks
Frozen campaign as run	102 (85.0%)	14	21
Structural repair alone	89 (74.2%)	10	10
Structural repair with progress uncertainty	86 (71.7%)	17	10

The structural repair alone reduces the count of extreme values but not the usable band: it removes eleven false certainties without producing gradations. The progress-uncertainty term is what widens the usable band, from 10 forecasts to 17. Neither is sufficient, and 71.7% of the probabilities remain degenerate.

E Satellite Capacity Estimation: Protocol and Data

This appendix records the operational detail behind the opening of Section I, so that its measurements can be audited or repeated.

Data Sources

All sources are public French open-data services, accessed over standard web map and feature services and cached on disk so that a re-run does not depend on service availability. Table 19 lists each one with the role it plays in the pipeline.

All geometry is handled in the national metric projection, in which the ground sample distance is exactly the bounding-box width in metres divided by the image width in pixels rather than an approximation.

Table 19: Data sources of the capacity pipeline

Source	Use
Orthophotography, 20 cm	Primary imagery for detection and segmentation
Orthophotography, 5 to 10 cm	Partial coverage, used only for the resolution control of Figure 29
LiDAR surface model	Building height where available, delivered as 32-bit float raster
National topographic database	Building footprints and, where populated, height
Classified installations register	Declared storage volume, used only to calibrate the fill factor

Calibrated Constants

Four constants turn geometry into a capacity figure. Each is calibrated rather than assumed, and Table 20 gives the sample behind each.

Table 20: Calibrated constants of the capacity model

Constant	Value	Basis
Fill factor	0.80	Median of declared volume over footprint $\times$ height across 122 regulated warehouses, IQR [0.52, 1.22]
Dock pitch	3.8 m	Median 3.83 m, coefficient of variation 9.6% over 19 independently measured facades
Default height	11.6 m	Median of the corpus, IQR 9.8 to 13.7 m, used only when the topographic database gives no height
Ground sample distance	0.25 m	Measured optimum for the pre-trained detector; at 0.50 m the model collapses

The coefficient of variation of 9.6% on the dock pitch is the result that makes transfer capacity generalisable: because the pitch is stable across sites, capacity depends only on how much facade is equipped, and no per-site measurement of the pitch is needed.

One constraint is enforced throughout and is worth stating because it is unusual. The regulatory database of classified installations is used only to calibrate the fill factor, never to infer a result for a specific site. Many warehouses fall below the classification thresholds and do not appear in it at all, so a system that inferred from it would silently fail on exactly the smaller sites a supply chain twin most needs to see.

Calibration of the Fill Factor

The fill factor relates declared storage volume to the geometric envelope. For each site in the regulatory register, the nearest building footprint is matched, the geometric volume is computed as footprint area times eaves height, and the ratio of declared to geometric volume is recorded. Ratios outside [0.05, 20] are discarded as either an aberrant declaration or a mismatched building. The production constant of 0.80 is attributed in the source to a median over 122 matched sites with an interquartile range of [0.52, 1.22].

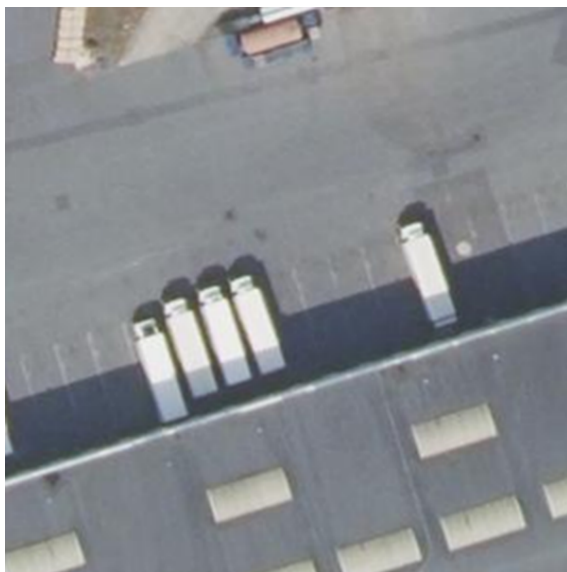
That run cannot be reproduced from the retained artefacts, and the discrepancy is recorded

here in full. The calibration script writes its detailed output to a file that is not present in the repository, so the only auditable sample is the 68-site corpus retained for other purposes. Recomputing the identical ratio over those 68 sites gives a median of 0.966, an interquartile range of $[0.80, 2.00]$, and 47% of sites inside the $[0.7, 1.4]$ band (Figure 32), against the 70% the calibration script itself prints as its acceptance target. The available evidence does not distinguish between a biased remnant sample and an understated production constant. Recollecting the register cleanly, with the departmental loop and pagination the script provides, and rederiving the factor, is the resolution.

One matching rule is worth recording because it changes the answer materially. Only the largest building within the search radius is matched, and neighbouring buildings are not summed: summing them moves the median ratio from 0.80 to 0.58, because a declared volume belongs to one warehouse while the footprints of a whole industrial estate do not.

The Resolution Control

Figure 29 shows the same equipped facade at 20 cm and at 8 cm. The trailers, the apron and the dark canopy band along the building edge are legible at 20 cm and sharper still at 8 cm, yet no individual door resolves inside the canopy band at either scale. This is the control that separates occlusion from insufficient resolution.



(a) 20 cm orthophotography



(b) 8 cm very-high-resolution imagery

Figure 29: The same equipped facade at 20 cm and at 8 cm

Approaches Tested and Rejected

Five approaches were tried before the reframing and abandoned, each for a measured reason, and Table 21 records them because the reasons constrain what a successful method can look like.

A related finding constrains the zero-shot approach that preceded the learned model. Parked trailers are not a proxy for docks: median trailer distance to the building reaches 126 m on large platforms, and at one site only 8 of 89 detected trailers lay within 26 m of the building, the

Table 21: Approaches tested and rejected

Approach	Measured outcome
Photometric thresholding of the apron	Recall of 5 of 6 on true aprons, but roads, an aeronautical tarmac and small-business forecourts all pass the same test
Vehicle density to reject car parks	Fails on a car park that was empty at the time of capture
Open-vocabulary detection from a text prompt	Confidence of 0.11 to 0.12, at the level of noise; the model is trained on ground-level imagery
Region growing from detected trailers	Returns 222 docks where 57 exist: the paved surface of an industrial estate is a single connected component, and the region spills onto neighbouring roofs of similar tone
Frequency analysis to count bays	Three real rhythms coexist on one facade, at 3.3 to 4.6 m for bays, 5.5 m for canopy posts and 8.9 to 10.7 m for portal frames. A frequency analysis returns a frequency, never its meaning

rest being in holding yards. With a proximity and perpendicularity filter applied, the zero-shot pipeline produced a capacity figure for only 36% of sites, 25 of 70 measured. Elsewhere it returned nothing, which is a gap in coverage rather than a wrong answer.

Annotation Protocol

The dataset comprises 442 georeferenced image chips, each retained at a single location, a single resolution and a single vintage, with a sidecar file recording its geographic transform. Of these, 351 form the training split and 91 the validation split.

Annotators mark one class, the continuous extent of facade served by a dock apron, and the following rules apply.

- Mark an equipped facade even where no vehicle is parked. The apron and canopy remain visible, and a facade annotated only when a vehicle is present teaches the model to detect vehicles.
- Do not annotate a trailer holding yard that is not adjacent to a wall.
- Do not annotate an aeronautical tarmac or a small-business forecourt. Both are paved and deep but serve no docks.
- A facade without docks is annotated empty. That is a useful negative, not an omission, and roughly 40% of the target sample is negative by design.
- Ignore everything above the facade line, which is roof. Rooflights and sawtooth roofs are perfectly periodic and are the principal source of false positives.
- Annotate every visible dock zone in a chip regardless of which building owns it. Ownership is resolved downstream by geometric clipping, so the model is never asked to infer it from pixels, and annotating a zone in one chip but not in the overlapping neighbour would give the model two contradictory labels for the same ground.

Sampling is stratified by footprint area across three bands, roughly 1 500 to 5 000 m², 5 000 to 20 000 m² and above 20 000 m², and across five site categories covering small enterprises, aerospace, refrigerated, very large and retail distribution. An unstratified random draw

over-represents large warehouses, and the model inherits that bias.

Annotation Integrity

One episode bears directly on research method. An automated third-party annotation process produced 442 annotations and reported its own output as verified and complete. The annotations were unusable: the process had placed a dock zone along every edge of every building without examining the imagery at all, then subdivided each zone at the nominal pitch, producing an implausible 88% positive rate.

All of it was discarded and the dataset re-annotated by hand. An audit of the manual work found 11 genuine problems across 687 annotated zones, a defect rate of 1.6%. A verification tool was then written and placed ahead of training, checking format, geometry and consistency against the project's own measured thresholds, so that no external annotation enters the pipeline unaudited. That tool caught a defect in its own first implementation, where distance was measured to the largest building in a scene rather than to all buildings, generating 26 false alarms on correct annotations of neighbouring buildings.

The transferable point is that a confident self-assessment from an automated process is not evidence, and that the cost of the verification step was small relative to the cost of training on corrupted labels.

Training Configuration

The model is a segmentation network trained from pre-trained weights at a 640 pixel input size, with rotation augmentation over the full circle, horizontal and vertical flips, and scale jitter. Rotation invariance is obtained by augmentation rather than by geometric rectification of the facade, which is what allows the model to accept any building shape and orientation without requiring a footprint as input; Figure 31 shows predictions on held-out imagery. Training ran on CPU throughout, with no GPU available, which is the practical reason the input size was not raised further and the context for the negative resolution result of Table 30.

A Systems Error Found in Deployment

Figure 30 shows the end-to-end output on one deployment site. The national topographic database fragments a single large warehouse into several polygons, so a 434 m dock apron was initially attributed in full to a 5 149 m² fragment, yielding an implausible 221 docks per hectare against a corpus median of 7.9. The fix was geometric rather than statistical: each predicted zone is now clipped so that a building receives only the portion facing its own facade, and the quality control was changed from a density threshold, which must be calibrated per site type, to the ratio of dock length to building perimeter, which is dimensionless and cannot legitimately exceed one.

Measured Facts Recorded for Reuse

- Empty bay contrast at verified locations: 0.18σ , 0.22σ and 0.01σ , against 5.3σ to 6.4σ at occupied bays on the same facades, over a full sweep of assumed apron depths from 0.3 m to 20 m.



Figure 30: End-to-end output on one deployment site. The tool's own banner reads 12 426 m² of footprint, three detected dock zones, 326 m of equipped facade and about 86 docks; the orange outline is the topographic footprint and the green fill the predicted zones.



Figure 31: Predicted dock zones on held-out imagery

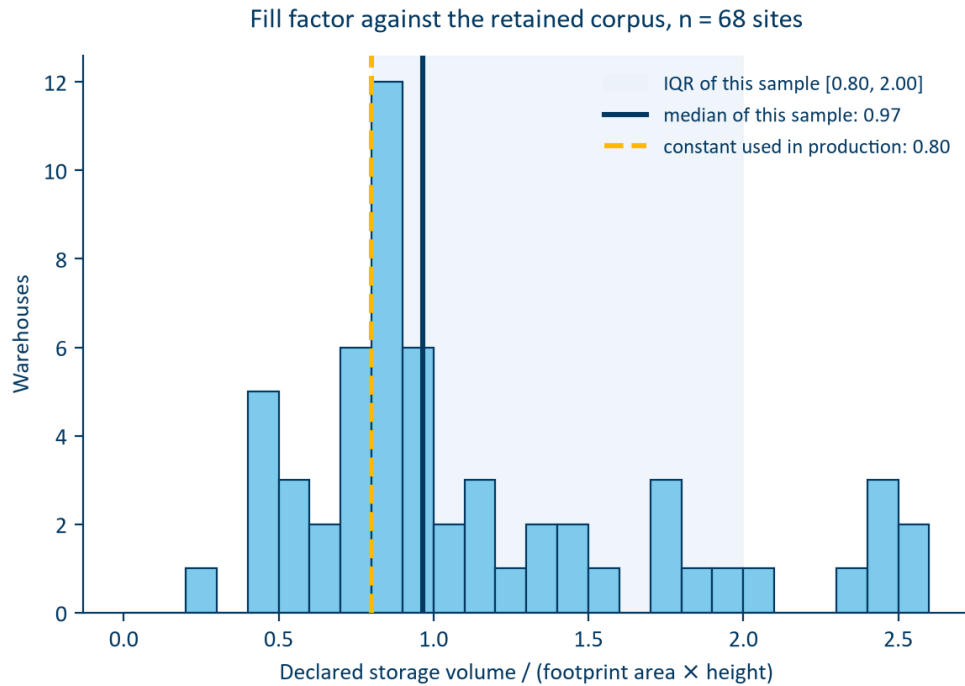


Figure 32: The fill factor against the auditable corpus

- Dock pitch coherence between facades of the same building: 0.03 to 0.05 m, which provides a free quality control.
- Roof overhang beyond the footprint edge: 0 to 14.4 m. This is why the footprint edge is not the start of the apron, and it was the cause of an earlier defect in which rooflights were counted as bays.
- Prompt sensitivity of the promptable segmentation model used for site sampling: a prompt naming an industrial warehouse returns 88 752 m², some 45% of the view, where a prompt naming a warehouse roof returns the 13 478 m² of the roof itself. The first is useful for locating a site, the second for measuring it, and they are not interchangeable.

Licensing Constraint

The detection and segmentation library used is distributed under a strong copyleft licence. This has no consequence for internal research use, but it becomes determinative the moment the capability is distributed or exposed over a network. The options are a commercial licence from the vendor or replacement of the backbone with a permissively licensed equivalent. This is recorded as a governance decision to be taken deliberately rather than discovered at deployment.

F A Worked Example, End to End

Section 3.2 gives the model as equations and Section 4 gives it as scores on a campaign. Neither shows a single number travelling through the whole chain, which is what a reader needs to check that the equations mean what the prose says they mean. This appendix

carries one node through every stage: the six blocks into a local urgency, that urgency along a three-node chain, the chain into a criticality ranking, and the result against a declaration.

Every figure below is produced by the production code rather than computed by hand, using the same functions the campaign ran, so the example cannot drift away from the system it illustrates. The scenario is illustrative and is not drawn from either campaign.

F.1 Stage 1: six blocks into a local urgency

The node is a deep-tier supplier whose next milestone falls in nine days against a lead time distributed $\mathcal{N}(11, 3)$, whose buffers are a third consumed, whose OEE stands at 0.80, and for which no node-level emissions data exists. Table 22 gives the resulting block values and the survival factor each contributes to Equation 4.

Table 22: The six blocks and their survival factors, all weights at $\omega_m = 1$

Block	u_m	$(1 - u_m)^{\omega_m}$	Reading
u_{time}	0.62	0.3800	Probability of missing the nine-day deadline
u_{cap}	0.35	0.6500	Buffers a third consumed
u_{perf}	0.20	0.8000	1 – OEE, with OEE at 0.80
u_{risk}	0.15	0.8500	Expected unavailability over the horizon
u_{cost}	0.28	0.7200	Operating cost drifting above nominal
u_{co2}	absent	1.0000	$\omega_m = 0$: the factor is neutral, not healthy
Product of survivals		0.120931	
$U_{r,\text{local}} = 1 - \text{product}$		0.8791	

The comparison that matters is with the aggregator the model rejects. A weighted average of the same five available blocks gives 0.3200. The two answers differ by more than half the scale on identical inputs, and they differ because they answer different questions: the average asks how stressed this node is on average, the noisy-OR asks how likely it is that at least one mechanism ruptures. Only the second is the quantity a coordinator needs.

The gap widens at the boundary. Let the deadline pass, so $u_{\text{time}} = 1$ and the node cannot become any later. The noisy-OR returns exactly 1.0000, because the factor $(1 - 1) = 0$ collapses the product whatever the other blocks read. The weighted average returns 0.3960, and would rank this node below a node that is mildly stressed on all six. That is the compensatory failure of Section 3.2.1 in a single pair of numbers.

F.2 Stage 2: propagation along a three-node chain

Place the node as the deepest tier of a chain $s_2 \rightarrow s_1 \rightarrow \text{customer}$, with upstream amplification $\beta_{s_2 s_1} = 0.70$ and $\beta_{s_1 c} = 0.50$. The two downstream nodes are locally calm, at $U_{r,\text{local}} = 0.30$ and 0.10. Equation 8 applied in topological order gives

$$\begin{aligned}
 U_{r,s_2} &= 0.8791, \\
 U_{r,s_1} &= 1 - (1 - 0.30)(1 - 0.70 \times 0.8791) = 0.7307, \\
 U_{r,c} &= 1 - (1 - 0.10)(1 - 0.50 \times 0.7307) = 0.4288.
 \end{aligned}$$

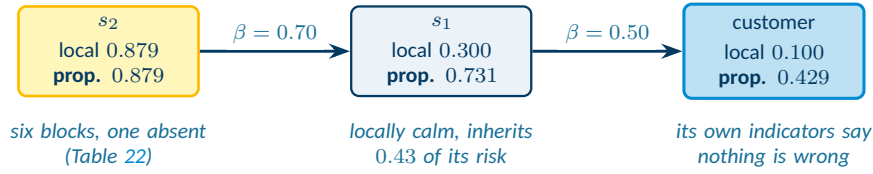


Figure 33: The three-node chain of Stage 2. Each box carries the node's own local urgency above and its propagated value below. The customer's local reading of 0.10 becomes 0.43 once two tiers of upstream risk arrive, which is the whole purpose of the propagation.

A node that is locally calm at 0.10 therefore carries a propagated urgency of 0.43, entirely inherited from two tiers upstream. This is the whole purpose of the propagation: the customer's own indicators say nothing is wrong, and something is.

The log-survival twin of Proposition 3.2.5 can be checked on the same figures. Equation 10 gives $\ell_c = 0.5601$ at the customer, and $1 - e^{-0.5601} = 0.4288$, which is $U_{r,c}$ to every digit shown. The twin is the same recurrence in another coordinate, exactly as the proposition states.

F.3 Stage 3: which node is the most critical

The criticality index of Section 3.2.5 shocks each active node to $U_{r,\text{local}} = 1$ in turn and measures the rise at the customer. On this chain:

Table 23: Effect of the worst local shock, measured at the customer

Shocked node	$U_{r,\text{local}}$	$U_{r,c}$ after	$\Delta U_{r,c}$	Rank
s_1	0.30 $\rightarrow$ 1.00	0.5500	+0.1212	1st
s_2	0.88 $\rightarrow$ 1.00	0.4555	+0.0267	2nd

The result is worth pausing on, because it is counter-intuitive and it is correct. The most urgent node, s_2 at 0.88, is the *least* critical of the two. Two effects combine. Its risk is already almost entirely realised in the baseline, so forcing it to 1 adds little that the customer was not already exposed to. And what it adds is damped twice, once by $\beta_{s_2 s_1} = 0.70$ and again by $\beta_{s_1 c} = 0.50$, while s_1 sits behind a single arc. Urgency and criticality are therefore different questions, and a coordinator who sorts the network by urgency alone will spend attention on the node that has least left to lose.

Proposition 3.2.5 can be seen on the same chain. Set every local urgency to 1.0, the state the campaign reaches on nine turns of nineteen. The baseline customer urgency is then 1, and shocking either node gives $\Delta U_{r,c} = +0.0000$ exactly. Both nodes tie at zero, the ranking carries no information, and the log-survival key of Proposition 3.2.5 is what separates them again.

F.4 Stage 4: the declaration, and the adequacy it produces

Finally, the deep supplier's own coordinator fills the weekly questionnaire. Table 24 scores two declarations against the computed $U_r = 0.8791$, using the default $\lambda_{\text{under}} = 2.25$, $\lambda_{\text{over}} = 1.0$ and $\alpha = 0.88$.

Table 24: Two declarations against the same computed urgency

U_d	F	H	penalty	A	Reading
0.83	0.0000	0.0491	0.1585	83.6	A lucid coordinator, slightly optimistic
0.40	0.0000	0.4791	1.1774	22.7	Hidden risk of nearly half the scale

The second row is the situation the whole instrument exists to surface. The node is at 0.88 and is reported at 0.40; nothing in the declaration is a lie, and the coordinator may sincerely believe it, which is Assumption 3.2. The adequacy score falls to 22.7 and the hidden risk $H = 0.479$ is what a supervising view would rank on. Note also what the first row shows: at $U_d = 0.83$ against $U_r = 0.88$ the score is 83.6 and not 95, because Proposition 3.2.6(iv) makes the curve steepest near zero. Small mis-declarations are penalised hard on purpose, since they are the ones still cheap to correct.

G Register of Modelling Assumptions

A model is the set of commitments it makes, and those commitments are scattered through Section 3 at the point where each first becomes necessary. This appendix collects them in one place so that a reader can weigh the whole set at once rather than reconstruct it, and so that the ones this work could not test are as visible as the ones it could.

Three statuses are used, and the distinction between the last two is the one that matters. *Tested* means the campaign produced evidence bearing on the assumption and Section 4 reports it. *Testable*, *untested* means an experiment within reach of this design would bear on it and was not run, usually for want of events or time; these are recoverable and Section 5.5 carries them. *Outside this design* means no experiment this instrument can run would settle the question, so the assumption is load-bearing and permanent until the design changes; these are the ones that bound what the thesis may claim.

G.1 The register

Read down the status column first. It is the shorter question, and it decides how much weight the rest of the row can carry.

Table 25: Every modelling assumption, where it is stated, and what is known about it

	Assumption	Status	If it fails
A1	The six blocks are independent competing causes of rupture (Assumption 3.1, Section 3.2.1)	Tested	Correlated blocks double-count a shared cause and the aggregate overstates urgency. The direction is conservative: it errs towards alarm, never towards complacency. Section 4.4 reports the observed inter-block correlations.

Table 25, continued

	Assumption	Status	If it fails
A2	The blocks are exchangeable up to their weights ω_m (Section 3.2.1)	Tested, and false in detail	u_{time} is a frequentist probability and u_{cost} an ordinal statement in cardinal dress; no transfer function makes them equally informative. The aggregate is therefore usable for ranking and for comparison against U_d , and is not a calibrated probability. This is the reason no Brier score is reported on U_r itself.
A3	The declaration is sincere (Assumption 3.2, Section 3.2.6)	Outside this design	Hidden risk stops measuring misperception and starts measuring what a declarant is willing to write down. The two demand opposite interventions and are indistinguishable in this data. Every behavioural reading in Section 4 rests on it.
A4	Language-model agents stand in for human declarants (Section 3.4.4)	Outside this design	Every statement about declaration behaviour applies to that population and not to consultants. The mechanical findings are unaffected, since they concern code paths rather than declarants.
A5	Lead times are lognormal by default, parameterised by moments (Section 3.2.1)	Testable, untested	The tail of the completion-time distribution is misstated and u_{time} with it. The law is configurable per node, and normal-clipped and triangular alternatives are implemented, so the sensitivity is measurable on frozen scenarios at low cost.
A6	Lead times are drawn independently across nodes (Section 3.2.1)	Outside this design	A common shock, an energy crisis or a regional closure, correlates delays that the simulation treats as independent, and the simulated completion distribution is too narrow. Nothing in the campaign scenario could separate this from ordinary variance.
A7	The arc coefficients β_{ji} and γ_{ik} are known constants (Section 3.2.4)	Testable, untested	Propagated urgency is misscaled. The coefficients are set by the scenario rather than estimated from data, and estimating them from observed co-movement of node states is the natural next study.
A8	Backup arcs carry neither propagation (Section 3.2.4)	Modelling choice, exercised	A network with a real alternative supplier reads as more exposed than it is. The choice is deliberate, since an unqualified backup is documentation rather than capacity, and the campaign scenario exploits it on purpose.

Table 25, continued

	Assumption	Status	If it fails
A9	The Prospect Theory parameters transfer to supply chain coordinators, $\lambda_{\text{under}} = 2.25$ and $\alpha = 0.88$ (Section 3.2.6)	Testable, untested	The adequacy score is miscalibrated as a governance instrument, though its ordering is unaffected, since A is monotone in the gap whatever the parameters (Proposition 3.2.6(ii)). Both are configurable per project and neither is hard-coded.
A10	Local urgency follows a stationary AR(1) (Section H.3)	Testable, untested	The projected urgency path is wrong in shape, and with it every p_h . Nineteen turns is short for a specification test, which is why the coefficient is clipped to $[0, 0.98]$ rather than trusted.
A11	Weeks are independent Bernoulli trials for the adverse-event rate (Section H.3)	Testable, untested	Clustered events, one incident producing three weeks of consequences, break independence and the Beta posterior understates uncertainty. The prior strength of $N_0 = 26$ dominates nineteen turns of data, which masks the effect here and would not on a longer campaign.
A12	The calibration service's outcome definition is the right target (Section 3.4.5)	Tested	Every score in this thesis moves. This is the one assumption whose failure was measured rather than argued: Section 4.7 scores identical forecasts against three definitions and finds a spread wider than any modelling decision taken in this work.

G.2 What the register says when read as a whole

Twelve assumptions, of which three are tested, five are testable and untested, one is a declared modelling choice, and three sit outside what this design can reach. That last group is the honest statement of the thesis's boundary, and it has a shape: A3 and A4 both concern the declarant, and A6 concerns correlation between nodes. What the instrument cannot see is people and common causes, which is to say the two things a supply chain is made of besides its arithmetic.

That is not a reason to distrust the measurements reported here. It is the reason they are reported as measurements of a mechanism rather than as estimates of a rate, and the reason Section 5.5 asks first for a human campaign rather than for a larger simulated one. A larger simulated campaign would tighten every interval in Section 4.8 and would move none of the three above.

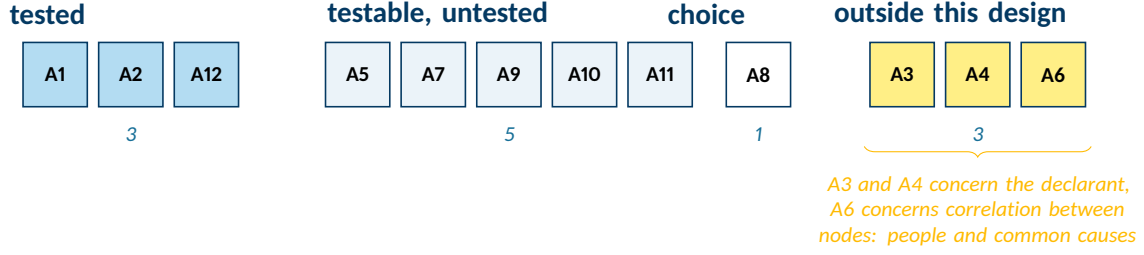


Figure 34: Twelve assumptions sorted by status. The three on the right are what bounds the thesis, and they have a shape: the instrument cannot see people, and it cannot see a cause acting on several nodes at once.

H Technical Supplement

The body states the model, the experiment and what was measured. This appendix holds the apparatus behind them: the symbol table, the elicitation and outranking methods in full, the specification of the forecast layer, and the cost of every operation. It is written to be consulted rather than read through.

H.1 Notation

Table 26 fixes every symbol used in this section and the next. Two conventions are worth stating before the table rather than after it. Subscript i always indexes a node and subscript m always indexes one of the six KPI blocks, so u_m and $U_{r,i}$ are quantities of different kinds despite the shared letter u . And $[x]_+ = \max(x, 0)$ denotes the positive part throughout.

The within-block term weights are written a_k for capacity and c_k for cost. An earlier draft wrote both as α_k and β_k , which collided with the Prospect Theory curvature α of Section 3.2.6 and with the arc amplification coefficient β_{ji} of Section 3.2.4. The letters differ here because the quantities are unrelated.

H.2 Elicitation and outranking, in full

The module `core/ahp.py` implements the Saaty eigenvector method for a four-criterion questionnaire administered through the weekly review page, and is the tool that produces $U_{d,\text{local}}$. The four criteria are operational impact, meaning the downstream consequence of failure; time window, meaning how long before the situation becomes irredeemable; downstream dependencies, meaning how many tasks are blocked waiting; and recoverability, meaning whether lateness can be caught up, inverted so that lower is worse.

The criteria set follows from two constraints. Cognitively, Saaty’s method degrades as the number of pairwise-compared criteria grows, with consistency hard to maintain beyond roughly seven items and each additional criterion adding $n - 1$ comparisons to the weekly workload (Saaty, 1990; Triantaphyllou; Yanase, 2024). Keeping $n = 4$ limits the questionnaire to six comparisons. Structurally, the four questions mirror how declared urgency is consumed downstream: impact and dependencies capture the severity dimension that propagation amplifies, the time window captures the slack dimension confronted with u_{time} , and recoverability captures the resilience dimension confronted with the recovery term of u_{risk} . Each declared criterion therefore has an operational counterpart in the computed signal, which is

Table 26: Notation used throughout the model and its analysis

Symbol	Domain	Meaning
Network structure		
$G = (V, E)$	DAG	Supply network, nominal arcs only; backup arcs are inert
n, e	$\mathbb{N}$	Node count $ V $ and arc count $ E $
$\text{Pred}(i), \text{Succ}(i)$	$\subseteq V$	Nominal predecessors and successors of node i
f	$\in V$	Final-customer node, at rank 0
n_{act}	$\mathbb{N}$	Number of active nodes, those neither DONE nor ABANDONED
Local urgency		
u_m	$[0, 1]$	Partial urgency of KPI block $m \in \{\text{time, cap, perf, risk, cost, co2}\}$
ω_m	$\mathbb{R}_{\geq 0}$	Aggregation weight of block m ; $\omega_m = 0$ encodes missing data
a_k, c_k	$[0, 1]$	Term weights inside the capacity and cost blocks, renormalised over available terms
$\kappa_{\text{late}}, \kappa_{\text{early}}$	$\mathbb{R}_{\geq 0}$	Planning-drift coefficients of the time block
$U_{r,\text{local},i}$	$[0, 1]$	Local real urgency at node i , before propagation
$U_{d,\text{local},i}$	$[0, 1]$	Local declared urgency at node i , from the questionnaire
Propagation		
$U_{r,i}, U_{d,i}$	$[0, 1]$	Propagated real and declared urgency at node i
β_{ji}	$[0, 1]$	Upstream risk amplification on arc $j \rightarrow i$
γ_{ik}	$[0, 1]$	Downstream demand attenuation on arc $i \rightarrow k$
s_i	$(0, 1]$	Survival at node i , $s_i = 1 - U_{r,i}$
ℓ_i	$\mathbb{R}_{\geq 0}$	Log-survival at node i , $\ell_i = -\ln s_i$
ε	10^{-9}	Regularisation floor: local urgency is clipped to $1 - \varepsilon$ in the twin
Criticality		
$\Delta U_{r,f}(i)$	$[0, 1]$	Rise of final-customer urgency when node i is shocked to $U_{r,\text{local}} = 1$
$\Delta \ell_f(i)$	$\mathbb{R}_{\geq 0}$	Same shock measured in log-survival; the tie-break under saturation
σ	$[0, 1]$	Shock severity drawn from the calibrated event distribution
Cognitive adequacy		
F, H	$[0, 1]$	False urgency $[U_d - U_r]_+$ and hidden risk $[U_r - U_d]_+$
$\lambda_{\text{under}}, \lambda_{\text{over}}$	$\mathbb{R}_{>0}$	Penalty weights on hidden risk and false urgency; 2.25 and 1.0
α	$(0, 1]$	Psychophysical curvature of the value function; 0.88
A	$[0, 100]$	Adequacy score, 100 meaning declared and computed urgency agree
Stochastic lead time		
L_i	$\mathbb{R}_{\geq 0}$	Lead time of node i , lognormal by default
S_i, C_i	$\mathbb{R}_{\geq 0}$	Start and completion time of node i , in project hours
d_i	$\mathbb{R}$	Deadline of the next active milestone of node i , in project hours
p_i	$[0, 1]$	Reported progress of that milestone; only $1 - p_i$ of L_i remains
R_i	$\mathbb{R}_{\geq 0}$	Production time already lost at node i , added without rescaling
t	$\mathbb{R}_{\geq 0}$	Current project time, the instant the simulation restarts from
N	$\mathbb{N}$	Monte Carlo draws; 10 000 by default, guarded by $Nn \leq 5 \times 10^7$

what makes the later comparison meaningful rather than a comparison of unrelated quantities. Larger criteria sets are deliberately routed to the Fuzzy BWM weighting of the KPI blocks instead.

The method runs in four steps, and each is given here because the consistency gate that Section 4.2 reports on is a property of the third.

The operator fills a reciprocal comparison matrix $A = (a_{ij})$ of order $n = 4$, where a_{ij} is the Saaty judgement of criterion i against criterion j on the scale $\{1/9, \dots, 1/2, 1, 2, \dots, 9\}$, with $a_{ii} = 1$ and $a_{ji} = 1/a_{ij}$ enforced at entry, so six judgements determine the sixteen entries.

The priority vector is then obtained by normalising each column and averaging each row,

$$\tilde{w}_i = \frac{1}{n} \sum_{j=1}^n \frac{a_{ij}}{\sum_{k=1}^n a_{kj}}, \quad w = \tilde{w} / \sum_i \tilde{w}_i. \quad (15)$$

This is Saaty's classical approximation of the principal eigenvector rather than the eigenvector itself, and the distinction is worth making rather than glossing: the two coincide exactly when A is perfectly consistent, and diverge as inconsistency grows, which is precisely the regime the gate below is meant to catch. The approximation is used because it is closed-form, needs no iteration and cannot fail to converge on an operator's half-filled matrix; the cost is that the reported $\lambda_{\max}$ is itself approximate.

Consistency is measured on that vector. The principal eigenvalue is estimated as the mean componentwise ratio $\lambda_{\max} = \frac{1}{n} \sum_i (Aw)_i / w_i$, and

$$\text{CI} = \frac{\lambda_{\max} - n}{n - 1}, \quad \text{CR} = \frac{\text{CI}}{\text{RI}(n)}, \quad \text{RI}(4) = 0.90. \quad (16)$$

The index has a direct reading: a perfectly consistent reciprocal matrix has $\lambda_{\max} = n$ exactly and $\text{CI} = 0$, and $\text{RI}(n)$ normalises against the consistency a matrix filled at random would show, so CR is the share of the inconsistency of pure noise that this operator has produced. For $n \leq 2$ any reciprocal matrix is consistent and CR is set to zero.

Declared urgency is finally the weighted average of the four criterion scores $s_j \in \{1, \dots, 9\}$ the operator gives alongside the comparisons, rescaled onto the unit interval by Equation 6 so that it is commensurable with the computed signal. The mapping matters for the comparison this thesis is built on. A declaration of $U_d = 0$ means every criterion scored the minimum of the Saaty scale, not that the operator declined to answer, and the two are distinguished in the data model exactly as a missing KPI block is distinguished from a healthy one.

The consistency ratio (Consistency Ratio (CR)) is the consistency index (Consistency Index (CI)) over Saaty's random index (Random Index (Saaty) (RI)) for a matrix of order n , $\text{CR} = \text{CI}/\text{RI}(n)$. If $\text{CR} > 0.10$ the interface flags the questionnaire as inconsistent and invites the operator to revise the most contradictory comparison before the assessment is accepted; Figure 35 shows that gate in the interface. Declared urgency is further smoothed with an exponential moving average, $\rho = 0.3$, across consecutive weekly assessments, absorbing transient stress spikes without erasing a sustained change in perception.

Fuzzy BWM and PROMETHEE II Network Outranking

The modules `mcda/fbwm.py` and `mcda/promethee.py` implement the two complementary methods of Figure 36: Fuzzy BWM offers a lower-effort elicitation route than crisp AHP for

Comparaisons par paires (AHP)
Curseur à gauche : le second critère domine ; à droite : le premier domine ; au centre : importance égale.

« Impact opérationnel » vs « Fenêtre temporelle »

-8
Importance égale

-8

+8

0

+8

0

« Impact opérationnel » vs « Dépendances aval »

-8
Importance égale

-8

+8

0

+8

0

« Impact opérationnel » vs « Récupérabilité »

-8
Importance égale

-8

+8

0

+8

0

« Fenêtre temporelle » vs « Dépendances aval »

-8
Importance égale

-8

+8

0

+8

0

« Fenêtre temporelle » vs « Récupérabilité »

-8
Importance égale

-8

+8

0

+8

0

« Dépendances aval » vs « Récupérabilité »

-8
Importance égale

-8

+8

0

+8

0

Interface terms

Comparaisons par paires (AHP)
Pairwise comparisons (AHP)

Impact opérationnel
Operational impact

Fenêtre temporelle
Time window

Dépendances aval
Downstream dependencies

Récupérabilité
Recoverability

Importance égale
Equal importance

Curseur à gauche / à droite
Slider left / right: which side dominates

Figure 35: The AHP questionnaire page, with its interface terms keyed into English. Six comparisons cover the four criteria.

weighting the six KPI blocks, and PROMETHEE II turns those weights into a network-wide, non-compensatory ranking of nodes.

Fuzzy BWM. The elicitation cost is the reason the method is there. AHP needs $n(n - 1)/2$ comparisons, which is fifteen for the six blocks; the Best-Worst Method needs $2n - 3$, which is nine, and it obtains them by asking the operator to name only the most important block B and the least important block W and then to judge B against every other block and every other block against W (Rezaei, 2015). The fuzzy variant replaces each judgement by a triangular fuzzy number (l, m, u) drawn from a five-point linguistic scale (S. Guo et al., 2017), so that the operator’s hesitation is carried through the arithmetic instead of being rounded away at the door. Table 27 gives the scale as implemented.

Table 27: The linguistic scale of the fuzzy Best-Worst Method, as implemented in `mcdaf/bwm.py`

Judgement	Triangular fuzzy number (l, m, u)
Equally important	$(1, 1, 1)$
Weakly more important	$(2/3, 1, 3/2)$
Moderately more important	$(3/2, 2, 5/2)$
Very much more important	$(5/2, 3, 7/2)$
Absolutely more important	$(7/2, 4, 9/2)$

Writing $\tilde{a}_{Bj}$ for the fuzzy judgement of the best block against block j and $\tilde{a}_{jW}$ for that of block j against the worst, the weights $\tilde{w}_j = (l_j, m_j, u_j)$ solve the minimax programme

$$\begin{aligned}
& \min_{\tilde{w}, \xi} \quad \xi \\
& \text{s.t.} \quad \begin{cases} |\tilde{w}_B \oslash \tilde{w}_j - \tilde{a}_{Bj}| \leq (\xi, \xi, \xi), & j \neq B, \\ |\tilde{w}_j \oslash \tilde{w}_W - \tilde{a}_{jW}| \leq (\xi, \xi, \xi), & j \neq W, \\ \sum_j R(\tilde{w}_j) = 1, \quad 0 < \epsilon \leq l_j \leq m_j \leq u_j, \end{cases}
\end{aligned} \tag{17}$$

where the fuzzy quotient uses the standard interval approximation $(l_1, m_1, u_1) \oslash (l_2, m_2, u_2) = (l_1/u_2, m_1/m_2, u_1/l_2)$ and R is the graded mean integration representation that defuzzifies a triangular number,

$$R(l, m, u) = \frac{l + 4m + u}{6}. \tag{18}$$

The objective ξ is the largest deviation between any elicited judgement and the ratio the solved weights actually imply, so it is a consistency measure of the same nature as the CR above: a small ξ says the operator's two comparison vectors are close to being realisable by a single weight vector. The programme is non-linear and is solved by sequential least squares from five seeded starting points, so a given questionnaire yields the same weights on every run; if no start returns a feasible solution the module falls back to uniform weights rather than raising, and the fallback is recorded. The output feeds ω_m in Equation 4 and the criterion weights of PROMETHEE II below, so one elicitation serves both the per-node score and the network ranking.

PROMETHEE II. Outranking answers a different question from scoring: not how urgent is this node, but which node should be looked at before which other. For each ordered pair of nodes (a, b) and each criterion j , the oriented deviation $d_j(a, b) = g_j(a) - g_j(b)$ passes through a preference function $P_j : \mathbb{R} \rightarrow [0, 1]$, non-decreasing and null on $d \leq 0$. The six Brans-Vincke forms are implemented, usual, U-shape, V-shape, level, linear with indifference, and Gaussian (Brans; Vincke, et al., 1986). The aggregated preference index and the three flows are then

$$\pi(a, b) = \sum_j w_j P_j(d_j(a, b)), \quad (19)$$

$$\phi^+(a) = \frac{1}{q-1} \sum_{b \neq a} \pi(a, b), \quad \phi^-(a) = \frac{1}{q-1} \sum_{b \neq a} \pi(b, a), \quad \phi(a) = \phi^+(a) - \phi^-(a), \quad (20)$$

over q alternatives. The outgoing flow ϕ^+ measures how far a dominates the rest, the incoming flow ϕ^- how far the rest dominates a , and the net flow ϕ gives a complete ranking, which is what distinguishes PROMETHEE II from the partial ranking of PROMETHEE I. The pairwise structure is where the $\Theta(kq^2)$ term of Table 28 comes from.

Two properties of the implementation belong in the record. Missing data is handled per pair rather than per node: for a pair and a criterion where either value is absent, P_j is set to zero in both directions and the weights are renormalised over the criteria present for that pair alone, so a node with an incomplete KPI set is still ranked rather than dropped, and if no criterion is present for a pair then $\pi(a, b) = \pi(b, a) = 0$. And PROMETHEE II is not independent of irrelevant alternatives: adding a node can invert the relative rank of two existing nodes. That is a known and documented property of net-flow methods (Triantaphyllou, 2000), it is registered as a descriptive property-based test rather than as a bug, and it is the reason the criticality index of Section 3.2.5 rather than the outranking is what hypothesis H3 is scored on.

The linguistic scale maps five qualitative judgements to triangular fuzzy numbers on a 1 to 4.5 scale, each carrying its own maximum consistency index. The consistency ratio is $CR_{BWM} = \xi^*/CI$, where ξ^* is the optimal consistency indicator returned by a constrained non-linear optimisation fitting the fuzzy weights to the elicited comparisons. If the optimisation fails to converge, the implementation falls back to uniform weights rather than raising an error, so a single malformed elicitation cannot break the weekly review. Fuzzy weights are defuzzified with the Graded Mean Integration Representation,

$$w_j = \frac{a_j + 4b_j + c_j}{6}, \quad (21)$$

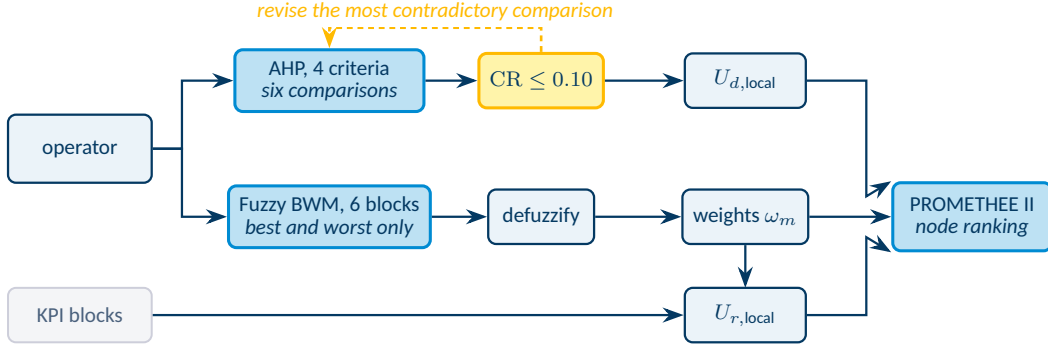


Figure 36: The two elicitation routes and what they feed. AHP weights four declared criteria behind a consistency gate that returns an incoherent questionnaire to the operator; Fuzzy BWM weights the six KPI blocks, and those same weights ω_m serve both the noisy-OR aggregation and the PROMETHEE II criteria, so the network ranking and the per-node score stay mutually consistent rather than resting on a separately calibrated weight set.

then renormalised to sum to one. The triangular bounds are written (a_j, b_j, c_j) rather than the more common (l_j, m_j, u_j) to avoid colliding with the risk blocks u_m .

PROMETHEE II ranks the active nodes of a project on the same six weighted urgency blocks that feed local real urgency. For each pair of active nodes and each block, the preference intensity is computed with a Type V linear preference function with an indifference threshold, and the aggregated preference index and net outranking flow follow the standard formulation:

$$\pi(a, b) = \sum_m \omega_m P_m(g_m(a) - g_m(b)), \quad \phi(a) = \phi^+(a) - \phi^-(a). \quad (22)$$

H.3 The forecast layer, specified

The results of this thesis turn on one component, so it is specified here rather than left implicit behind the phrase used in the scoring protocol below. The forecast service projects every active node over one to four weeks from that node's own persisted weekly history, and emits the probability p_h that an adverse outcome occurs within the horizon.

Three quantities are estimated from the node's own weekly history and then simulated forward, and p_h is the fraction of trajectories in which the outcome occurs. Each is given here in full, because the results of Section 4 are results about this estimator and about nothing else.

The urgency path. Local real urgency is modelled as a mean-reverting first-order autoregression on the weekly series $x_1, \dots, x_T$ of that node,

$$x_k = \mu + \varphi(x_{k-1} - \mu) + \sigma \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, 1), \quad (23)$$

with μ the sample mean of the series. The coefficient φ is the ordinary least-squares slope of x_{t+1} on x_t and σ the residual standard deviation,

$$\varphi = \frac{\sum_t (x_t - \bar{x})(x_{t+1} - \bar{z})}{\sum_t (x_t - \bar{x})^2}, \quad \sigma^2 = \frac{1}{T-2} \sum_t \hat{\varepsilon}_t^2, \quad (24)$$

with the standard errors that the outer draw below consumes, $\text{se}(\varphi) = \sigma / \sqrt{S_{xx}}$ where $S_{xx} = \sum_t (x_t - \bar{x})^2$, and $\text{se}(\sigma) = \sigma / \sqrt{2(T-2)}$.

Three guards apply, and each is a modelling decision rather than a numerical convenience. The slope is clipped to $[0, 0.98]$: the upper bound keeps the process strictly stationary, so a projection cannot diverge over four weeks, and the lower bound at zero forbids negative autocorrelation, which on an urgency series would mean a node alternating between calm and crisis every week for no physical reason. The residual scale is floored at 0.01 before the standard errors are computed, so a node whose history happens to be flat still carries some forward uncertainty rather than announcing its current state with certainty. And a degenerate series, one with S_{xx} at machine zero, falls back to $\varphi = 0$ with residuals taken about the mean, which is the honest reading that the history contains no usable dynamics.

The event rate. Each week of history is a Bernoulli trial, a success being a week in which at least one non-reverted event of gravity critical or defect is logged, the same gravities the calibration service uses. With k successes in T weeks the rate τ has the conjugate Beta posterior

$$\tau \mid k, T \sim \text{Beta}(\alpha_0 + k, \beta_0 + T - k), \quad \alpha_0 = \tau_0 N_0, \quad \beta_0 = (1 - \tau_0) N_0, \quad (25)$$

with a prior of mean $\tau_0 = 0.03$ and strength $N_0 = 26$ pseudo-observations, so $\alpha_0 = 0.78$ and $\beta_0 = 25.22$. The strength is the repository convention of half a year of weekly observations, and it is deliberately heavy relative to the campaign: with nineteen turns of history at most, the prior still carries more weight than the data, so an early node with one bad week is not immediately declared high-risk. That is a conservative choice in one direction and a sluggish one in the other, and Section 4.6.1 reports which of the two the campaign actually paid for.

The lead-time law. The moments of the lead-time distribution are drawn through a parametric bootstrap on the same $N_0 = 26$ pseudo-observation convention, so their uncertainty shrinks as $1/\sqrt{N_0}$ rather than being treated as known exactly.

The nested draw, and what the interval covers. These three sources are parameters, not observations, so simulating trajectories at their point estimates would report only the sampling noise of the simulator and would call a forecast precise whenever the simulator was cheap. The rollout therefore nests two loops: $K = 20$ outer draws of the parameters themselves, each carrying $S = n/K$ inner trajectories, for n trajectories in total. Writing $\hat{p}_k$ for the outcome frequency within outer draw k , the estimator and its two variance components are

$$\hat{p} = \frac{1}{K} \sum_{k=1}^K \hat{p}_k, \quad \hat{V}_{\text{mc}} = \frac{1}{K} \sum_{k=1}^K \hat{p}_k (1 - \hat{p}_k), \quad \hat{V}_{\text{par}} = \frac{1}{K-1} \sum_{k=1}^K (\hat{p}_k - \hat{p})^2. \quad (26)$$

This is the law of total variance applied to the parameter vector θ ,

$$\text{Var}(\hat{p}) = \underbrace{\mathbb{E}_{\theta} [\text{Var}(\hat{p} \mid \theta)]}_{\text{Monte Carlo, within a draw}} + \underbrace{\text{Var}_{\theta} [\mathbb{E}(\hat{p} \mid \theta)]}_{\text{parametric, between draws}}, \quad (27)$$

and it is what the two braces of Figure 37 denote. The first component is reducible by simulating more trajectories; the second is not, and shrinks only with more weeks of history.

The reported interval combines them asymmetrically, and the asymmetry is deliberate:

$$\text{se}_{\text{mc}} = \sqrt{\hat{V}_{\text{mc}}/n}, \quad \text{se}_{\text{tot}} = \sqrt{\hat{V}_{\text{par}} + \hat{V}_{\text{mc}}/n}, \quad \text{IC}_{80} = \hat{p} \pm 1.2816 \text{se}_{\text{tot}}, \quad (28)$$

clipped to $[0, 1]$ and tagged in the data model with the coverage it claims. The Monte Carlo term enters divided by n , because it is the standard error of an estimator and shrinks with the simulation budget. The parametric term enters undivided, and that is the point: the quantity of interest is p itself and not $\mathbb{E}[p]$, so the parametric spread is a dispersion the forecast genuinely has rather than an estimation error the reader could average away. An interval built the other way would narrow as $1/\sqrt{K}$ and would report a confident forecast whenever twenty draws happened to be requested. The campaign runs at $n = 500$, so $K = 20$ outer draws carry $S = 25$ trajectories each.

What this component is not. A second and separate path exists in the codebase, a frozen product model that standardises a feature vector, scores it through a logistic regression and passes the result through an isotonic calibrator. It consumes the rollout probability above as one of its input features. It is named here only so that the boundary is unambiguous: it is not the component scored anywhere in this thesis, and every figure in Section 4 comes from the nested rollout of Equation 26, at five hundred trajectories, as the recorded forecast logs confirm.

The estimator is deliberately tied to the target. What p_h estimates is exactly the adverse outcome of the calibration service, a missed milestone or a non-reverted event of gravity critical or defect; the third condition of that definition, node abandonment, is not simulated, because only active nodes are projected and abandonment is an outcome to anticipate rather than an input state. The forecast is therefore the prospective estimator of the quantity the calibration service measures retrospectively. Section 4.7 shows how much rests on that alignment.

Uncertainty is decomposed rather than lumped, through the nested draw of Figure 37: the variance of p_h splits into a Monte Carlo component within outer draws and a parametric component between them, and the reported interval is labelled by the sources it covers. This matters for the reliability analysis of Section 4.6, since an interval that hides which uncertainty it represents cannot be read.

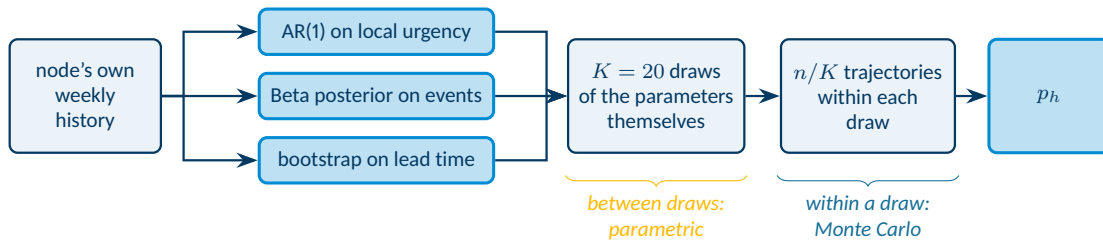


Figure 37: The nested rollout. Three quantities are estimated from the node's own history, then drawn K times through their own uncertainty, the autoregression coefficient and residual scale through their least-squares standard errors, the event rate through its Beta posterior, the lead-time moments through the bootstrap, and each draw carries its own trajectories. The two braces are the two variance components the reported interval distinguishes.

Two disciplines are enforced in the code rather than by convention. The service never writes, so a forecast cannot perturb the state it forecasts. And counterfactual comparisons, which

price a corrective action by rerunning two branches on common random numbers, are named throughout as an effect according to the model and never as a measured effect, because both branches are simulated and neither is an observation.

H.4 Computational complexity and measured scalability

The campaign graphs are small, eight nodes on the first chain and fifteen on the second, so nothing in the results depends on performance. The question this subsection answers is a different one: whether the design would survive a real programme, where a tier-one supply network runs to hundreds or thousands of nodes. Two things are needed for that, a complexity that does not grow faster than the graph and a measurement confirming the constant is small.

Table 28 gives the first. Every entry follows from Proposition 3.2.4(iv), since each operation is a fixed number of topological passes over the graph, and the only quadratic term in the system is the pairwise outranking of PROMETHEE II, which runs over criteria and alternatives rather than over the network.

Table 28: Complexity of the principal operations, in a graph of n nodes and e arcs

Operation	Time	Remark
Topological order	$\Theta(n + e)$	Kahn; nominal arcs only
Full propagation, U_d and U_r	$\Theta(n + e)$	Two passes, one per direction
Incremental propagation	$\Theta(C + E(C))$	Restricted to the affected cone C
Log-survival twin	$\Theta(n + e)$	One logarithm per node and per arc
Local urgency, six blocks	$\Theta(1)$ per node	Fixed number of KPI terms
Adequacy score	$\Theta(1)$ per node	Closed form, Equation 13
Monte Carlo lead time	$\Theta(N(n + e))$	$\Theta(Nn)$ space, guarded at 5×10^7
Criticality index	$\Theta(n_{\text{act}}(n + e))$	Doubled by the twin; no incremental form
Probabilistic criticality	$\Theta(S(n + e))$ per node	Vectorised over the S draws
PROMETHEE II outranking	$\Theta(k q^2)$	q alternatives on k criteria

Table 29 gives the second. The figures are wall-clock means from the benchmark suite, run on a reproducible stress graph of 1 000 nodes and 1 244 arcs over twenty-five ranks, generated from a fixed seed so that successive runs measure the same work. The machine is the development laptop described in Section 3.6, an AMD Ryzen 7 PRO 8840HS with sixteen threads and 11.7 GB of memory, under CPython 3.12.1, with no GPU involved at any point. Each row reports the mean over the number of rounds the harness chose, together with its standard deviation.

Three readings matter. A full propagation of a thousand-node network costs 6.3 ms, so the recomputation triggered by a weekly review is imperceptible and the interactive what-if of Section 3.3 is genuinely interactive rather than nominally so. The incremental pass is 6.1 times faster than the full one, which is the empirical form of Proposition 3.2.4(iii): the cone of a single changed node is a small fraction of the graph, and everything outside it is read from cache without approximation. And the Monte Carlo layer at 479 ms is two orders of magnitude above the analytic path, which is why it is an opt-in per project rather than the default, and why the memory guard exists at all: at ten thousand draws over a thousand nodes the simulation holds 10^7 completion times in memory at once.

Table 29: Measured performance on a 1 000-node, 1 244-arc stress graph

Operation	Mean (ms)	SD (ms)	Rounds
Incremental propagation, one node changed	1.04	0.25	472
What-if shock on a rank-25 supplier	4.99	1.06	224
Full propagation, U_d and U_r	6.34	1.12	129
Dashboard network figure	43.7	1.0	5
Evaluate every node and persist	105.2	16.2	3
Monte Carlo lead time, $N = 10\,000$	478.8	33.9	3
Load the graph from the registry	782.1	39.8	3

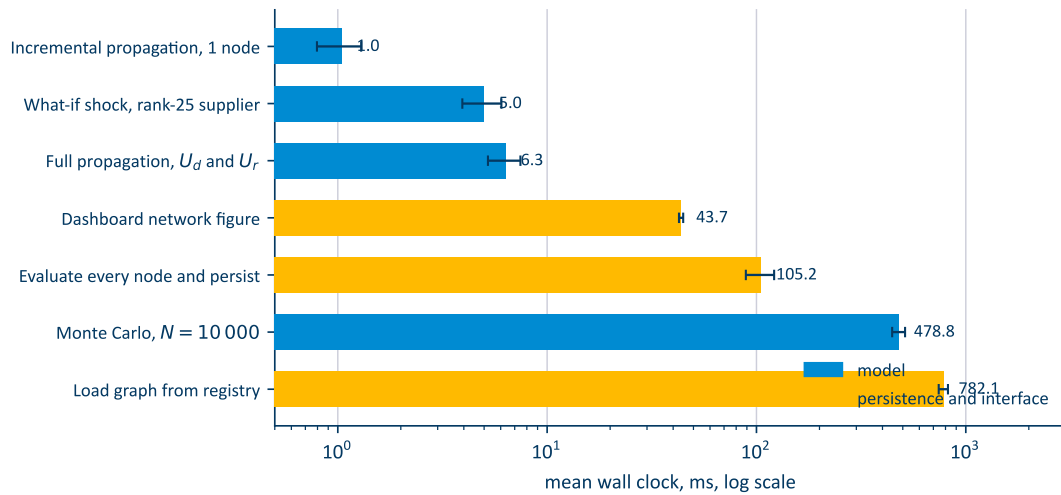


Figure 38: The same seven operations on a log scale, model work in azure and persistence or interface work in ochre. Three orders of magnitude separate an incremental propagation from a graph load, and the operations a user waits on repeatedly are the fast ones.

The one operation that dominates is loading the graph from the registry at 782 ms, and it is a persistence cost rather than a model cost. It is paid once per session, not per recomputation, which is the reason it was left alone: optimising it would improve a number nobody waits on twice.

I Observed Capacity: An Opening

Section 5.4 records that the satellite work is reported as an opening and claimed as no part of the contribution: nothing joins it to the urgency model. It is set out here for that reason, with its protocol and data in Appendix E.

Every KPI reaching the six blocks arrives either through a manual weekly review or through a scripted event, so a node's score is never fresher than its last submitted review, and the capacity block in particular depends on the same operator whose declaration the adequacy score exists to audit. The independence that Section 4.11 identifies as the precondition for the whole measurement is therefore not structurally guaranteed.

A second system, developed alongside the main work, addresses one part of that dependency by estimating the physical capacity of a warehouse from public satellite imagery, which is an

observation no operator supplies and none can bias. It is an opening rather than a second contribution: the two systems have not been connected, and no claim is made that they have. The pipeline takes a coordinate, retrieves the building footprint from the French national topographic database, retrieves 20 cm orthophotography and, where available, a LiDAR surface model, and derives a storable volume as $0.80 \times \text{footprint} \times \text{height}$ and a transfer capacity as the equipped facade length divided by a dock pitch of 3.8 m. Both constants are calibrated rather than assumed, and Appendix E gives each with the sample behind it, together with the one that does not survive audit: the fill factor of 0.80 is attributed in the source to a 122-site calibration whose detailed output was not retained, and recomputing it over the 68 sites still on disk gives a median of 0.97. The volumetric figures below carry that unquantified bias; the dock figures, which depend on the pitch instead, do not.

I.1 The Physical Limit and the Reframing

The original objective was to count loading docks. That objective is unreachable, and establishing why is the most transferable result of this work.

An empty dock door under a canopy produces no measurable signal from directly overhead. Measured contrast at verified empty bays is 0.18σ , 0.22σ and 0.01σ across a full sweep of assumed depths, while occupied bays on the same facades register at 5.3σ to 6.4σ . The measurement is sensitive enough; the signal is absent. Repeating the test on 8 cm very-high-resolution imagery leaves the doors equally invisible, which shows the cause to be occlusion by the canopy roof rather than insufficient resolution. A gain of $2.5\times$ in ground sample distance recovers no additional door: what improves with resolution is everything except the quantity of interest. Any method claiming to count empty doors one by one from a nadir view is recovering an artefact.

That finding splits the problem in two. Counting each door individually is not observable at nadir, and no amount of resolution changes it. Recognising that a facade is equipped for docking, and delimiting its extent, is observable, because the canopy, the apron and the bay rhythm remain visible whether or not a vehicle is present. The second formulation is learnable, and it is sufficient, because capacity is a function of equipped length rather than of individually identified doors. Figure 39 is the direct evidence: three facades of 135 m, 105 m and 15 m on which the automatic trailer detector found no vehicle at all, where individual doors are not resolvable and the bay markings and canopy edge are.

One reservation is recorded honestly. The proportion of sites showing no usable dock signature was measured on a subsample that required vehicles to be present to establish ground truth, and is therefore biased towards sites with vehicles. Five zero-shot approaches were tried before the reframing and each was abandoned for a measured reason; Appendix E records them, because the reasons constrain what a successful method can look like.

I.2 Learned Dock-Zone Segmentation

The reframed problem was addressed by training a segmentation model to delimit equipped facade zones directly. Table 30 gives the progression across four training runs, read at the best epoch of each by mask mAP50-95.

Figure 40 makes the cause visible: quadrupling the training set from 86 to 351 images roughly

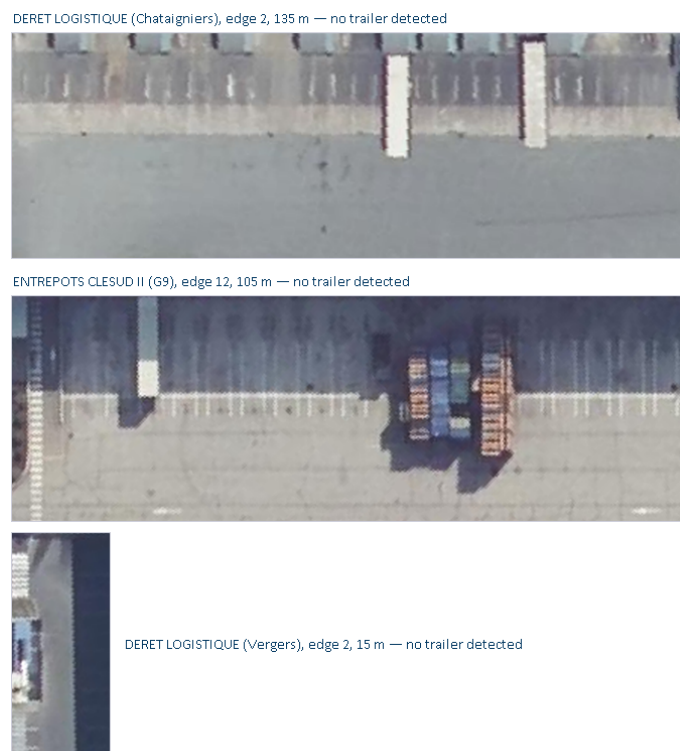


Figure 39: Equipped facades on which no trailer was detected

Table 30: Segmentation training progression

Run	Train / val	Image size	Precision	Recall	mAP50	mAP50-95
1	86	640	0.461	0.404	0.363	0.155
2	203 / 50	640	0.635	0.539	0.610	0.283
3	351 / 91	640	0.702	0.606	0.618	0.305
4	351 / 91	1280	0.652	0.547	0.570	0.272

doubled mask mAP50-95. Resolution did not: run 4 is identical to run 3 except for a doubled input size, and it is worse on every metric, so at a fixed data budget and a CPU-only training budget the additional pixels cost more than they bought.

What is measured matters as much as the score. Standard detection metrics score shape overlap, but the operational quantity is capacity, which depends only on the length of the equipped zone: a polygon drawn too deep, or with imprecise edges, still yields the correct dock count. The project therefore scores itself on the error in the derived dock count rather than on mAP, and that business metric is the one used to decide whether a model is good enough to deploy. On the reference run it reports a global bias of -35% , a median error of 62% on images containing docks, and 89% of true negatives correctly left empty, which is a system that is directionally useful and quantitatively not yet trustworthy.

The pipeline runs end to end on a geographic area, retrieving footprints, presenting each to the trained model, assigning predicted zones to the correct building and writing capacity figures. One defect found during that deployment is recorded in Appendix E because it illustrates the difference between a model error and a systems error, as is the annotation

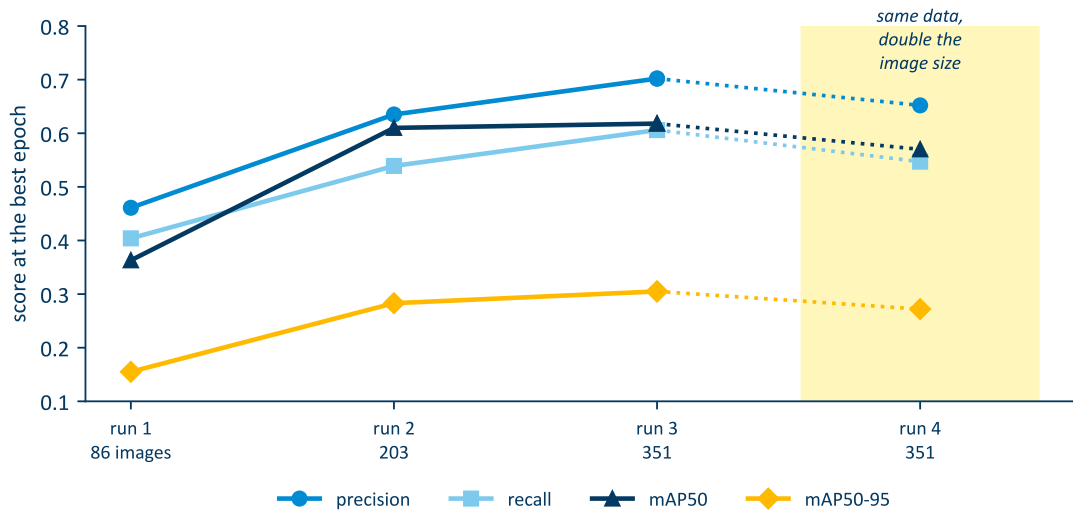


Figure 40: The same four runs plotted. Runs 1 to 3 raise the training set at a fixed image size and every metric rises with it. Run 4 holds the set and doubles the image size instead, and every metric falls. The binding constraint was annotated data rather than resolution, which is a conclusion about the corpus and not about the architecture.

episode in which an automated third-party process reported 442 annotations as verified and complete while having placed a zone along every edge of every building without examining the imagery at all.

I.3 What This Can and Cannot Feed

The system supplies a storable volume, a pallet-position estimate and, where a dock zone is detected, a transfer capacity and an occupancy ratio. Those map onto the capacity block of Equation 4 and onto nothing else: it says nothing about lead time, cost or emissions. Coverage is metropolitan France, bounded by the public data services it draws on. The trained segmentation model is not yet wired into the production capacity path, which still uses the zero-shot trailer method and its 36% coverage. And the imagery is a snapshot, so the resulting figure is a structural capacity rather than a live occupancy.

What it does change is the provenance of one input. A capacity figure derived from imagery is not supplied by the operator whose declaration the adequacy score audits, which is precisely the independence Section 4.11 identifies as the precondition for the measurement to mean anything. Figure 41 states where that input would enter and, equally important, which blocks it would leave untouched.

I.4 Reference tables

The tables below define the terms of the model equations, list the campaign's network and events, and record the two literature comparisons. Each is referred to from the body at the point it applies.

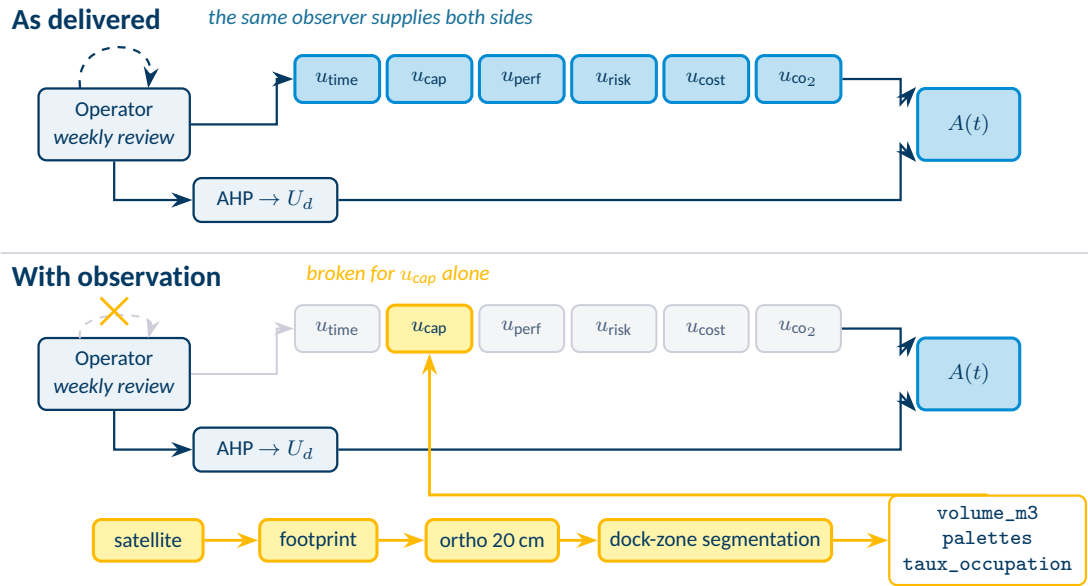


Figure 41: Where satellite-observed capacity would enter the real-urgency model, and which blocks it does not reach. Above, every block and the declaration itself originate with the same observer, which is the circularity of Section 4.11. Below, one block of six is fed by an observation no operator supplies, so the circular dependency is cut for that block; the other five, greyed, still arrive through the weekly review.

Table 31: DISCO technology stack

Component	Technology / Value
Backend API	Spring Boot + Java 21
Database	PostgreSQL (Azure)
Frontend	Angular
Authentication	Player key / Azure AD SSO OAuth2
Scores computed	CO ₂ , Cost, Risk per supplier and itinerary, on [1, 10]

Table 32: Aggregation and growth models found in the literature. Figure 3 plots what each one does.

Model	Formula	Where it appears
Linear in time	$U_0 + \lambda t$	Deadline-sensitive scheduling (K. Chen et al., 1996)
Hyperbolic in time	$K/(t_c - t)$, bounded	Temporal discounting (Steel et al., 2006); crash precursors (Johansen et al., 2001)
Weighted average of blocks	$\sum_m \omega_m u_m$	Composite indicators (Greco et al., 2018)
Noisy-OR over blocks	$1 - \prod_m (1 - u_m)^{\omega_m}$	Combining independent causal evidence (Pearl, 1988)

Table 33: Temporal Motivational Theory parameters

Symbol	Concept	Operational role
$V(t)$	Temporal motivational value	Dynamic task salience
U	Objective utility	Task strategic importance
E	Expectancy of success	Probability of task completion
$D(t)$	Delay to deadline	Remaining safe execution window
Γ	Impulsiveness parameter	Operator sensitivity to delay
k	Sensitivity scaling parameter	Non-linear scaling of delay discounting

Table 34: The five families of precursor, and what each contributes

Family	Why it precedes rupture	Sources
Slack	The margin between a realistic lead time and the committed deadline. The only indicator whose exhaustion is irreversible: a task with no slack left cannot be recovered	(Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013)
Capacity saturation	Adds waiting time a nominal lead time does not show, so a loaded system carries a risk its announced schedule conceals	(Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024)
Resource performance drift	Monitored through composite proxies, most often OEE, because direct performance data is rarely available on the supplier side	(Mokhtar et al., 2019; Hlioui et al., 2017)
Adverse-event exposure	Treated probabilistically, as failure probability times recovery duration times impact severity	(Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020)
Cost signals	Symptoms of strain rather than causes: paying air freight to protect a deadline is evidence the system is already compensating	(Heckmann et al., 2015; Mokhtar et al., 2019)

I.5 Summary tables of the conclusion

The three tables below summarise, in one place, what each research question returned, what each contribution rests on, and what each item of future work takes. Each is referred to from Section 5.

J The Repair and Its Re-measurement

The campaign identified one defect and specified one repair, which was then carried out and re-measured against the identical frozen campaign. The declarations, the scenario and the event script are unchanged and the same 120 forecasts are recomputed by the modified model, so every difference below belongs to the model and to nothing else.

Table 35: The four methods on the three axes that matter for a weekly-cadence instrument, with n criteria and q alternatives

Method	Judgements	Returns	Compens.	Known weakness
AHP (Saaty, 1980)	$n(n-1)/2$	criterion weights, with a consistency ratio	yes	rank reversal on adding an alternative; sensitive to cognitive noise during elicitation (Dyer, 1990; Triantaphyllou; Yanase, 2024)
BWM (Rezaei, 2015)	$2n-3$	criterion weights	yes	anchors on a best and a worst the operator must be able to name
Fuzzy BWM (S. Guo et al., 2017)	$2n-3$	fuzzy weights, defuzzified	yes	non-linear programme; needs a feasible start, and consistency conditions were established only later (Liang et al., 2020)
PROMETHEE II (Brans; Vincke, et al., 1986)	$q(q-1)$ pairs per criterion	a complete ranking of alternatives	no	not independent of irrelevant alternatives: adding an alternative can invert two existing ranks (Triantaphyllou, 2000)

Table 36: Selection filter applied to each candidate block

Criterion	Question asked of each candidate indicator
Causal link	Does the indicator have a plausible mechanism linking it to delivery rupture?
Observability	Can it be measured or credibly estimated from client, supplier or planning data?
Actionability	Can an operator act on it, or use it to decide?
Non-redundancy	Does it add information not already carried by a retained indicator?

Table 37: Terms of the propagation equations

Component	Type	Description
$U_{d,i}, U_{r,i}$	Dependent variables	Propagated declared and real urgency at node i
$U_{d,local,i}, U_{r,local,i}$	Input variables	Local urgency at node i after status-rule adjustment
γ_{ik}	System parameter	Downstream demand attenuation coefficient on arc $i \rightarrow k$
β_{ji}	System parameter	Upstream risk amplification coefficient on arc $j \rightarrow i$
$Succ(i), Pred(i)$	Structural sets	Nominal successors and predecessors of node i

Table 38: Terms and parameters of the adequacy equations

Component	Type	Description	Default
F	Dependent variable	False urgency, from over-declaration	n/a
H	Dependent variable	Hidden risk, from under-declaration	n/a
λ_{under}	System parameter	Penalty weight on hidden risk	2.25
λ_{over}	System parameter	Penalty weight on false urgency	1.0
α	System parameter	Psychophysical curvature	0.88
A	Dependent variable	Adequacy score, 100 meaning perfect alignment	[0, 100]

Table 39: The eight HÉLIOS campaign nodes

Node	Role	Rank	Real-world anchor
Orbitalys	Satellite prime, final customer	0	Fictional satellite programme, contractual delivery pressure
AvioSys Intégration	Avionics integrator	1	Qualified-stock squeeze on avionics tier-1 suppliers
Électis EMS	Electronics manufacturing services	2	EMS plants placed under allocation, 2021
CompoDis Europe	Component distributor	3	Distributor allocation and spot-market premiums
TransGlobal Fret	Freight operator	3	Suez Canal blockage, West Coast port congestion
NovaFab Semiconductors	Foundry, expected bottleneck	4	Composite of the Texas winter-storm freeze and the Taiwan drought
Meridian Semi	Secondary foundry, inert backup arc	4	A foundry sold out through 2023, the announced plan B that delivers nothing
SilPure Materials	Raw wafer supplier	5	Roughly 60% of world wafer supply, polysilicon price roughly tripling in 2021

J.1 The Repair That Was Recommended Does Not Work

Section 4.11 proposed making the calibrated events write a lead-time impact. Implemented and measured, that repair fails, and the reason is the one Figure 42 draws: a completion date estimated as a full production cycle beginning now multiplies the lead time by the fraction of work remaining, so a quantity routed into the lead time is scaled by that fraction too. On NovaFab, the node carrying the scenario's accident, the four-week probability fell from 0.846 to 0.000 on a milestone that was in fact missed.

The completion model was rewritten accordingly, identically in the three independent estimators the system carries for the same quantity, the analytic u_{time} block, the Monte Carlo forecast rollout and the PERT lead-time simulator:

$$C = t + (1 - p)L + R, \quad (29)$$

where p is the declared progress, L the lead time and R the time already lost. Five families of capacity-halting event, machine failure, accident, strike weighted by the share of workforce involved, material shortage weighted by criticality, and cyber incident, now accumulate their

Table 40: The twelve calibrated engine events

Turn	Node	Event type	Real-world anchor
T3	NovaFab	Demand spike	Automotive reorders on sold-out slots, Nov. 2020
T5	CompoDis	Supplier delay	Distributor moves to allocation, Jan. 2021
T6	NovaFab	Accident (critical)	Texas winter-storm fab freeze, Feb. 2021
T7	NovaFab	Demand spike	Reorders after a competitor's fab fire, Mar. 2021
T7	TransGlobal	Transport disruption	Suez Canal blockage, six days, Mar. 2021
T10	NovaFab	Material shortage	Taiwan drought, ultrapure-water rationing, Jun. 2021
T12	Électis	Capacity loss	Southeast-Asia backend lockdowns, Aug. 2021
T12	TransGlobal	Transport disruption	Shift to saturated air freight, Aug. 2021
T13	CompoDis	Demand spike	Bullwhip over-ordering, Sep. 2021
T14	TransGlobal	Transport disruption	West Coast port congestion, Oct. 2021
T14	CompoDis	Tariff increase	Allocation spot premiums, Oct. 2021
T16	SilPure	Tariff increase	Polysilicon price pass-through, Dec. 2021

Table 41: Final per-node snapshot of the arm A project

Node	U_d	U_r	A	F	H
Orbitalys	0.409	1.000	15.3	0.000	0.591
AvioSys Intégration	0.727	0.698	95.2	0.029	0.000
Électis EMS	0.914	0.965	82.9	0.000	0.052
CompoDis Europe	0.882	1.000	67.5	0.000	0.118
TransGlobal Fret	0.819	1.000	56.0	0.000	0.181
Meridian Semi	0.841	0.998	60.1	0.000	0.157
NovaFab Semiconductors	0.912	0.915	98.5	0.000	0.003
SilPure Materials	0.876	0.304	48.9	0.572	0.000

Table 42: The time block at turn 6

Node	u_{time}
Orbitalys	7.3×10^{-9}
AvioSys Intégration	1.2×10^{-8}
Électis EMS	0.0
CompoDis Europe	0.99996
TransGlobal Fret	1.7×10^{-4}
Meridian Semi	2.9×10^{-9}
NovaFab Semiconductors	0.0216
SilPure Materials	0.0

Table 43: What each research question returned

Question	Answer
RQ1 multi-block real urgency	Specified, implemented and exercised over two chains. The construction works and is exactly decomposable, which is what allowed its own defect to be found: five blocks behaved as intended and the time block was structurally unable to respond to a disruption, because the event engine wrote to no quantity it reads. That path has since been built and measured (Section 4.10). The aggregation is only as informative as its inputs, and the campaign that produced the negative result ran with three of the six blocks inert.
RQ2 adequacy formulation	Bounded, interpretable and reconstructible term by term. Its validity splits by granularity. At node level it discriminates and does so early: Section 4.5 reports six consecutive turns on which the flagged set is a subset of the eventually failing set, with no false alarm, ahead of any public sign of the crisis. At the granularity of a probability attached to a node-week it was not demonstrated, and Section 4.6.1 attributes that to one repairable cause rather than to the formulation. Usable today for triage, not yet for probability.
RQ3 prioritisation and criticality	AHP and Fuzzy BWM elicitation, PROMETHEE II outranking and systematic shock simulation combine into a working ranking that placed the expected bottleneck at the top. Two behaviours bound it. The criticality index degenerated when many nodes saturated at once, which a log-survival sort key has since addressed (Section 4.10); the outranking inherited a perfect correlation between two of its criteria, which the tool flagged on its own.
RQ4 validation, and the effect of display	Operationalised as a persistent twin and exercised through two pre-registered campaigns. Five of six hypotheses are not confirmed and H5 on urgency contagion is confirmed. On the second half of the question the answer is unambiguous within its population: at first exposure no declarant on either chain revised upward, and across all exposed turns of the second chain upward revisions outnumber downward ones 26 to 9. The display transfers authority to the model rather than sedating the operator.

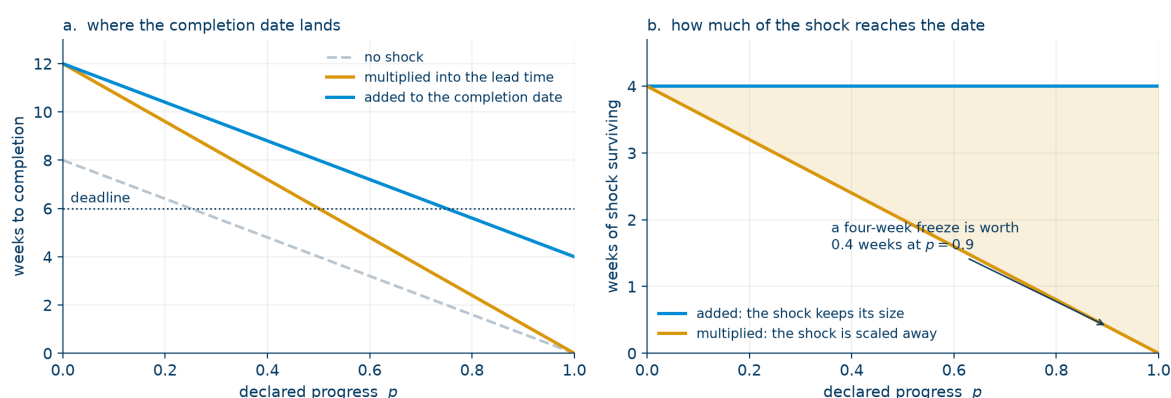


Figure 42: The two repairs on one milestone, with a nominal lead time of eight weeks and four weeks lost to a shock. Routing the lost time through the lead time (ochre) scales it by the work remaining, so a four-week freeze is worth 3.6 weeks at a tenth complete and 0.4 weeks at nine tenths; adding it to the completion date (azure) keeps its size throughout. The milestones on which the two disagree most are those nearest their deadline, which are precisely those on which a warning has value.

Table 44: Each contribution, the measurement behind it, and what bounds it

	Measurement	Bound
C1	3 nodes flagged over the 6 turns before the crisis is public; 3 of 3 later miss an <i>originally committed</i> deadline; 0 false alarms; worst case flagged 8 turns ahead; one flag on a node declaring $U_d = 0.001$	4 positives, Wilson $[0.44, 1.00]$; signal weakens once the crisis is public, so the scope is early warning
C2	noisy-OR real urgency, bidirectional DAG propagation, consistency-gated declaration, asymmetric penalty: specified, implemented, operable, exactly decomposable	fills the combination Section 2.14 finds absent; a claim about the reviewed literature, not about optimality
C3	Brier skill -3.4 to -6.9 ; 21 confident forecasts, 0 materialised. Cause: the accumulated-delay field written non-zero <i>exactly zero times</i> ; block numerically zero at 6 of 8 nodes mid-crisis	a count, so sample size does not bear on it; after repair $-0.553 \rightarrow -0.216$ at 4 weeks, badly wrong to less wrong
C4	8 of 8 declarants revised down or were confirmed, 0 up, on first exposure, 2 turns before the critical event; both arms carry numerically identical forecasts	attributable to the display, for a population of agents; human extension unestablished (Assumption 3.2)
C5	naive target: base rate $5.0\% \rightarrow 38.7\%$, apparent skill $\times 5$, discrimination unchanged. Original-commitment target: AUC $0.485 \rightarrow 0.654$, skill $-3.36 \rightarrow -0.55$	the three targets disagree by more than any modelling decision in this work; no figure is quotable without its definition

effective duration into R . Loss of storage capacity deliberately does not, since it removes space rather than production hours.

The mechanical effect is unambiguous. Where the accumulated-delay field was written a non-zero value exactly zero times across the whole of arm A, after the repair NovaFab accumulates 154.7 engine hours at its first shock and 371.2 over the campaign, carried in 27 of its 41 snapshots.

J.2 Two Further Defects Exposed by the First

With the time block responsive, a second defect became measurable. The milestone-miss probability returned exactly 0 or exactly 1 in 85.0% of the 120 forecasts, leaving 14 anywhere in between: a switch rather than a probability, as two declarants recorded in free text before it was measured. The cause is that the rollout drew the lead time at random while taking declared progress as an exact value. Declared progress is the least reliable input the system receives and belongs to precisely the class of quantity the framework exists to doubt, so the

Table 45: Each item, what it costs, and what it unblocks

Item	What it takes	Cost	
Before the framework can be evaluated fairly			
F1	Separate misperception from discretion	Instrument the escalation channel, so the interval between recognising and communicating becomes observable; or read the free-text notes the system already collects, in which second chain declarants stated an intent to delay contacting customers while marking near-maximal severity in the same submission	none for the second route
F2	Widen the evidence base	More chains, longer horizons, more milestones. Three routes without new science: replay documented disruptions on the same harness; instrument a live chain through the weekly review, already supported; generate synthetic scale for the calibration layer, keeping the real campaigns as the evaluation set	scenario authoring
F3	Run the human campaign, outcome definition pre-registered	Protocol, hashed data pack and scripts are ready. Run against the <i>repaired</i> model. Arm C in its agent version met 5 milestones of 11 on original commitment against 5 of 11 for arm A: it changed <i>which</i> milestone was met, saving one and replanning one that would have held. That is the quantity to power for	eight participants, nine days
And then			
F4	Fill the indicators, rebuild the calibration label on the product's target	Coverage rose 25.6% $\rightarrow$ 49.1% on completion from public figures, activating all six blocks and exposing a model bug undetectable on an empty database. The post-hoc calibrator labels an event at exactly $t + 1$ where the product reads a miss over $(t, t + 4]$, which costs 0.721 $\rightarrow$ 0.580 of four-week AUC. Blocked on the synthetic factory recording milestone truth	concrete, blocked
F5	Smooth the milestone alarm	It moves from 0.05 to 1.00 in a single turn: correct, and unusable as advance warning	low
F6	Make the forecast state its own ignorance, and control the display	Make the horizon salient so a low four-week figure cannot read as general safety; state the risk classes not covered. Arm B is not blinded, and turns carrying a deliberately uninformative forecast would separate the effect of the content from the effect of being given a number at all	low
F7	Connect the observed capacity channel	Wire the trained segmentation into the production capacity path in place of the trailer-span method; re-measure the no-signature rate on an unbiased draw; recollect the regulatory corpus to re-derive the fill factor. The LiDAR surface model is the most promising unexplored route, the canopy step sitting several metres below the main roof, which would correct the 0 to 14.4 m overhang now displacing the apron boundary	medium
F8	Broaden elicitation, couple the platforms	Multi-expert AHP aggregating several judgements per node; couple SupplyScore with the DISCO platform it runs alongside. Both depend on what a repaired system shows	medium

forecast layer was treating as certain a figure the adequacy score is built to suspect. Progress is now drawn per replicate, with a standard deviation widened node by node in proportion to that node's own measured hidden risk H , so the system's adequacy measurement feeds its own forecast.

The third defect was in the data rather than the model, and it would have invalidated any calibration performed before it was found. Indicator coverage on the campaign database was 25.6%: 24 of the 42 indicators were empty on all eight nodes, the carbon and cost blocks were inert everywhere, and only three of the six urgency blocks contributed. The lead time itself was present on five nodes of eight, and the remaining three received a risk floor of zero through the documented fallback for a missing lead time, which is an absence of data presenting itself as a reassuring prediction. Completing the indicators from public figures of the 2020 to 2022 shortage raised coverage to 49.1% and made all six blocks active. One model bug was undetectable until then: a per-node recovery capability, a 26-week aerospace requalification time, was being read as production time already lost, pinning that node's time block at 1.0 from the first turn. An empty database conceals the errors of the model that consumes it, which is a methodological point independent of this framework.

J.3 The Re-measurement

Table 46 scores the recomputed forecasts against the original-commitment definition of Section 4.7, at all four horizons, on the same censored samples of 112, 104, 96 and 90 observations. The middle column of each pair isolates the structural repair; the right-hand column adds the progress-uncertainty term.

Table 46: The frozen campaign rescored after repair, original-commitment target

Horizon	Brier skill			AUC		
	before	structure	+ uncertainty	before	structure	+ uncertainty
1 week	−1.278	−0.646	−0.687	0.736	0.793	0.729
2 weeks	−0.690	−0.401	−0.367	0.636	0.671	0.634
3 weeks	−0.654	−0.360	−0.286	0.641	0.657	0.602
4 weeks	−0.553	−0.272	−0.216	0.654	0.721	0.584

The structural repair improves both axes at every horizon without trading one for the other, which is the clean result of this thesis's second measurement campaign. The progress-uncertainty term is a trade rather than a gain, and the AUC movement it costs is not interpretable. With 26 positive observations the cluster-bootstrap interval on the four-week AUC runs from 0.21 to 0.87 and contains every value in its row, and the same objection that bounds the original discrimination claim of Section 4.6 bounds this one. The term was retained on the calibration evidence and on the argument that a declared quantity should not enter a probabilistic forecast as a certainty, not on the AUC.

Three qualifications belong beside the table. The Brier skill remains negative at every horizon, so the forecast is still worse calibrated than a forecaster that announces the base rate and stops; the repair moves the system from badly wrong to less wrong, not to useful. The milestone probability remains binary in 71.7% of cases against 85.0% before, so the over-confidence is reduced rather than removed. And the post-hoc calibration layer that Section 4.11

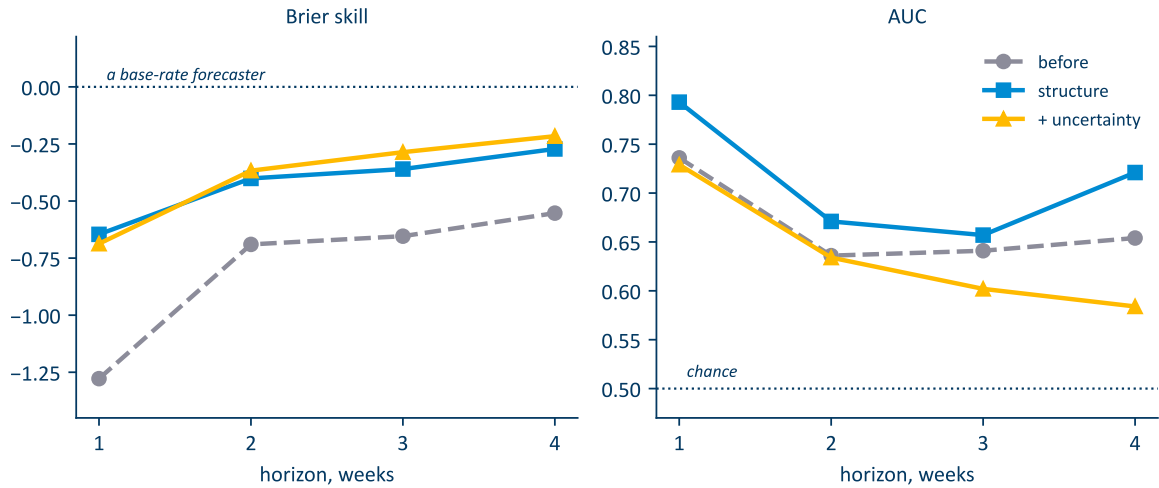


Figure 43: The same three model versions on both metrics. The structural repair lifts calibration and discrimination together. The progress-uncertainty term then lifts calibration further at every horizon beyond one week and lowers discrimination at every horizon, which is a trade rather than a gain and is why it was retained on the calibration evidence rather than on the AUC.

recommended was trained, measured, and deliberately not connected: on the synthetic factory that produced its training data it improves the Brier skill by 30.8% and the AUC from 0.792 to 0.850, but applied to the campaign it improves calibration at every horizon while dropping the four-week AUC from 0.721 to 0.580. Its training label is an event striking the node at exactly $t + 1$; the product displays a missed milestone or a critical event cumulated over $(t, t + 4]$. The layer cannot rank the cases driven by the milestone channel because it has never seen one. This is the trap of Section 4.7 encountered from the opposite direction: a calibration fitted against the wrong target degrades a system that works, exactly as an evaluation against the wrong target can condemn one. Connecting the layer requires the training label to be rebuilt on the product's target, which requires the synthetic factory's snapshots to record milestone truth, which they do not; that is the next concrete step rather than an open research question.

J.4 A Third Arm, and What It Does Not Show

The repaired system was then exercised through a full nineteen-turn campaign in closed loop, in which the same eight agents no longer only declared but also selected actions from a fixed catalogue, applied to the chain by the mutation service and recorded in the intervention journal: promote a backup arc, replan a milestone, expedite a shipment, raise capacity, revise a declaration, or do nothing. One hundred and fifty-two declarations were recorded with no unresolved consistency failure. The anti-contamination protocol of Section C.2 was tightened for this arm: every message reaching an agent is written by the tool into that agent's own sandbox, and a failed consistency check is answered with the interface's own guidance rather than with a coordinator's instruction.

The headline figure flatters the framework and should not be used: six of eleven milestones were missed in the frozen scenario and none was left unfinished in the closed loop, which reads as six saved. Against the original commitments it is not, and Figure 44 gives the honest count. The saved milestone is an electronics series delivered on its original date that had slipped in

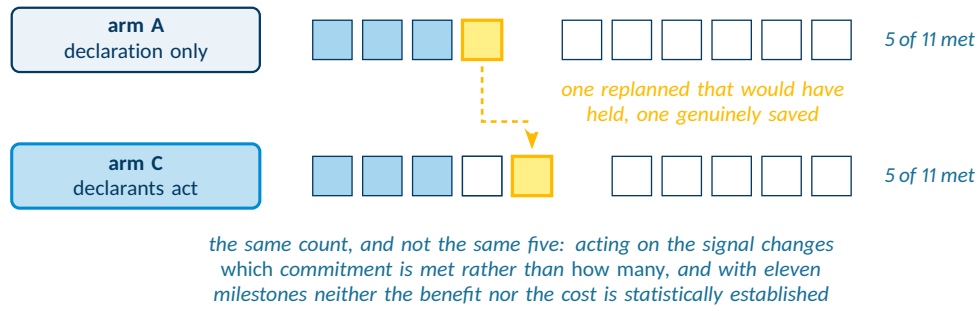


Figure 44: Milestones met on their original commitment, arm A against arm C on the repaired system. The totals are identical and the sets are not: the two ochre squares are the swap. That is the effect a human campaign would need to be powered to detect.

the frozen truth. Beyond the swap, four late deliveries became replannings announced in advance rather than slips discovered at the deadline, which has value to a customer without being a save. Both the benefit and the cost are reported as facts of one campaign rather than as results.

Two observations from that arm do carry. The arm B disarming reproduced exactly, eight agents of eight at first exposure. And the causal chain the campaign showed to be broken now closes: where arm A's foundry manager recorded that the tool was holding a low position through a widely reported crisis and took no action, the repaired run moved the same node's four-week probability from 0.088 to 0.214 at the shock turn and the agent replanned its milestone. One reserve on internal validity must be stated: the model changed twice during that campaign, at turns 15 and 16, so turns 16 to 18 are not comparable with what precedes them. Every behavioural observation reported above is drawn from turns strictly earlier than 15.

J.5 A Hypothesis That Became Measurable Again

The criticality index degenerates under saturation, which forced H3 to be judged on the ten informative turns of nineteen and the other nine to be excluded. That exclusion rule is no longer necessary. The ranking now carries a log-survival key, $\ell(p) = -\ln(1 - p)$ computed on an ε -regularised twin of the same propagation, which orders identically to ΔU_r outside saturation, a property-based invariant with a test to match, and keeps separating nodes inside it.

Re-measured on the repaired run, the two criteria decompose cleanly on the same data. The old criterion alone leaves 10 informative turns with NovaFab top-1 on 9 of them; adding the $\Delta \ell$ key leaves 17 informative turns of 19 with NovaFab top-1 on 16, a share of 94% against a pre-registered threshold of 80%. Holding the criterion fixed instead and changing only the model, the frozen campaign gives 70% and the repaired one 90%, so completing the indicators moves the figure as much as the new key does.

This is reported as exploratory and does not re-decide H3, for the same reason given for H4 in Section 5.3: a hypothesis re-run against a changed model no longer tests what the register asked, and two things changed at once here. Only the comparison of the two criteria on one run isolates a single cause. The defensible claim is the narrow one: the degeneracy that made nearly half the campaign unreadable is largely gone, and the quantity H3 tests now sits above

its threshold on the repaired system.